Monopolistically Skewed Business Cycles*

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March 9, 2026

Abstract

The cross-sectional distribution of firm growth rates shifts systematically over the business cycle: mean growth falls sharply in recessions, dispersion rises in both large contractions and expansions, and skewness is procyclical. Critically, skewness swings are twice as large for large firms as for small ones, a pattern we call the size gradient of skewness. We provide the first joint explanation of all three moments and the size gradient through a single mechanism: weak restrictions on demand imply that firms' optimal policies are log-concave in shocks, and concavity rises with market power. One aggregate shock, transmitted through heterogeneous firm exposures and concave policies, shaped by our demand restrictions simultaneously generates all four patterns. Empirical evidence using impulse responses to identified aggregate shocks confirm that growth and skewness comove tightly, with the skewness response concentrated among large firms and traced to market power through industry concentration and profitability sorts. Cyclical asymmetry is an endogenous outcome of market structure: rising concentration amplifies business cycle downturns through the left tail.

Keywords: business cycle, skewness, market power
JEL Codes: D21, E32, L11

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1 Introduction

This paper documents and explains how the distribution of firm growth rates evolves over the business cycle. Using quarterly Compustat data on U.S. public firms, we show that mean growth, dispersion, and skewness move systematically in response to aggregate shocks, and that these movements differ sharply between large and small firms.

Empirically, firm outcomes display pronounced distributional movements over the business cycle. The variance of firm outcomes rises in recessions (Bloom, 2009), skewness varies over the cycle (Salgado et al., 2025), and the cyclicality of these outcomes differs systematically by firm size (Crouzet and Mehrotra, 2020). These facts have motivated increasingly rich business-cycle models, often introducing multiple aggregate disturbances in order to discipline different features of the cross-sectional distribution. These range from first-moment productivity shocks, to second-moment uncertainty shocks, and third-moment tail-risk shocks. This proliferation raises the question of whether higher-order shock processes are truly needed, or whether their apparent importance reflects the absence of a structural propagation mechanism that links first-moment shocks to higher-moment outcomes. We take the latter view. We show that a single aggregate shock, combined with heterogeneous firm exposures (as in Davis et al., 2025) and standard properties of product demand, is sufficient to generate the observed comovement in mean growth, dispersion, and skewness. The mechanism operates through transparent economic primitives — demand elasticity and market power — making it interpretable, testable, and policy-relevant.

We begin by documenting three empirical regularities about the first three moments of firm growth rates that, to our knowledge, have not been explained jointly in a micro-founded model. First, growth fluctuates around a trend but falls sharply in recessions, with no comparably sharp expansions.[1] Second, dispersion rises with the absolute deviation of growth from trend, increasing in both strongly negative and strongly positive growth episodes (Table 1, Panel A). Third, skewness is procyclical: left-skewed[2] when growth is falling and right-skewed when growth is strong (Salgado et al., 2025, see also Table 1, Panel A). Crucially, this cyclicality is not uniform across the firm size distribution: the amplitude of skewness swings is roughly twice as large for large firms as for small ones (Table 1, Panels B and C). We call this the *size gradient of skewness*. It increases monotonically with firm size and we trace it to market power: *(i)* empirically, the gradient correlates strongly with profitability and industry concentration, and *(ii)* theoretically, a higher market power index increases the curvature of firm output under realistic sufficient conditions.[3]

[1] This asymmetry has been documented at least since Sichel (1993).

[2] We use the term left-skewed (right-skewed) synonymously with negatively skewed (positively skewed).

[3] The connection between size and market power has also been shown to hold empirically (e.g. Autor et al., 2020)

Our theory provides a demand-side microfoundation for these facts. The key mechanism is that if demand satisfies a strengthened version of Marshall's Second Law of Demand (MSLD), meaning that demand becomes more elastic as prices rise, firms' optimal policies map shocks into log output in a concave fashion. This condition is not demanding: it holds for most commonly used demand systems away from the CES benchmark. We link curvature of log output policies to the implied cross-sectional growth rates. Concavity then implies negative growth skewness on impact of an adverse aggregate shock and positive skewness during the recovery. Indeed, the close match between our theoretical predictions and the data constitutes strong evidence that firm production policies are log-concave.

Formally, we show that MSLD together with an increasing pass-through rate (IPT) condition is sufficient for the curvature of firms' log-output responses to become more pronounced as market power rises. Under IPT, marginal-cost increases pass through more strongly into prices for firms whose cost is already high. This *monotone concavity* provides a direct explanation for the size gradient of skewness. We also show that CES demand is a knife-edge case in which growth rates are unskewed. Away from CES, asymmetries emerge generically. Since CES is rejected in many industries, asymmetric growth-rate dynamics should be expected as a common feature of business-cycle fluctuations.

These theoretical results translate directly into distributional predictions once firms differ in their exposure to aggregate shocks. The framework then rationalizes the co-movement of mean growth, dispersion, and skewness. The theory is agnostic about what generates the disturbance. Commodity-price movements, pandemics, or disruptions to critical suppliers within production networks all enter through the same reduced-form impulse. While our baseline interpretation is a cost shock, the analysis carries over to an isomorphic environment in which shocks shift demand. Uncertainty shocks can also be incorporated mechanically and produce dynamics similar to those generated by aggregate shocks.

Figure 1 illustrates the transmission mechanism in four steps. An aggregate impulse in Panel (a) hits firms according to their shock exposure, generating a cross-sectional distribution of firm-level shocks in Panel (b). This distribution shifts in location and scale over the cycle: its variance rises in a U-shape around trend. Concave firm policies then map this shifting shock distribution into log-production choices, whose first differences deliver the growth distribution in Panel (c). Panel (d) traces the implied dynamics of mean growth, dispersion, and skewness as the aggregate shock hits and decays. The size gradient of skewness emerges because large-firm policies are more concave, hence their skewness swings are larger.

This concavity of firm policies is not assumed but derived. Our demand-based microfoundation complements recent empirical work documenting that concave firm policies generate distributional dynamics through labor market frictions, as in Ilut et al. (2018).

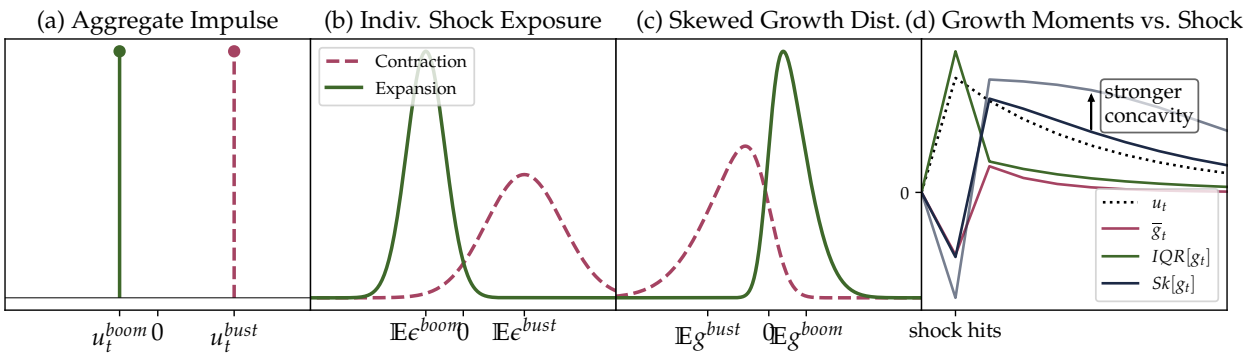
While Ilut et al. (2018) establish that concave policies matter empirically, they do not explain where the concavities come from. We show that concavity emerges naturally from product market primitives: firms with market power facing demand that satisfies MSLD optimally respond asymmetrically to shocks. The degree of asymmetry rises with market power, which is why large-firm policies in Panel (d) are more curved.

Since we ground concavity in demand structure, our theory applies at the firm level. Production-network explanations of growth skewness across industries or other aggregates, as in Dew-Becker and Vedolin (2021), may still apply. However, grounding the mechanism in demand has four advantages: *(i)* Parsimony—we require minimal structure on shocks to match a demanding set of stylized facts. *(ii)* Universality—we connect pass-through rates and markups, familiar objects in industrial organization, to business cycle propagation. *(iii)* Testability—business cycle features are observable and firms can be sorted along several dimensions, yielding a rich set of testable predictions. *(iv)* Policy relevance—competition policy affects not just allocative efficiency but business cycle amplification through the curvature channel. Conversely, curvature of firm policies can serve as a telltale sign of market power at play.

Note that a supply-side explanation of skewness would have to account for the *size gradient* by requiring that large firms face *stronger* more strongly convex cost functions than small firms. Yet, the common view is the opposite: large firms enjoy better access to finance, more diversified supplier and customer networks, and lower adjustment frictions. Thus, a demand-side mechanism is the natural candidate.

Guided by the theory, we then test three predictions using quarterly `Compustat` data on U.S. public firms from 1983 to 2024, spanning more than forty years and multiple recessions. The tests focus on three implications: aggregate shocks generate tight co-movement between mean growth and skewness; this comovement is stronger for firms

Figure 1: Distributional Shifts in Theory



Note: The figure shows the transmission of an aggregate impulse, u_t, in (a), into a heterogeneous shock distribution, $\epsilon_{it} = \widetilde{\lambda}_i u_t$ in (b). In (c), the shock distribution is mapped into a skewed growth rate distribution by a log-concave mapping. Plot (d) shows the co-movement of growth moments (means $\overline{g}$, interquantile range IQR, Kelley skewness Sk) after an AR(1) aggregate shock predicted by our theory.

with greater shock exposure; and the size gradient of skewness reflects market power rather than size per se.

First, we estimate local projections for six identified aggregate shocks, monetary, oil, credit, uncertainty, sentiment, and TFP. Each shock produces a statistically significant decline in the skewness of firm sales growth. The effect peaks four to six quarters after impact and dissipates within ten quarters. Across shocks, the correlation between the skewness and growth impulse responses ranges from 0.89 to 0.98, indicating that the comovement between mean growth and skewness is not specific to a particular recession episode or identification strategy. Consistent with the theory, the skewness response is concentrated among large firms and is close to zero for small firms.

Second, we exploit heterogeneity in firms' pre-pandemic risk-factor disclosures as a source of differential exposure to the COVID shock. Firms with higher exposure experience a significantly larger decline in skewness at the onset of the pandemic. The negative difference remains sizable for roughly four quarters after 2020Q1 and followed by excess positive skewness along the recovery. Both observations are predicted by the impact-and-recovery pattern of the model.

Third, we study cross-sectional variation across industries and firms. The amplitude of skewness fluctuations is strongly increasing in average industry concentration across NAICS industries. Sorting firms by profitability rather than size yields a similar gradient: highly profitable firms display substantially larger cyclical movements in skewness. These patterns are consistent with profitability and concentration proxying for market power and with the theoretical prediction that greater market power amplifies skewness dynamics.

We confirm these three predictions in the main empirical analysis. The appendix reports three additional results. The stylized facts replicate in a disjoint sample of listed non-US firms, a growth decomposition shows that aggregate components are the key driver of skewness, and a simple simulation of the framework matches the data well.

Our findings carry implications for business cycle propagation and modeling. First, they demonstrate parsimony: we explain countercyclical variance, procyclical skewness, and size-dependent amplification through first-moment shocks alone. Second, because the curvature channel maps any mean-shifting shock into higher-moment responses, policies conventionally perceived as aggregate mean-shifters—monetary interventions, commodity-price movements, trade disruptions—automatically carry the distributional implications usually associated with uncertainty or tail-risk shocks. Third, to our knowledge, this is the first paper to connect the largely theoretical IO and trade literature on demand-system properties to distributional business cycle dynamics. Market structure affects business cycle propagation through higher-order properties of demand elasticity and how markups fluctuate in response to shocks. Fourth, the sufficient conditions we identify—MSLD and IPT—are empirically verifiable properties of demand systems, offer-

ing concrete guidance for practitioners building quantitative heterogeneous-firm models on what demand-system properties to impose. Finally, our results yield strong evidence that firms adjust output in a log-concave fashion. Since we can confirm our empirical hypotheses in the data, we also strengthen the case for a strong version of MSLD within a large universe of firms.

Literature Our paper contributes to literature on distributional business cycle dynamics across three dimensions: mean, dispersion, and skewness.

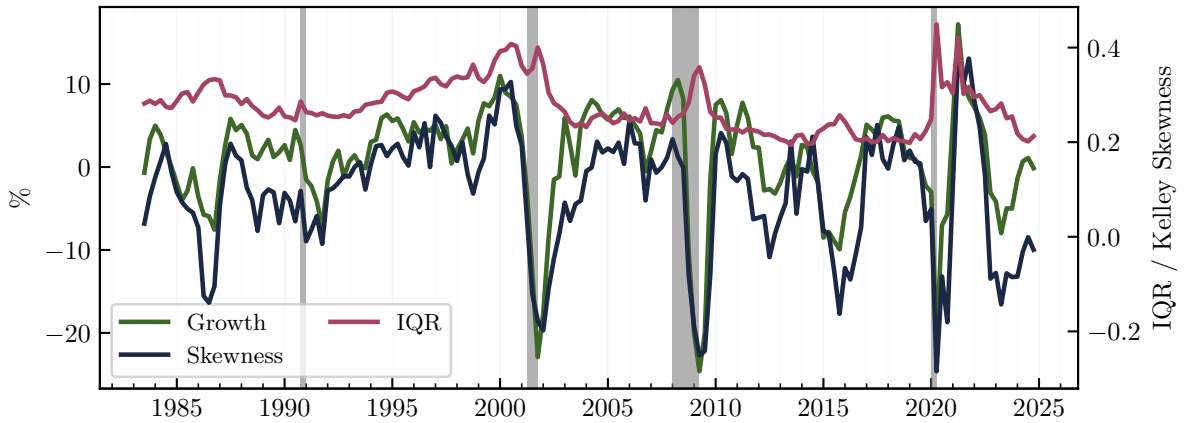
Mean dynamics. That business cycles are asymmetric—deep recessions versus mild booms, rapid contractions versus gradual expansions—has been documented since Sichel (1993), with further analysis in McKay and Reis (2008), Morley and Piger (2012), Berger et al. (2020), and Dupraz et al. (2025). Granular and network-based models (Gabaix, 2011; Carvalho and Grassi, 2019; Acemoglu et al., 2012, 2017; Di Giovanni et al., 2014) show how firm-level disturbances generate aggregate volatility and higher moments. Crouzet and Mehrotra (2020) document that large firms have more stable growth in levels; we extend this to higher moments, showing that large firms *amplify* cyclical skewness.

Dispersion. Two approaches explain countercyclical dispersion. The uncertainty-shock literature, initiated by Bloom (2009) and extended in Christiano et al. (2014); Vavra (2014); Arellano et al. (2019); Schaal (2017); Bloom et al. (2018), models time-varying volatility of firm-level shocks. An alternative, which we follow, emphasizes heterogeneous exposures to common shocks: Davis et al. (2025) show that firm-level exposures extracted from 10-K filings account for much cross-sectional variation. Related work on endogenous dispersion includes Bachmann and Moscarini (2011), Kuhn (2014), Decker et al. (2016), Straub and Ulbricht (2019), and Kehrig (2015).

Skewness. Closest to our work, Ilut et al. (2018) show that concave employment responses produce negative skewness, but do not microfound the concavity. Salgado et al. (2025) document procyclical skewness, interpreting it through "skewness shocks." Kamepalli et al. (2025) nest exogenous and endogenous skewness sources; Dew-Becker and Vedolin (2021) generate left-skewed aggregate fluctuations through input complementarities; Dew-Becker (2024) show that cross-sectional skewness is procyclical while aggregate time-series skewness is largely acyclical. We show that symmetric shocks combined with demand-side propagation suffice to generate all of these patterns.

Demand theory and market power. Our mechanism builds on concepts from demand theory and IO—Marshall's Second Law, superelasticities, and pass-through rates (Feenstra, 2003; Mrázová and Neary, 2017; Parenti et al., 2017; Matsuyama and Ushchev, 2020, 2022)—with empirical support from Hausman (1981), Mayer et al. (2021), Berman et al. (2012), and Amiti et al. (2019). We derive their implications for business cycle dynamics. Since our mechanism relies on a size gradient, we build on the well-documented link

Figure 2: Moments of the Growth Distribution



Note: The figure shows mean growth rates as well as the quantile versions of the second and third moment of the growth distribution.

between firm size and market power (Atkeson and Burstein, 2008; Melitz and Ottaviano, 2008; Edmond et al., 2015; Parenti, 2018; Boar and Midrigan, 2024; De Loecker and Warzynski, 2012; Autor et al., 2020; Hall, 1986).

2 Stylized Facts, Stylized Mechanisms

Before turning to the formal theory, we establish three empirical regularities about firm growth distributions and sketch the mechanisms that connect them.

Consider a cross-section of firms, each with policy $Q^\star$ mapping a cost shock $\widetilde{\epsilon}_{it} = \epsilon_{it} + \xi_t$ to time-t log-output:

$$\log q_{it} = Q^\star(\epsilon_{it} + \xi_t \mid \epsilon_{-i,t} + \xi_t) + at = Q^\star(\epsilon_{it}) - b\,\xi_t + a\,t, \tag{1}$$

where ξ_t is the cross-sectional mean of $\widetilde{\epsilon}_{it}$ (an aggregate shock common to all firms $i \in I$) and $a \geq 0$ is a secular trend. Policies are decreasing in the firm's own shock. We assume demand is homothetic, so that the aggregate shock ξ_t shifts all firms' log-output uniformly (proportionately to some constant factor $b > 0$) rather than entering through $Q^\star$.[4] The centered component $\epsilon_{i,t}$ therefore satisfies $\mathbb{E}[\epsilon_{i,t}] = 0$. We also assume that a firm's policy does not depend on other firms' shocks beyond the aggregate, justifying the second equality in (1). Growth rates are then

$$g_{i,t} = g(\epsilon_{i,t}, \xi_t \,|\, \epsilon_{i,t-1}, \xi_{t-1}) = a + \Delta Q^\star(\epsilon_{i,t}) - \Delta\xi_t.$$

The dynamics of the cross-sectional distribution over the business cycle depend on prop-

[4]For proof and extensive discussions on monopolistic competition with homothetic demands, see Parenti et al. (2017). We provide a full model of monopolistic competition for which growth rates take the form of eq. (1) in Supplemental Appendix B.

erties of $Q^\star$. To understand how, we revisit the first three moments of the growth distribution in our 1983–2022 sample of `Compustat` firms.

Means Let $\overline{g_t}$ denote the time-t average growth rate in our sample. An increase in the aggregate shock ξ_t leads to negative average growth: the economy falls below its potential log-output $Q^\star(0) + at$, and growth remains negative as long as cost pressures keep increasing. Conversely, positive growth occurs when costs decrease or rebound toward pre-recession levels.[5]

Figure 2 reveals the striking asymmetry: average growth is relatively stable at around 3–5% annually, with occasional sharp slumps that are not followed by symmetric positive recoveries. The pattern is consistent with a world in which ξ_t jumps up during a recession and slowly decays afterward—the "deepness" and "steepness" asymmetry of business cycles documented since Sichel (1993).

Stylized Fact 1

Mean growth rates usually fluctuate smoothly around trend but occasionally drop sharply at the onset of recessions.

Dispersion The dispersion of growth rates also shifts over the business cycle. One reason is that the variance of ϵ_t itself may increase, which mechanically raises the variance of $Q^\star(\epsilon_t)$.[6] A second possibility is that homotheticity is violated, so that policies are evaluated as $Q^\star(\epsilon_{i,t} + \xi_t)$.[7] A shift in ξ_t then moves the location at which $Q^\star$ is evaluated; if the policy is steep at the new location, dispersion increases.

However, the second explanation predicts a monotone pattern: dispersion falls in booms (when ξ_t is small, i.e. costs are low) and rises in recessions (when ξ_t is large).[8] We reject this using Panel A of Table 1. The first regression reveals only a weak negative relationship between dispersion and average growth. But once we condition on whether growth is above or below its long-term average, a U-shaped relationship emerges: dispersion increases as growth rates move farther from trend in either direction.

This pattern is robust and replicates within samples split by size into 'small' (bottom 90% of sales distribution) and 'large' firms (Panels B, C). Interestingly, Ilut et al. (2018) find stronger countercyclicality in dispersion for firms with more concave policies. We see the same in Panels B and C: growth dispersion is more countercyclical for large firms. Our

[5] Not all recessions are followed by a full recovery to trend. The recovery from the 2007/2008 financial crisis stopped below previous output trend levels; the recovery from the 2020 COVID crisis continued, if not overshot, pre-crisis trends.

[6] Scaling the variance of ϵ_t by a factor ϕ scales the variance of $Q^*(\epsilon_t)$ roughly as $\phi(Q^*)'(0)^2$, holding the time-t mean of ϵ_t at 0.

[7] This is the approach taken by Ilut et al. (2018).

[8] The opposite pattern—dispersion rising in booms and falling in recessions—is clearly at odds with Figure 2 and with Bloom (2009).

Table 1: Cyclicality of Higher Moments of the Growth Rate Distribution

Parameter	s_t (IQR) Estimate	SE	k_t (Skewness) Estimate	SE
Panel A: Aggregate				
$m_t = \alpha + \beta \overline{g_t} + u_t$				
β	-0.057	(0.136)	1.743***	(0.088)
$m_t = \alpha + \beta^- \overline{g_t} \mathbf{1}\{\overline{g_t} < \overline{g}\} + \beta^+ \overline{g_t} \mathbf{1}\{\overline{g_t} \geq \overline{g}\} + u_t$				
β^-	-0.763***	(0.122)	1.589***	(0.106)
β^+	0.262**	(0.103)	1.813***	(0.104)
Panel B: Small Firms (bottom 90% of time-t size distribution)				
$m_t = \alpha + \beta \overline{g_t} + u_t$				
β	-0.036	(0.133)	1.600***	(0.084)
$m_t = \alpha + \beta^- \overline{g_t} \mathbf{1}\{\overline{g_t} < \overline{g}\} + \beta^+ \overline{g_t} \mathbf{1}\{\overline{g_t} \geq \overline{g}\} + u_t$				
β^-	-0.713***	(0.134)	1.456***	(0.096)
β^+	0.234**	(0.104)	1.657***	(0.101)
Panel C: Large Firms (top 10% of time-t size distribution)				
$m_t = \alpha + \beta \overline{g_t} + u_t$				
β	-0.233	(0.143)	3.104***	(0.303)
$m_t = \alpha + \beta^- \overline{g_t} \mathbf{1}\{\overline{g_t} < \overline{g}\} + \beta^+ \overline{g_t} \mathbf{1}\{\overline{g_t} \geq \overline{g}\} + u_t$				
β^-	-0.879***	(0.071)	2.493***	(0.400)
β^+	0.516***	(0.133)	3.812***	(0.450)
Observations	162		162	

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

microfoundation attributes stronger concavities to larger firms, a connection we develop formally below.

Stylized Fact 2

Dispersion follows the absolute deviation of growth from its trend, becoming large in contractions and expansions alike.

Skewness We use a quantile-based skewness measure for the empirical results of this paper. Skewness of a random variable X with quantiles X_r, $r \in (0, 1)$, is defined as in Groeneveld and Meeden (1984):

$$\mathrm{Sk}_r[X] := \frac{X_r + X_{1-r} - 2X_{0.5}}{X_{1-r} - X_r} \in (-1, 1). \tag{2}$$

The measure compares the average of the upper and lower tails to the median. Values below zero indicate a longer left tail (negative skewness); values above zero indicate a longer right tail. For a visual example, see the left-skewed distribution in Figure 1(c). The measure is empirically robust and nests Kelley skewness for $r = 0.1$, which we use

throughout. For theoretical approximations, we also use the cumulant-based skewness:

$$\gamma_1[X] \equiv \mathbb{E}[(X - \mathbb{E}X)^3]/(\mathbb{V}[X])^{\frac{3}{2}}, \tag{3}$$

which is more tractable analytically.

Skewness is strongly procyclical: it fluctuates together with the mean (Figure 2 and Salgado et al. (2025)). Concavity of $Q^\star$ explains why. Consider the simple case where the economy was on trend in $t-1$ and firms receive cost shocks in t with positive mean. Then $g_{i,t} = Q^\star(\epsilon_{i,t}) - Q^\star(0) - \xi_t$, which is negatively skewed if $Q^\star$ is concave. A concave policy thus explains the downward jump of skewness when a recession hits. If output then recovers in $t+1$, skewness flips sign and becomes positive, since $g_{i,t+1} \approx -g_{i,t}$.

The key driver of skewness in growth rates is *shock dispersion*. Because ϵ_t is dispersed while $\epsilon_{i,t-1}$ and $\epsilon_{i,t+1}$ are not (in this example), the sign and magnitude of skewness in $Q^\star(\epsilon_{i,t})$ determine the skewness of $g_{i,t}$. More generally, the shock period with larger cross-sectional variance dominates. Combined with stylized fact 2—variance rises when growth departs from trend—and stylized fact 1—aggregate shocks cause sharp declines and smooth recoveries—this yields negative skewness in recessions and positive skewness in recoveries.

Our mechanism differs from that of Ilut et al. (2018), who model skewness fluctuations through changes in the curvature of the firm policy at different mean shock locations. Under homotheticity, however, location shifts are absorbed into ξ_t and are immaterial for skewness. Our approach instead keeps the policy fixed and generates skewness dynamics through changes in shock dispersion.

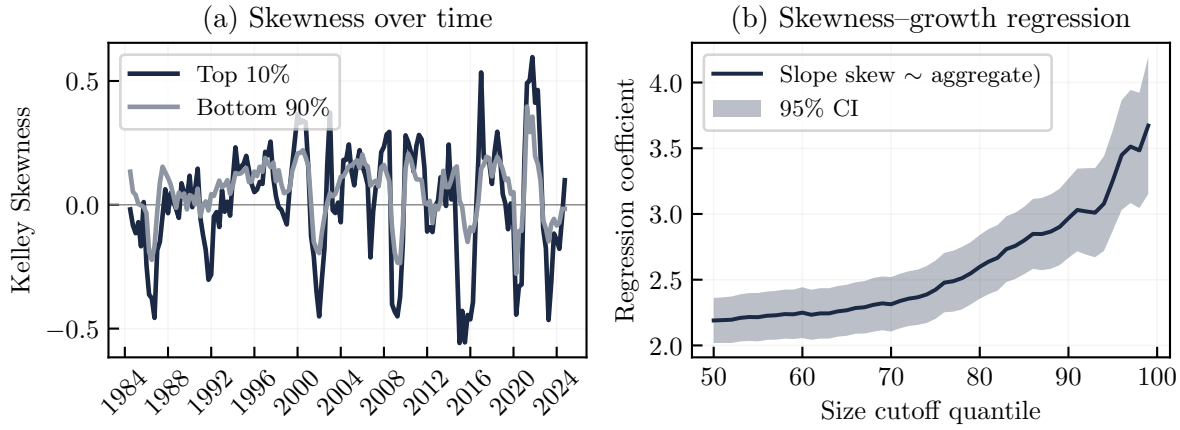
Figure 2 also shows a small positive long-run level of skewness. This is consistent with skewed shocks or a positively skewed trend component (e.g., from innovation by a few firms each period).[9] Since we focus on cyclical fluctuations rather than secular trends, the long-run level is not a primary object of study. Skewed shocks would mean that the exact relationships in our analytical results hold only approximately, but do not threaten the mechanism.

We find highly robust differences in skewness dynamics for small v. large firms. Panels B and C of Table 1 show that the amplitude at which skewness changes over the cycle is significantly stronger for large firms. This pattern is also striking in Figure 3 (a), does not depend on the size cutoff (Figure 3 (b)), and holds in a disjoint sample of non-US firms (see Appendix, Figure 11).

Since these differences replicate in samples split by *profitability* and persist when comparing high-HHI to low-HHI industries, we attribute them to *market power* (see Section 5.4). In our microfoundation, differences in skewness amplitudes arise from differences in firm-level market power, which manifest as differential curvature of policies.

[9] See for example the literature on quality-ladder models or recently Berlingieri et al. (2025).

Figure 3: Size-dependent skewness



Note: Size groups are defined based on average real sales over previous three years. The standard deviation of Kelley skewness for large firms is about 0.23 — more than twice the corresponding value of 0.11 for small firms.

Stylized Fact 3

Skewness is strongly procyclical. Its amplitude over the cycle is greater for large than for small firms.

Modeling choice Our reduced-form structure differs from Ilut et al. (2018) and Straub and Ulbricht (2019), who assume $g_{i,t} = f(s_{i,t})$ where $s_{i,t}$ is a time-t signal and f is a concave mapping. There, moment fluctuations are driven by shifts in the *location* of $s_{i,t}$. We prefer our formulation for three reasons. *(i)* It accommodates aggregate shocks that do not shift the average firm's signal location, staying closer to a general equilibrium characterization. *(ii)* It imposes a time dependence in growth rates, making clear that signals cannot logically be i.i.d. *(iii)* It can be microfounded by deriving $Q^\star$ from a firm problem, linking distributional business cycle dynamics to demand theory. For an additional discussion on what dynamics a non-homothetic model like in Ilut et al. (2018) would imply and how it would align with the data, see Remark 1 in Section 3.

3 Firm Policies and the Growth Distribution

The stylized facts reveal a systematic relationship between firm size and the cyclical properties of growth rate skewness. To explain these patterns, we investigate how the moments of growth rate distributions interact with shocks within a reduced-form structure: firms are exposed to aggregate shocks and adjust output according to their policy function. We show that even minimal structure can reproduce the complex higher-moment dynamics shown in Figure 1 (d).

3.1 Shocks and the Moments of Growth

Consider the following underlying structure of shocks $\widetilde{\epsilon}_t$. Let there be a continuum of firms, $i \in I$. Firm i receives cost shocks through its individual exposure $\widetilde{\lambda}_i$ to an aggregate factor u_t, and through a purely idiosyncratic, symmetrically distributed, i.i.d. component $\kappa_{i,t}$ with zero mean and variance $\mathbb{V}[\kappa_{i,t}] = \sigma^2_{\kappa,t}$. Let $\overline{\lambda} = \mathbb{E}[\widetilde{\lambda}] \geq 0$, and define the normalized exposures as

$$\lambda_i \equiv \widetilde{\lambda}_i - \overline{\lambda},$$

with symmetric probability density Λ and $\mathbb{V}(\lambda) = \int \lambda^2 \Lambda(\lambda)\, \mathrm{d}\lambda = 1$ without loss of generality. The shock hitting firm i's marginal cost is then given by

$$\widetilde{\epsilon}_{i,t} = \kappa_{i,t} + \underbrace{\lambda_i u_t}_{=\epsilon_{i,t}} + \underbrace{\overline{\lambda} u_t}_{=\xi_t}, \qquad \overline{\lambda}, u_t \geq 0. \tag{4}$$

It scales marginal cost by $\exp(\kappa_{i,t} + \epsilon_{i,t} + \xi_t)$. Although we frame shocks as cost shocks for concreteness, it will be clear from the theoretical model that the shock structure is isomorphic to shocks scaling demand.[10]

Note that the aggregate shock is *adverse*. Because $\overline{\lambda} \geq 0$ and $u_t \geq 0$, the aggregate component can only drive costs above their baseline level. The shock process is therefore consistent with an economic environment that is either at rest ($u_t \approx 0$) or in contraction ($u_t > 0$), in line with stylized fact 1.

As above, a firm policy $Q^\star$ maps $\epsilon_{i,t}$ and ξ_t into log-output. Substituting the factor structure into eq. (1) and taking first differences yields growth rates:

$$g_{i,t} = a + \Delta Q^\star(\kappa_{i,t} + \lambda_i u_t) - b\,\overline{\lambda}\,\Delta u_t. \tag{5}$$

With this structure, we can approximate the moments of g_t under different shock regimes. Proposition 1 characterizes the growth distribution in terms of means, dispersion, and skewness, given idiosyncratic and aggregate shocks. Note that we compute $\mathbb{E}, \mathbb{V}, \gamma_1$ all cross-sectionally, hence conditionally on time-t.

Proposition 1 (Approximate Growth Rate Moments)**.** *Let all moments be finite, κ_t, λ be symmetrically distributed for all t, and $\kappa_t \overset{D}{\sim} \kappa_{t-1}$ up to a difference in variances. Let $Q^\star_1 \equiv (Q^\star)'(0)$ and $Q^\star_2 \equiv (Q^\star)''(0)$, and denote $\mathbb{V}[\lambda] = \sigma^2_\lambda = 1$. Then, up to a second-order approximation:*

$$\mathbb{E}[g_t] - a \simeq \frac{Q^\star_2}{2}(\Delta\sigma^2_{\kappa,t} + \Delta u_t^2 \sigma^2_\lambda) - b\,\overline{\lambda}\Delta u_t. \tag{6}$$

[10] The reader may note for the remainder of this paper that all subsequent results also hold for shocks to the scale of the inverse demand function. Such shocks generate a monopolist's problem that is isomorphic to the one at hand.

For the variance,

$$\mathbb{V}[g_t] \simeq (Q_1^\star)^2 \cdot \left(\sigma^2_{\kappa,t} + \sigma^2_{\kappa,t-1} + (\Delta u_t)^2 \sigma^2_\lambda\right). \tag{7}$$

For the moment-based skewness measure:

$$\gamma_1(g_t) \simeq \frac{Q_2^\star}{|Q_1^\star|} \frac{3}{2} \frac{\Delta \mathbb{V}(\kappa_t^2) + 4(\Delta u_t)(u_{t-1}\sigma^2_{\kappa,t-1} + \sigma^2_{\kappa,t})\sigma^2_\lambda + (\Delta u_t^2)(\Delta u_t)^2 \mathbb{V}[\lambda^2]}{\left(\sigma^2_{\kappa,t} + \sigma^2_{\kappa,t-1} + (\Delta u_t)^2 \sigma^2_\lambda\right)^{\frac{3}{2}}}. \tag{8}$$

Proof of Proposition 1. We prove the proposition using standard techniques with a second degree Taylor approximation of g_t around 0, and calculating higher order moments, discarding all terms higher than order 2. For a comprehensive proof, see Lemma 3 and its proof in the appendix. All subsequent proofs are in the appendix. ■

We use Proposition 1 to discuss the dynamics of each moment in response to aggregate and uncertainty shocks, relating them to the stylized facts. A key requirement to generate higher moment dynamics in line with the stylized facts is the concavity of $Q^\star$. To see that non-linearities are fundamental, consider a linear $Q^\star$. This is the classical case in which an increase in the cost causes a proportional decrease in output. Since then $Q_2^\star \equiv (Q^\star)''(0) = 0$, increases in idiosyncratic uncertainty $\sigma^2_{\kappa,t}$ and changes in the aggregate shock u_t have no indirect effects on growth, but only through the mean exposure $\overline{\lambda}$ to the aggregate shock (eq. 6). In addition, there is no effect on growth rate skewness by eq. (8). It is therefore clear that $Q_2^\star \neq 0$; furthermore for uncertainty to have a negative effect on growth, and for skewness to be pro-cyclical, one requires $Q_2^\star < 0$.

Mean First, consider the mean. Mechanically, an increase in the aggregate shock raises mean costs by $\overline{\lambda}\Delta u_t$. By homotheticity, the mean shift affects all firms symmetrically and reduces growth. Additionally, there is a concavity effect: firms with large positive exposures reduce their log-output disproportionately more than firms with small or negative exposures. This asymmetry is captured by a negative second derivative, $Q_2^\star < 0$. Consequently, an increase in shock dispersion—whether mediated by Δu_t^2 or by a rise in σ_κ—has a negative contribution to growth. This is a well-known result: concavity of the policy function makes shock dispersion matter for aggregate growth.

Dispersion Second, consider the variance. Since a steeper policy implies larger responses to shocks, the variance scales in $(Q_1^\star)^2$. Beyond this, it is driven by the variance of shocks in t and $t-1$, but there is a stark difference between aggregate and uncertainty shocks. Without idiosyncratic shocks and with no change in the aggregate shock ($\Delta u_t = 0$), there would be no dispersion among growth rates. This follows because firms retain their exposures and thus face the same marginal cost in t and $t-1$. All else equal, they make the same production choices in both periods (or all grow at the same deter-

ministic rate a), so growth rates are identical. Therefore, the variance of g scales in the *squared change* of the aggregate shock. By contrast, since purely idiosyncratic shocks are independent across time and firms, *both* variances $\sigma^2_{\kappa,t}$ and $\sigma^2_{\kappa,t-1}$ contribute.

The contribution pattern of shock variances has important implications for the co-movement of moments over the cycle. The aggregate exposure structure with $u_t \geq 0$ implies precisely the U-shaped relationship between mean and dispersion documented in stylized fact 2. An increase in u_t generates both negative aggregate growth and higher variance. Since $(\Delta u_t)^2$ enters the variance as a squared term, a *decrease* in the aggregate shock also raises dispersion. In contrast, this clean relationship breaks down for uncertainty shocks. For example, consider the sequence $(\sigma_{\kappa,t-1}, \sigma_{\kappa,t}, \sigma_{\kappa,t+1}) = (0, 1, 1)$, holding $a, u = 0$ fixed throughout. A straightforward calculation shows that $\mathbb{E}[g_t] < 0 = \mathbb{E}[g_{t+1}]$ while $\mathbb{V}[g_t] < \mathbb{V}[g_{t+1}]$. This positive correlation between mean and dispersion contradicts the regression results underlying stylized fact 2.

Skewness Third, consider skewness, measured by the cumulant-based measure $\gamma_1[\,\cdot\,]$ of eq. (3). The key insight concerns how the mechanism—concave $Q^\star$ inducing left-skewed log-production $\widehat{q}_t$—carries over to growth rates. Figure 4 illustrates the relationship between concavity and skewness. Panel (a) shows that a concave, decreasing policy $Q^\star$, when applied to a shock ϵ_t, induces negative skew in the transform $\widehat{q}_t$. Greater concavity produces stronger negative skew, as seen by comparing the blue and red mappings. Panel (b) shows that a larger variance of ϵ_t also exacerbates left-skew. Both the curvature and variance effects are reflected in $\gamma_1[g_t]$.

The curvature effect is captured locally by the second derivative normalized by the absolute value of the first, $\rho[Q^\star](x) \equiv (Q^\star)''(x)/|(Q^\star)'(x)|$, evaluated at 0 in our formula: $\rho[Q^\star](0) = \frac{Q^\star_2}{|Q^\star_1|}$. A function f is concave if $\rho[f]$ is negative, and is (locally, around x) 'more concave' than another function g if $\rho[f](x) < \rho[g](x)$. Stronger concavity can be understood in the intuitive sense of Figure 4 (a), where the blue function bends below the red curve. We formalize this notion in the next section. Crucially, $\frac{Q^\star_2}{|Q^\star_1|}$ scales $\gamma_1[g_t]$ in the proposition. While the second term depends on the shock dispersion of both shock types and their respective dynamics, $\frac{Q^\star_2}{|Q^\star_1|}$ is a straightforward scaling of the amplitude of skewness, leaving its dynamics unaffected. Therefore, firms that differ only in the concavity of their policy exhibit highly correlated skewness dynamics that differ in amplitude—*precisely as in the small versus large firm comparison of stylized fact 3.* We build on this empirical observation in subsequent sections, providing a microfoundation for why larger firms have more concave production policies. We show that this pattern is not merely an empirical curiosity, but a predictable consequence of market power.

The variance effect depends strongly on the type of shock. In general, an increase in dispersion for either shock type causes negative skewness, while a decrease causes positive skewness. Since increases in dispersion are associated with negative growth for

both shock types, skewness is procyclical. This captures the essence of stylized fact 3. However, there are quantitative differences between the two shock types.

To illustrate, consider an aggregate shock decaying at rate β: $(u_0, u_1, u_2) = x \cdot (0, \beta^0, \beta^1, ...)$ with $x > 0$, setting idiosyncratic shocks to 0. Substituting into the formula, γ_1 scales in x, is negative on impact, and decays exponentially. Specifically, skewness drops sharply on impact, *rebounds* to a positive value, and then decays:

$$(\gamma_{1,1},\, \gamma_{1,2},\, \gamma_{1,3},\, ...) \propto \big((-1),\, \beta^0(1+\beta^{-1}),\, \beta^1(1+\beta^{-1}),\, ...\big).$$

This pattern coincides with the impulse response of the mean in both sign and rate of decay. Alternatively, to characterize an uncertainty shock, consider a similar impulse to the standard deviation of idiosyncratic shocks. The corresponding skewness response is

$$(\gamma_{1,1},\, \gamma_{1,2},\, \gamma_{1,3},\, ...) \propto \Big((-1), \beta^0 \frac{1-\beta^2}{\sqrt{1+\beta^2}}, \beta^1 \frac{1-\beta^2}{\sqrt{1+\beta^2}}, ...\Big).$$

While the decay is also exponential, the skewness rebound is at least four times stronger for the aggregate shock than for the uncertainty shock.[11] One would therefore expect the empirical impulse response function (IRF) of skewness to an aggregate shock to display a stronger rebound than that to an uncertainty shock. The most direct comparison of this kind in our paper is between the *oil* and *uncertainty* shock IRFs in Figure 5 in the empirical section, which confirms this prediction.

Since both shock types induce a robust positive correlation between skewness and the mean (in contrast to the relationship between dispersion and the mean), it is unsurprising that this relationship is extremely tight in the data. We conclude that the joint dynamics of mean, variance, and skewness of growth rates are well captured by *(i)* adverse aggregate shocks with heterogeneous exposures and *(ii)* concave firm policies. Uncertainty shocks may play a role, but cannot alone account for the dynamics of dispersion. Moreover, if large firms have more concave policies, the size gradient can also be explained.

Modeling Choice II The structure imposed by eqs. (1) and (4) can match a wide range of distributional business cycle facts. However, it does not succeed at all margins: the asymmetry in the the relationship between dispersion and mean growth, which shows in the first column of Table 1 is not readily explained by our reduced form model. Nevertheless, it is not obvious that the main competing assumption, non-homotheticity, offers a way to match this fact. In Remark 1 we argue that non-homotheticity would not help in explaining this particular pattern in growth rates, even though it does look promising at first glance.

[11] For $\beta \in (0,1)$, consider $\frac{\left[\frac{1-\beta^2}{\sqrt{1+\beta^2}}\right]}{1+1/\beta} =: R(\beta)$. One checks that $R'(\beta) = -(\beta^3 + 2\beta - 1)/(1+\beta^2)^{3/2} = 0$, holds at $r \approx 0.45$. So, R has a unique maximum at $R(\beta) \approx 0.226$.

Remark 1 (Non-Homotheticity and the asymmetric U-shape of Dispersion vs. Mean)**.** *We note that the asymmetry of dispersion during expansions vs. recessions: Especially for small firms, dispersion increases much more with negative than with positive growth rates (Table 1B). One reason may be a violation of homotheticity.*

The homothetic structure of our baseline model implies that the aggregate shock $\xi_t = \overline{\lambda} u_t$ *shifts all firms' log-output by a common amount, leaving the cross-sectional distribution's shape unchanged. Under non-homothetic demand, this need not hold: the location of the shock distribution itself shifts with aggregate conditions, so that the mapping from* ϵ *to* $\widehat{q}$ *depends on the aggregate state. Figure 4c illustrates the consequences. It imposes the factor structure of eq. (4), but relaxes the requirement that* $u \geq 0$ *for this exercise. If in* $t-1$ *output was at the steady-state, then* $\mathbb{V}[g_t] = \mathbb{V}[Q^\star(u_t\widetilde{\lambda})]$*. A (i) positive shock* $u > 0$ *(blue) and (ii) a symmetric negative shock* $-u$ *(orange) imply the same variance in the cost shocks. Yet, the variance increase of growth rates is much larger for* $u > 0$*, because the policy function becomes steeper for higher cost. This reading appears to capture this particular asymmetric expansions of dispersion, however, it dismisses the fact that shocks are mean-reverting: if the shock were to jump back to* $u = 0$ *in period* $t+1$*, the cross-sectional variance of in either scenario would be the same as at time* t*, but now scenario (i) is associated with positive growth and (ii) with negative. On net, in a regression of* s_t *and* $\overline{g}_t$ *the asymmetry would be muted. Moreover, the non-homothetic approach does not account for fluctuation in growth skewness without additional assumptions. This is self-evident from the quantiles in Figure 4c.*

In the following sections, we briefly revisit the interaction of concavity with our empirical, quantile-based skewness measure, and then provide a microfoundation for concave $Q^\star$ and the excess concavity of large-firm policies.

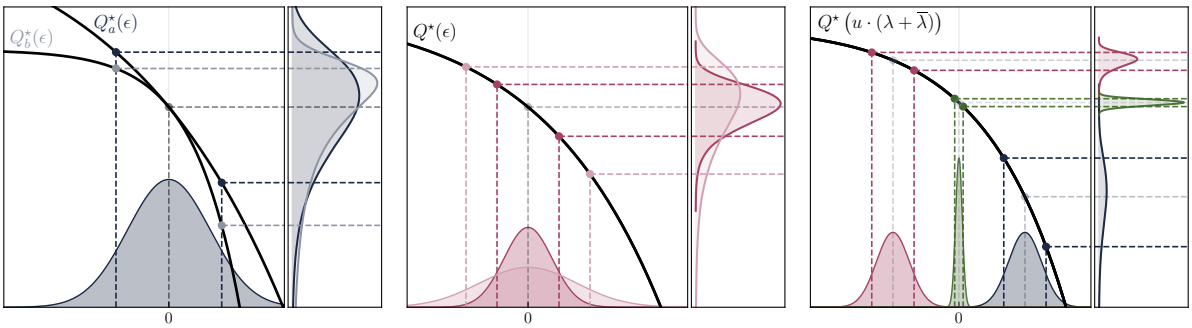
3.2 Concavity and Quantile Measures of Skewness

The results for the cumulant-based skewness measure carry over to the quantile measure used in our empirical work, Sk_r, in a *global* sense. To establish this point, we formalize *relative concavity* and confirm that it is the key mathematical concept governing differences in the amplitude of $\mathrm{Sk}_r[g_t]$.

The natural way to transform an unskewed random variable into a negatively skewed one is to pass it through a concave function f. Intuitively, applying an additional concave transformation s induces stronger left-skew, because the composition $g := s \circ f$ is more concave than f alone. The formal mathematical notion is '*relative concavity*' (Palmer, 2003). A function f is concave relative to g if it is a concave transformation of—and thus 'more concave than'—g.

Definition 1 (Relative Concavity)**.** *Consider two strictly monotone functions* f *and* g*.*

Figure 4: Skewness in log-Output: Concavity, Shock Variance, and Non-Homotheticity



(a) Concave policies (b) Shock variance (c) Non-homotheticity

Note: Panel (a) shows how a symmetric input (shock) distribution ϵ is mapped into skewed log-output $\widehat{q} = \ln q$ through concave production policies. Since $Q_b^\star$ is more concave than $Q_a^\star$ ($Q_b^\star \prec Q_a^\star$), the induced distribution is more left-skewed. Panel (b) shows how increasing the variance of the input distribution amplifies skewness in outcomes. Panel (c) illustrates a non-homothetic setting in which the location of the shock distribution shifts with the aggregate state. Different colors correspond to different aggregate states; the resulting output distributions differ in location and dispersion, with dispersion responding asymmetrically to adverse versus favorable shocks. Dashed lines indicate the 10th, 50th, and 90th percentiles.

*f **is concave relative to** g if there exists a strictly increasing, strictly concave function s such that $s \circ f = g$. We write $f \prec g$.*[12]

Under differentiability conditions, the relation $f \prec g$ is equivalent to the inequality $\rho[f] < \rho[g]$.[13] Relative concavity is therefore consistent with the arguments about the local curvature effect in the preceding section.

Negative skewness of a transformed random variable is exacerbated when the transformation becomes more concave (i.e., additional concave transformations are applied). We show that the local dynamics of Proposition 1 for skewness carry over to Sk_r:

Proposition 2 (Global Impulse-Response of Kelly-Skewness)**.** *Let $Q^\star$ be decreasing and strictly concave, factor exposures λ be symmetrically distributed, and there be no idiosyncratic shocks ($\kappa_t = 0$). Consider the impulse-response of the growth rate skewness if $(u_0, u_1, u_2, \dots) = (0, \beta^0, \beta^1, \dots)$. Then,*

$$\mathrm{Sk}_r[g_1] < \mathrm{Sk}_r[g_0] = 0 < \mathrm{Sk}_r[g_t] \quad \forall t > 1, \quad \textit{and} \quad \lim_{t\to\infty} \mathrm{Sk}_r[g_t] \to 0. \tag{9}$$

Additionally, suppose that policy functions are parametrized by $\alpha \in [0, 1]$, such that a larger index implies stronger concavity, i.e. $Q_\alpha^\star \prec Q_{\alpha'}^\star$ iff $\alpha > \alpha'$. Then, the amplitude of skewness increases in α:

$$|\mathrm{Sk}_r[q_{t,\alpha}]| > |\mathrm{Sk}_r[q_{t,\alpha'}]| \quad \forall \alpha > \alpha'. \tag{10}$$

Proposition 2 describes the impulse-response path of cross-sectional growth skewness

[12] Likewise, f is convex relative to g if there exists a convex, strictly increasing transformation s such that $s \circ f = g$. We write $f \succ g$, accordingly.

[13] Suppressing the argument indicates that the inequality holds globally.

to a single impulse to the aggregate shock u_t, echoing Proposition 1. At baseline, growth rates are unskewed (abstracting from heterogeneous growth trends a). On impact, skewness turns negative. It then reverses sign in $t \geq 2$ and decays back to zero. The amplitude of this response is governed by the degree of concavity of the policy function.

Relative concavity not only allows us to formulate a global version of Proposition 1. We can also use the notion to make statements about welfare. Remark 2 shows that log-concave output policies cause a loss in welfare if the social welfare function is at least as concave as the natural logarithm. Accordingly, stronger concavities in production policies exacerbate the welfare loss.

Remark 2 (Welfare Implications of Concavity)**.** *Log-concave policy functions have direct welfare implications. Consider a social planner with social welfare function* $W = \mathbb{E}\sum_t \beta^t \Upsilon(y_t)$*, where* Υ *is at least as concave as the natural logarithm, i.e.* $\Upsilon \prec \ln$ *or* $\Upsilon = \ln$*. Then, if* $Q^\star$ *is strictly concave,*

$$W = \sum_t \beta^t \mathbb{E}\Upsilon(\exp(Q^\star(\widetilde{\epsilon}_t))) = \sum_t \beta^t \mathbb{E}\Upsilon(\exp(Q^\star(\widetilde{\epsilon}_t))) < \sum_t \beta^t \Upsilon(\exp(Q^\star(\mathbb{E}\widetilde{\epsilon}_t))) = \sum_t \beta^t \Upsilon(\bar{y}_t),$$

where $\bar{y}_t$ *is the potential (zero-shock) output. Hence, welfare lies below its potential output benchmark.*

Additionally, indexing W *and* $Q^\star$ *by index* α*, we have that stronger concavity implies more welfare losses:*

$$\alpha > \alpha' \implies Q_\alpha \prec Q_{\alpha'} \implies W_\alpha < W_{\alpha'}.$$

We now turn to microfounding the two key assumptions that drive our transmission mechanism: concave policy functions and increasing concavity of large-firm policies. For the latter, we exploit the well-documented positive association between firm size and market power. We provide interpretable and testable conditions on demand functions for our microfoundation.

4 A Demand-Side Microfoundation

4.1 Set-Up and Firm Problem

Firms produce with convex cost $c(q;\epsilon) = q^\eta e^\epsilon$, where $\eta > 1$ denotes the cost convexity parameter and q denotes output. The term e^ϵ is a stochastic cost shifter, where ϵ is drawn from a symmetric distribution with zero mean and finite variance, and is observed by the firm at the time of production. Note that ϵ allows for a factor structure with idiosyncratic shocks as in the preceding section.

We adopt an iso-elastic cost function in order to focus on how competition shapes

growth rate skewness through the demand side.[14] A simple, iso-elastic cost function serves as the technological constraint, while the inverse demand function is kept as general as possible.[15] Since our empirical sample contains firms that are large by global standards and have a very low exit rate ($< 1\%$), firm exit and sample attrition are unlikely to drive results, and we abstract from exit decisions.

Monopolist The firm faces inverse demand $p(q)$. Assumption 1 on p ensures that the firm's profit-maximizing output is unique and the problem is well-behaved.

Assumption 1 (Regularity Conditions). *The inverse demand function p satisfies: $p : \mathbb{R}_+ \to \mathbb{R}_+$ (non-negative domain and range), $p \in \mathcal{C}^3(\mathbb{R}_+)$ (differentiability), $p' < 0$ (decreasing in quantity) and $\frac{\partial^2}{\partial q^2} \ln p \leq 0$ (log-concavity). By convention, $\mathbb{R}_+ = [0, \infty)$. Furthermore, marginal revenue, $\mathfrak{mr}(r) \equiv \frac{\partial}{\partial q} qp(q)$ satisfies $\mathfrak{mr}(a) > c$ for some $0 < a < \infty$.*

The monopolist's problem is

$$\max_{q \geq 0} qp(q) - c(q; \epsilon) \tag{11}$$

The problem admits an interior solution $q^\star > 0$ if and only if $q^\star$ satisfies the first-order condition. Lemma 1 characterizes the solution. The optimality condition takes the form of a Lerner (1934)-style condition. We define the elasticity operator $\mathcal{E}$ as $\mathcal{E}f(x) \equiv \frac{f'(x)}{f(x)} x$ for any differentiable function f that is either strictly positive or strictly negative.

Lemma 1 (Solution of Firm Problem). *The solution, of the monopolist's problem is unique, interior (positive) and implicitly given as the solution to the first-order condition equating marginal cost to marginal revenue:*

$$c'(q; \epsilon) = \underbrace{p(q)\left(1 + \mathcal{E}p(q)\right)}_{\equiv \mathfrak{mr}(q) \text{ (marginal revenue)}} \overset{\widehat{q} = \ln q}{\Longleftrightarrow} \ln \circ \mathfrak{mr}\left(e^{\widehat{q}}\right) - \left[\ln \eta + (\eta - 1)\widehat{q}\right] = \epsilon. \tag{12}$$

Optimal log-output, *$Q^\star(\epsilon)$, is a decreasing function of ϵ.* ***Marginal revenue***, *$\mathfrak{mr}$, is defined on some open interval $D \subset [0, a]$. The* ***elasticity of inverse demand*** *(not in absolute values) satisfies $\mathcal{E}p \in (0, -1)$ on D. The* ***mark-up*** *is given by $\mu(q) \equiv (1 + \mathcal{E}p(q))^{-1}$.*

We are interested in the properties of the firm's policy $Q^\star$. Its shape is determined directly by the inverse demand function p. We therefore introduce our first shape restriction on p: *Marshall's second law of demand* (*MSLD*).

Property 1 (MSLD). *We say that* ***Marshall's Second Law of Demand*** *(MSLD) holds if for all $q \in D$, the elasticity of inverse demand is increasing: $|\frac{\partial}{\partial q}\mathcal{E}p(q)| > 0$. We say it only holds weakly, if the inequality is weak.*

[14] All subsequent results also hold for shocks to the scale of the inverse demand function, which generate an isomorphic monopolist's problem.

[15] All our results would also apply to a setting in which all firms face marginal cost that is constant in q, subject to a capacity constraint for price-taking firms such that $q = \infty$ is not a feasible outcome.

MSLD states that the absolute elasticity of demand increases with price, so $|(\mathcal{E}p)'| > 0$, or equivalently, that low-cost firms set higher mark-ups. This property has been the subject of considerable empirical and theoretical scrutiny. It is prevalent in the trade literature, and key insights of seminal papers such as Krugman (1979) rest on it. Recent evidence by Mayer et al. (2021) strongly supports it. Matsuyama and Ushchev (2022) show that *MSLD* is instrumental for rationalizing incomplete pass-through and strategic complementarities in pricing, which arise when firms reduce their mark-ups in response to higher competitive pressures.

Melitz (2018) strengthen *MSLD* to *MSLD'*, which we refer to as the '*strong second law of demand*':

Property 2 (MSLD'). *We say that* ***Marshall's Strong Second Law of Demand*** *(MSLD') holds if for all $q \in D$, the elasticity of marginal revenue is increasing: $\frac{\partial}{\partial q}|\mathcal{E}\mathfrak{mr}(q)| > 0$. We say it only holds weakly, if the inequality is weak.*

MSLD' states that the absolute elasticity of *marginal revenue* decreases as output decreases, so $|(\mathcal{E}\mathfrak{mr})'| > 0$. While *MSLD* is equivalent to the average elasticity of marginal revenues being increasing, i.e., $\int_0^q |\mathcal{E}\mathfrak{mr}(q)| \, \mathrm{d}q > 0$, *MSLD'* requires this to hold at the margin as well, making it only slightly stronger.

Both *MSLD* and *MSLD'* presuppose that the pricing power of firms varies with firm size. These properties collapse if demand is iso-elastic or if firms lose pricing power altogether. The natural benchmark for the monopolist is therefore the behavior of a price-taking firm.[16]

Price-Taker A price-taker chooses output q_{pt}, taking the market price $\overline{p}$ as given. The first-order condition, written in logs, $\epsilon = \ln \overline{p} - \ln \eta - (\eta - 1)\widehat{q_{\mathrm{pt}}}$, implies that $\widehat{q_{\mathrm{pt}}} = Q^{\star}_{\mathrm{pt}}(\epsilon)$ is linear in ϵ.[17]

Alternative Cost Functions We adopt an iso-elastic cost function in order to focus on the effects of competition on growth rate skewness. While cost functions could also contribute to skewness, it is harder to rationalize a dependence of skewness on firm size through cost functions alone. We discuss mechanisms for generating cross-sectional skewness driven by the cost structure of firms in more detail in Appendix A.

4.2 Concave Responses to Shocks

Monopolist v. price-taker The monopolistic firm endogenously adjusts its quantity because a shift in unit cost implies a different profit-maximizing log-output $\widehat{q} = Q^{\star}(\epsilon)$. Propositions 1 and 2 establish that growth rate moments satisfy the stylized facts if $Q^{\star}$

[16] Note that a price taking firm does not necessarily operate in a perfectly competitive market.

[17] Going forward, we use *pt*-subscripts to refer to variables calculated in the price-taker equilibrium.

is strictly concave. In Lemma 2, we show that a concave log-output policy—and hence skewed outcomes—is a direct consequence of *MSLD'*.

Lemma 2 (Concavity of Output)**.** *Consider the solution of the firm problem $Q^\star(\epsilon)$ (log-output as a function of the shock). Let H be the left side of the first-order condition in eq. (12) (so $Q^\star = H^{-1}$). Then*

$$Q^\star \text{ is concave} \iff H \text{ is concave} \iff \text{MSLD' holds.} \tag{13}$$

The relationship between concavity of log-output and the shock is straightforward to establish and has been documented before (see Melitz, 2018).[18] Intuitively, because marginal revenue becomes more elastic under *MSLD'* at higher quantities, it decreases more rapidly in relative terms for firms with high q—that is, firms that have received a negative cost shock—than for firms that have received an adverse cost shock. As a result, upward adjustments of log-quantity are dampened while downward adjustments are encouraged.

We are now ready to state the first comparative prediction for high- and low-market-power firms.

Corollary 1. *The cross-section of firms with market-power features a skewness response according to Proposition 2; the price-taking firms have an unskewed growth-rate response.*

Corollary 1 states that the firm with market power exhibits a growth rate response whose skewness is characterized by Proposition 2, consistent with stylized fact 3. Also in line with Proposition 2, the amplitude of the skewness response increases in market power in this binary comparison: the price-taking firm has a linear policy and thus unskewed growth rates.

Next, we show that a simple parametrization of market power by an index α implies concavity increasing in α under realistic conditions.

Parametrized Market Power We introduce a simple and flexible parametrization of the inverse demand function that allows us to vary market power continuously. Write the inverse demand function as $p(q) = p^*(q)^\alpha \bar{p}^{1-\alpha}$ for some inverse demand function p^*, some fixed price $\bar{p}$ and parameter $\alpha \in [0, 1]$. For low values of α, the firm faces highly elastic demand and has little influence over price. For α close to 1, the firm faces lower elasticity determined through p^*.

We aim to generalize Corollary 1 under slightly stronger conditions to guarantee that concavity is monotone in α. Monotonicity of concavity in market power is the key property that generates the size gradient of skewness described in the stylized facts. We

[18] Note that Lemma 2 can be relaxed slightly: *MSLD'* does not need to hold globally, but only on the support of $Q^\star(\epsilon)$. While similar local relaxations can be made to all results in this paper, we omit these additional remarks.

define it formally in Property 3 and then develop the reasoning that yields economically interpretable sufficient conditions, culminating in this section's main result, Proposition 3.

Property 3 (Monotone Concavity)**.** *Let $\widehat{q}_\alpha$ be the output produced by a firm with market power $\alpha \in [0,1]$. We say* ***Monotone Concavity*** *holds if larger α induces stronger concavity. That is, $Q^\star_\alpha \prec Q^\star_{\alpha'}$ iff $\alpha > \alpha'$.*

Monotone concavity is a global property in the sense that it concerns all degrees of market power. To identify characteristics of p^* that imply monotone concavity, we first establish a condition that allows comparing the concavity of log-output across firms with different market power indices. To this end, index the mark-up by α: $\mu_\alpha(q) = (1+\alpha\mathcal{E}p^*(q))^{-1}$. The elasticity of inverse demand facing such a firm is $\mathcal{E}p = \alpha\mathcal{E}p^*$. The firm's first-order condition in $\widehat{q}$ now depends on α explicitly:

$$H_\alpha(\widehat{q}) \equiv \alpha \ln p^*(e^{\widehat{q}}) + (1-\alpha)\widehat{p} + \ln\left(1+\alpha\ \mathcal{E}p^*(e^{\widehat{q}})\right) + [\widehat{\eta} - (\eta-1)\widehat{q}] = \epsilon. \tag{14}$$

We require that $Q^\star_\alpha \prec Q^\star_{\alpha'}$ if and only if $\alpha > \alpha'$. Since $Q^\star_\alpha$ is the inverse of the first-order condition, the same ordering must hold for H_α. Consider the only two terms in eq. (14) that are nonlinear in $\widehat{q}$. The concavity of only one of them—the inverse mark-up—is affected by changes in market power. We need changes in the concavity of this term to be decisive for the overall concavity of H_α. It is therefore natural that a sufficient condition is

$$\frac{1}{\alpha}\ln\big(\underbrace{1+\alpha\mathcal{E}p^*(\cdot)}_{1/\mu_\alpha}\big) \prec \ln\left(p^*(\cdot)\right). \tag{15}$$

In words, the log-inverse mark-up is concave relative to the log-inverse demand function. Economically, this means that the firm's mark-up reacts to shocks with pronounced convexity: the firm raises its mark-up in response to an adverse cost shock far more than it reduces it after a favorable cost shock. This strong mark-up reaction becomes the dominant contributor of concavity to the first-order condition. Because the inverse mark-up is governed by market power α, this parameter also becomes the key determinant of concavity of $Q^\star_\alpha$. Note that this is a second order condition: the price would also fluctuate with constant mark-ups in response to a cost shock. To generate concavity in log-output we need the proportion of the cost increase which is translated into a price increase itself to move.

Eq. (15) is a local condition for a fixed α; if it holds, it allows ranking the concavity of $Q^\star_{\alpha'}$ against $Q^\star_\alpha$ by comparing α to α'. We note that the relation in eq. (15) automatically holds for α if it is true for some smaller $\alpha' < \alpha$. That is, if it holds for low-market-power firms, it is also satisfied by high-market-power firms. Mathematically, this follows from simple algebra: one shows that $\frac{1}{\alpha}\ln\left(1/\mu_\alpha\right)$ is a strictly concave transformation of

$\frac{1}{\alpha'} \ln\left(1/\mu_{\alpha'}\right)$ whenever $0 < \alpha' < \alpha < 1$. Hence, the concavity of $\frac{1}{\alpha} \ln\left(1/\mu_{\alpha}\right)$ is increasing in market power. Since the right side of eq. (15) does not depend on α, and '$\prec$' is a transitive relation, it suffices to verify that the log-inverse mark-up is concave relative to the log-inverse demand even for firms with the lowest degree of market power, i.e., as $\alpha \downarrow 0$:[19]

$$\mathcal{E}p^*(\cdot) \prec \ln\left(p^*(\cdot)\right). \tag{16}$$

As argued, this condition yields monotone concavity. However, this condition is difficult to interpret economically in its current form. Fortunately, it is equivalent to a more interpretable (albeit not commonly considered) property: the *superelasticity of inverse demand* being increasing. The superelasticity of inverse demand is the elasticity of the elasticity function: $\mathcal{E}^2 p^*$. We define this property in Property 4 and abbreviate it to *ISID*.

Property 4 (ISID)**.** *Let $\mathcal{E}^2 p(q) \equiv \mathcal{E}(\mathcal{E}p)(q)$ be the* ***superelasticity of inverse demand*** *(that is the elasticity of the elasticity function). An inverse demand function, p, (locally) satisfies an increasing superelasticity of inverse demand (**ISID**) if $\mathcal{E}^2 p(q)$ is strictly increasing in q.*

ISID turns out to be a key property for generating monotone concavity. It implies *MSLD'* for any given α and can thus be seen as a further strengthening of Marshall's second law. Yet, despite its centrality for characterizing the curvature of demand, the superelasticity is an object about which it is difficult to form prior beliefs. *A priori*, we have no strong intuition about whether $\mathcal{E}^2 p^*$ should be constant, increasing, or decreasing in quantity. To make this concept more concrete, we note that it relates to a more interpretable metric: pass-through rates.

The *pass-through*, τ, is defined as the share of a cost increase that is passed on to customers in equilibrium. Formally, $\tau = \frac{d \log p}{d \log \bar{c}}$ (at cost level $\bar{c} \equiv e^{\epsilon}$) equals one plus the elasticity of the mark-up with respect to the cost-shifter $\bar{c}$. We formalize the property that pass-through is increasing in Property 5:

Property 5 (IPT)**.** *Let the* ***pass-through*** *be the share of a cost increase that is passed on to customers in equilibrium given by $\tau(\bar{c}) = 1 + \frac{d \log \mu}{d \log \bar{c}}$. An inverse demand function p features increasing pass-through (**IPT**) if $\frac{\partial}{\partial \bar{c}} \tau(\bar{c}) > 0$.*

Pass-through simplifies the empirical treatment of *ISID* and monotone concavity significantly. One need only assume the second law of demand, *MSLD*, and increasing pass-through rates, *IPT*, to conclude that *ISID* holds. In turn, both *MSLD'* and monotone concavity follow. This chain of implications is the main result of this section. Since both *MSLD* and *IPT* are empirically identifiable properties supported by evidence, we have arrived at conditions sufficiently strong to yield the sharp theoretical prediction of monotone concavity, while remaining interpretable. We summarize this in Proposition 3.

[19] Here, one notes the convergence of the LHS to the exponential function.

Proposition 3 (Sufficient Conditions for Monotone Concavity). ***Increasing pass-through rates (*IPT*) and* MSLD *are sufficient** conditions to guarantee that monotone concavity of $Q^\star$ holds. Formally, the following implications are true:*

$$ISID \iff (MSLD' \wedge \text{eq. (16)}) \implies \text{Monotone Concavity} \tag{17}$$

$$IPT \wedge MSLD \implies ISID. \tag{18}$$

Equation (17) of Proposition 3 states that under *ISID* the model predicts a market power gradient of concavity. This monotone concavity property translates into an amplitude of $\mathrm{Sk}_r[g_t]$ that depends on market power and, by proxy, firm size. Equation (18) establishes that increasing pass-through rates and Marshall's second law suffice. Therefore, under *IPT* and *MSLD*, our model satisfies the conditions of Propositions 1 and 2, and hence rationalizes all stylized facts.

Corollary 2. *Let $Q^\star$ be decreasing and strictly concave, factor exposures, λ, be symmetrically distributed, and there be no idiosyncratic shocks ($\kappa_t = 0$). Suppose, MSLD and IPT hold. Then,*

- $\mathrm{Sk}_r[g_t]$ *increases in $\overline{g}_t$ (is procyclical),*
- *the amplitude of skewness increases in market power: $\frac{\partial}{\partial \alpha}|\mathrm{Sk}_r[g_t]| > 0$.*

Corollary 2 ties our empirical results to theoretical properties of p. If the sufficient conditions are satisfied, the model predicts a policy function that exactly reflects stylized fact 3. Then, given the shock structure of Section 3.1, stylized facts 1 and 2 also hold. Therefore, *we jointly rationalize all stylized facts within our model.*

What makes this result powerful is the insight that patterns of comovement in moments of the growth distribution can arise endogenously. The amplification mechanism operates through concavity; it is simple and applies to a wide range of settings. Importantly, the model is parsimonious. Instead of exogenous shocks to variance or skewness, and instead of plausible yet ad hoc constraints on the production technology (e.g., time-to-build, customer acquisition), we begin with a foundational profit maximization problem. Analysis of this problem goes a long way: it reveals that firm policies and therefore growth-rate dynamics can be traced back to deep properties of the demand curve.

Before moving on to test direct implications of our theory in the data, we discuss the properties *ISID*, *MSLD*, and *IPT* more broadly, point out robustness and limitations of our theory, and address the question confronting every applied researcher: "*How far does one get with CES demand?*"

4.3 Discussion

Evidence for Increasing Superelasticity and Increasing Pass-Through Are increasing superelasticities empirically plausible? It depends. For example, the family

of *linear inverse demand* functions has an increasing superelasticity, while the constant pass-through demand family (CoPaTh) of Matsuyama and Ushchev (2020) violates the property. (We give an example in Supplemental Appendix B.2.) In general, since *ISID* is implied by a strictly increasing pass-through plus *MSLD*, it suffices to examine the evidence for *IPT*.[20] Recently, Baqaee et al. (2024) have, using a non-parametric calibration of Matsuyama and Ushchev (2022)'s H.S.A. demand system, provided evidence in support of *IPT* (and, additionally, of *MSLD*). Other recent empirical work also supports *IPT* (cf. Berman et al. (2012) and Amiti et al. (2019)).

Yet, *ISID* is not a property that has historically received much empirical attention as a demand-side restriction. To illustrate, note that the widely used Kimball (1995) aggregator in the parametrization of Klenow and Willis (2016) has a globally constant superelasticity. Its special case of CES has a superelasticity of zero.

Analogously, the homothetic translog aggregator by Feenstra (2003), which is popular in the trade literature and for mark-up estimation in IO (Ridder et al., 2026), violates *IPT* à priori (see also Matsuyama and Ushchev (2022)). Crucially, this makes conventionally estimated mark-ups using translog a tainted measure for explaining the size gradient of skewness. Since other demand systems that have been explored for mark-up estimation (CES, Kimball) also have this shortcoming, we use more robust yet less sophisticated approaches to tie market power to skewness amplitude in the empirical section.

CES turns out to be a special case also when it comes to skewness. A CES-style demand function $p^*(q) = Aq^{-\chi}$ for $A > 0, \chi \in (0,1)$, implies a constant inverse demand elasticity. Therefore, the first order condition is linear in $\widehat{q}$ and no degree of market power induces skewness. Note here that the irrelevance of market power for skewness is obvious: In this special case, our formalization of market power implies that p is also a CES inverse demand function. Another insight is that CES inverse demand is indeed the *only inverse demand function* for which growth rates are globally unskewed. This is formalized in Proposition 4.

Proposition 4 (the Special Case of CES Demand)**.** *The only non-constant inverse demand function which satisfies the regularity conditions (Assumption 1) for which growth rates are globally unskewed is of CES type, i.e.* $p^*(q) = Aq^{-\chi}$ *with* $A > 0, \chi \in (0,1)$.

Why Growth Rates? In principle, we could benchmark our theory against log-real sales $\widehat{q}_t$ directly. Why not compute the cross-sectional distribution of $\widehat{q}_t$ at any point in time and assess its skewness? There are three problems. First, the distribution of $\widehat{q}_t$ is essentially the size distribution. Attempting to hold market power fixed by restricting to size bins trivializes the distribution of $\widehat{q}_t$. Second, if one used generously wide size bins instead, the size distribution itself, which is heavily right-skewed in general, would obscure

[20] Matsuyama and Ushchev (2022) refer to weak IPT even as "Marshall's 3rd Law of Demand", implying that IPT is a natural restriction on the demand function.

any contribution to cross-sectional left-skewness in $\widehat{q}_t$ caused by shocks. Third, any shock requires a comparison of the 'shocked' state to a baseline, and the natural baseline is either the previous period or some long-run steady state. The former corresponds to a time-series growth rate $\widehat{q}_t - \widehat{q}_{t-1}$, while the latter describes growth relative to the steady state, $\widehat{q}_t - \widehat{q}_{|\epsilon=0}$. However, in the presence of idiosyncratic growth trends, measurement of $\widehat{q}_{|\epsilon=0}$ becomes elusive.

Supply-Side Skewness We do not preclude the existence of other sources of skewness in growth rates. For example, the input distributions of shocks may be systematically skewed (Salgado et al., 2025). Alternatively, the cost function may be log-convex, which can serve as an additional contributor to left-skewness of $Q^\star(\epsilon) - Q^\star(0)$ once a shock hits. We outline two mechanisms for this 'supply-side skewness' in Supplemental Appendix A: capacity adjustments and customer acquisitions. However, these complications do not offer a natural explanation for why skewness should differ with market power.

General Equilibrium and Example To motivate eq. (1), specifically the fact that ξ_t can be moved outside of $Q^\star$ while ignoring other firms' actions $\epsilon_{-i,t}$ entirely, we appeal to homotheticity. In Supplemental Appendix B we provide a general equilibrium industry model to justify our choice.

Additionally, we provide a full example of Proposition 1 under CoPaTh H.S.A. aggregation. We explicitly confirm concavity of firm policies under CoPaTh, which confirms the predictions of Lemma 2 (since CoPaTh satisfies MSLD'). Finally, we show that ISID may fail for the CoPaTh parametrization and that transforming the CoPaTh inverse demand by $(\cdot)^\alpha$ to reflect market power yields an inverse demand function which fails MSLD' for a range of extreme values. Nevertheless, one can show in our example that even though the sufficient conditions fail, concavity around the steady state still increases in the market power parametrization.

5 Empirical Evidence

Having established the theoretical link between heterogeneous exposures, market power and procyclical skewness, we now test these predictions empirically. Using quarterly Compustat data on US public firms, we first describe the data, then estimate impulse responses to identified aggregate shocks, exploit heterogeneous pandemic exposures as a natural experiment, and examine the role of concentration in shaping skewness outcomes.

Table 2: Summary Statistics for Compustat Data

	Full Compustat	Cleaned Sample	QFR
Assets (mln. USD)	4,005	1,884	43.2
Sales (mln. USD)	400.2	414.8	10.8
Sales Growth (%)	7.2	7.6	0.63
Q(Sales Growth)$_{0.25}$ (%)	-7.8	-7.5	-25.3
Q(Sales Growth)$_{0.75}$ (%)	20.7	21.1	26.6
Net Leverage (%)	26.9	12.4	20.0
Short-term debt (%)	75.9	8.7	33.0
Obs./quarter	6,337	4,828	6,122
Unique firms	22,103	17,327	–
ρ(Sales Gr., GDP Gr.)	0.64	0.55	–
ρ(Sales Gr., Skew)	0.85	0.88	–

Note: Statistics for QFR are for the manufacturing subset of Crouzet and Mehrotra (2020) from 1977Q3–2014Q1 and directly taken from tables 1 and 3 of their paper; the values are unweighted averages across size bins. The Compustat statistics are for 1983Q3–2014Q1. The reported values for assets and sales are in 2009 USD. Compustat values are deflated using the GDP deflator (GDPDEF), rebased to 2009.

5.1 Data

Our analysis uses quarterly data on US public firms from Compustat between 1983 and 2024, covering more than 40 years of data.[21] The long sample period enables us to cover multiple recessions and draw general conclusions about growth dynamics over US business cycles. We focus, as before, on sales. Let $q_{i,t}$ be firm i's real sales in quarter t. Year-on-year real sales growth is $g_{i,t} = \ln(q_{i,t}/q_{i,t-4})$. Our main business cycle indicator is aggregate real sales growth, constructed as the size-weighted average of existing firms' growth rates:[22]

$$g_t = \frac{\sum_i g_{i,t} q_{i,t-4}}{\sum_i q_{i,t-4}}.$$

This definition only considers firms present in both t and $t-4$, abstracting from entry and exit dynamics. Our main skewness measure is the Kelley skewness defined in eq. (2) with $r = 0.1$, applied to the empirical CDF of growth rates at each date t.

In order to put the firm characteristics obtained from Compustat into perspective, Table 2 compares the full Compustat sample against our cleaned data, together with summary statistics from the Quarterly Financial Reports (QFR), which Crouzet and Mehrotra (2020) use to construct a representative sample of US manufacturing firms. The

[21] Details on sample construction, variable definitions, and data source can be found in Appendix C. The data cleaning filters out roughly a quarter of the observations from raw Compustat. Despite this restriction, the time series of cross-sectional skewness is very similar before and after cleaning (Figure A.2), and Appendix D confirms that procyclical skewness is robust across skewness measures and cleaning procedures.

[22] We treat production and sales as equal, hence ignoring inventories.

average Compustat firm is considerably larger than the average QFR firm, both in assets and sales. Furthermore, the Compustat sales growth distribution has a larger mean, lower variance and is more positively skewed. Comparing the full Compustat sample to the cleaned version, the correlation between skewness and aggregate sales growth is virtually identical across both Compustat samples (0.85 vs. 0.88). Note, finally, since all firms in the sample are large relative to the universe of US firms, extensive-margin movements are limited and any bias from firm exit should be negligible.

5.2 Aggregate Shocks Cause Growth-Skewness Comovement

Generalizing the factor structure of equation (5) to accommodate multiple aggregate shocks, firm-level growth rates can be written as

$$g_{i,t} = a + \Delta Q^{\star}_{\alpha_i}\big(\kappa_{i,t} + \boldsymbol{\lambda}_i^T \boldsymbol{u}_t\big) - b\,\bar{\boldsymbol{\lambda}}^T \Delta \boldsymbol{u}_t + w_{i,t}, \tag{19}$$

where $Q^{\star}_{\alpha_i}$ denotes the log-output policy of a firm with market power α_i, the vectors $\boldsymbol{\lambda}_i$ and $\boldsymbol{u}_t$ collect firm-specific factor exposures and aggregate shocks of potentially different nature, $\bar{\boldsymbol{\lambda}}$ is the cross-sectional mean of exposures, and $w_{i,t}$ is a zero-mean i.i.d. disturbance.[23] This representation maps the theory directly into data. By Propositions 1 and 2, concavity of $Q^{\star}_{\alpha_i}$ implies that contractionary aggregate shocks generate negative cross-sectional skewness in firm growth. By Corollary 2, this skewness response is stronger for firms with greater market power, provided the sufficient conditions hold. The model therefore delivers two empirical predictions: aggregate shocks should generate tight comovement between mean growth and skewness, and this comovement should be systematically stronger among large firms, which operate with greater market power (higher α).

Empirical framework. We test these predictions in an impulse-response framework by estimating the responses of skewness and growth to six identified aggregate shocks: monetary, oil, credit, uncertainty, sentiment, and TFP shocks. These shocks differ in economic origin, identification, and sample coverage, allowing us to assess whether the predicted growth–skewness comovement is robust across a broad set of aggregate shocks. Specifically, we estimate these responses using local projections (Jordà, 2005):

$$y_{t+h} = \alpha_h + \beta_h \text{shock}_t + \sum_{\ell=1}^{L} \gamma'_{\ell,h} \text{controls}_{t-\ell} + e_{t+h}, \tag{20}$$

for horizons $h = 0, \ldots, 10$ and with up to $L = 2$ lags of controls. The coefficients β_h trace out the impulse response at each horizon. The outcome variable y is either cross-sectional

[23] The trend a can be motivated from the secular growth component discussed in Section 3.1, which turns into an aggregate trend in outcomes if demand systems are homothetic.

Table 3: Local projection specifications

Shock	Reference	Controls (lagged)	Sample period
Monetary	Bu et al. (2021)	Real GDP, GDP deflator, Shadow Rate, EBP	1994Q1 – 2019Q4
Oil	Baumeister and Hamilton (2019)	Real GDP, GDP deflator, Oil price	1983Q3 – 2019Q4
Credit	Gilchrist and Zakrajšek (2012)	Real GDP, GDP deflator, EBP	1983Q3 – 2019Q4
Uncertainty	Ludvigson et al. (2021)	Real GDP, GDP deflator, VXO	1983Q1 – 2015Q4
Sentiment	Lagerborg et al. (2023)	ICE, real GDP, uncertainty, Real stock prices	1983Q3 – 2019Q4
TFP	Ben Zeev and Khan (2015)	Real GDP per capita, real stock prices per capita, labor productivity	1983Q3 – 2012Q1

Note: All specifications include lags of the dependent variable and the shock series as controls and are estimated with two lags. 'ICE' is the University of Michigan Index of Consumer Expectations. Uncertainty is measured as the 12-month Jurado et al. (2015) uncertainty index.

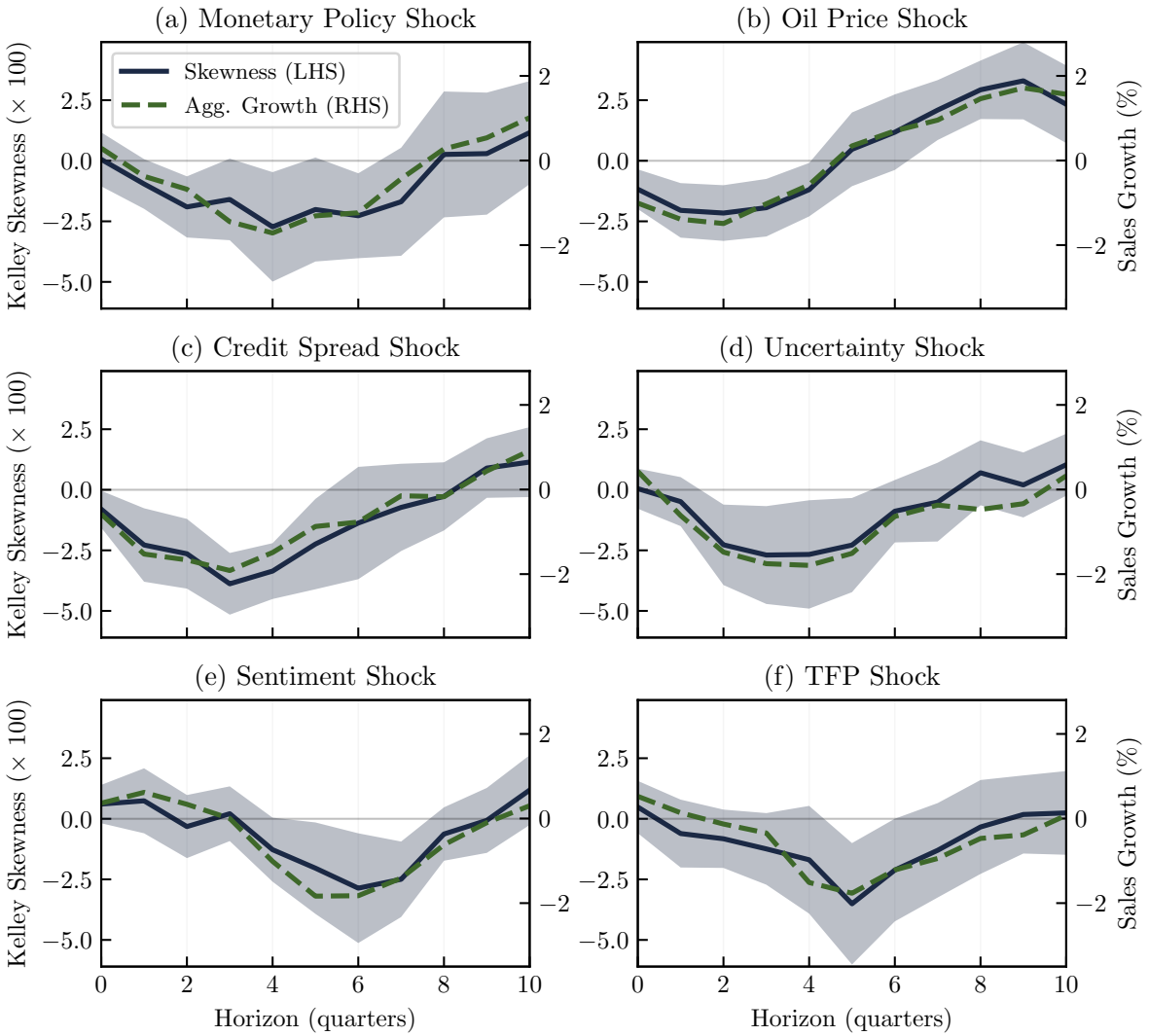
skewness or aggregate sales growth, computed either for the full sample or separately for large and small firms. Table 3 summarizes the specifications, and Appendix D.3 provides details on the shock series and robustness checks.

Aggregate evidence. Figure 5 reports the impulse responses of cross-sectional skewness (dark blue solid line, left axis) and aggregate sales growth (green, dashed line, right axis) to the six identified aggregate shocks. A clear pattern emerges across all specifications: contractionary shocks lower cross-sectional skewness, and aggregate sales growth moves closely alongside it. Following an adverse one-standard-deviation shock, the skewness index falls by about 0.02 to 0.06 points, with the response typically peaking after 4 to 6 quarters and fading within about 10 quarters. The decline is largest for credit shocks and smallest for oil shocks. Across shock series, the correlation between the impulse responses of growth and skewness ranges from 0.89 to 0.98. Aggregate shocks therefore appear capable of (1) inducing significant movements in skewness and (2) generating strong comovement between sales growth and skewness, in line with our theoretical predictions.

Note that these results closely mirror the reduced-form evidence in Figure 2. Cross-sectional skewness moves closely with aggregate growth across many US recessions (including the Covid recession), suggesting that procyclical skewness is a robust business cycle fact not specific to particular types of recessions. It is therefore encouraging that different types of shocks — all considered potentially important drivers of the US business cycle — induce this pattern.

Heterogeneity by firm size. We now turn to the second empirical prediction, that the skewness effect of aggregate shocks should be stronger among larger firms, which operate with greater market power. To test this implication, we split the local projections in Equation (20) by firm size and compute skewness indices for large and small firms separately. Figure 6 shows a clear shows a clear size gradient of skewness. Across all

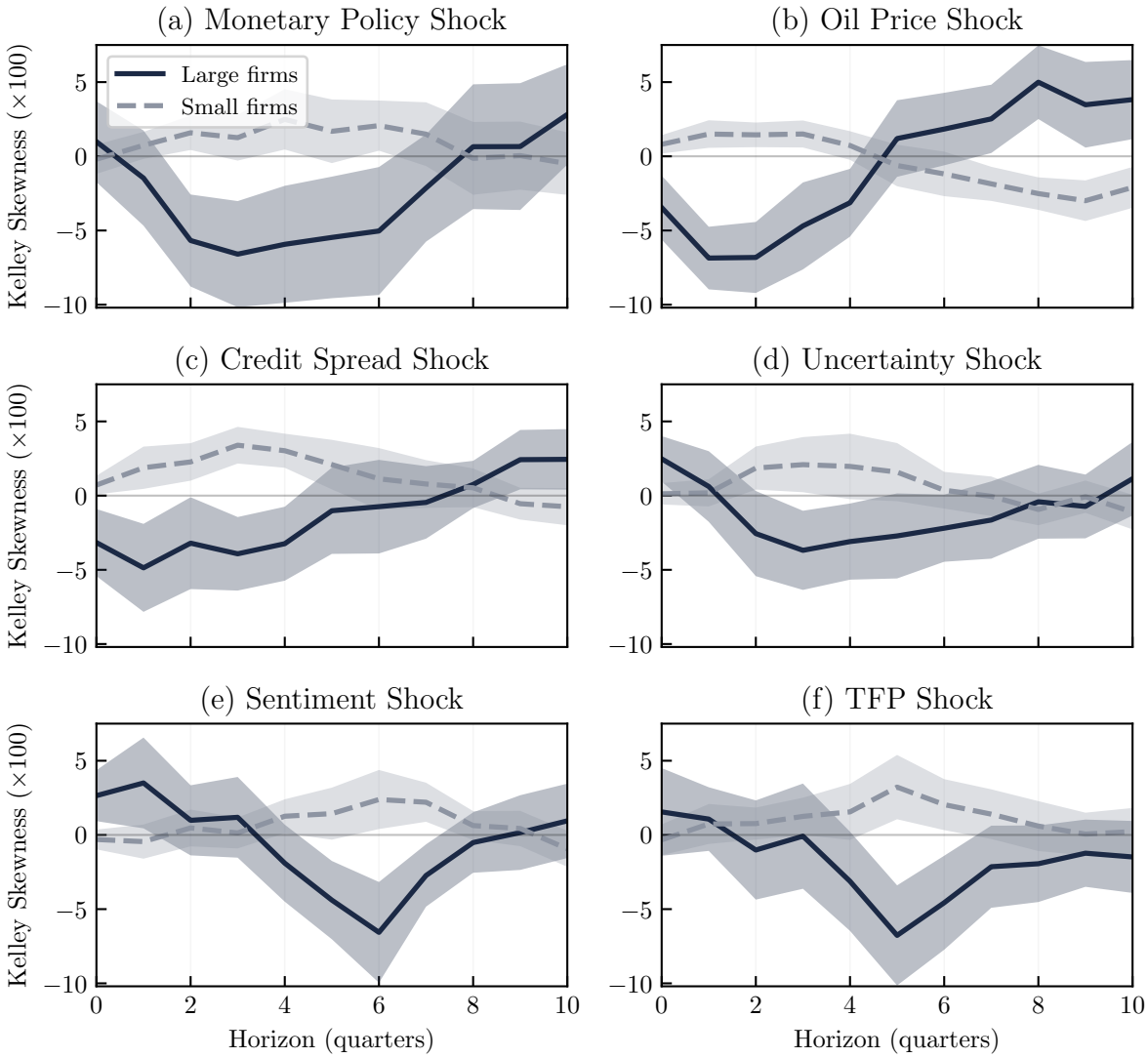
Figure 5: Comovement of growth and skew after aggregate shocks



Note: The 90% confidence bands are based on Newey-West standard errors. Shock magnitudes are normalized to be one standard deviation. The signs of the sentiment and the TFP shock are reversed to be contractionary.

shocks, the skewness index of large-firm growth rates (dark blue solid lines) declines by about 0.04 to 0.06 points, while the corresponding effect for small firms (dashed gray lines) remains close to zero. This pattern maps directly into the mechanism of the model. Firms with greater market power have more concave output policies, and greater concavity implies a stronger skewness reaction to a common aggregate disturbance. Since large firms operate with higher market power, their growth-rate skewness responds much more strongly to aggregate shocks. In Appendix D.2, we show that the same size pattern emerges from a bottom-up approach to impulse responses, confirming the robustness of these findings.

Figure 6: Skewness response of small and large firms

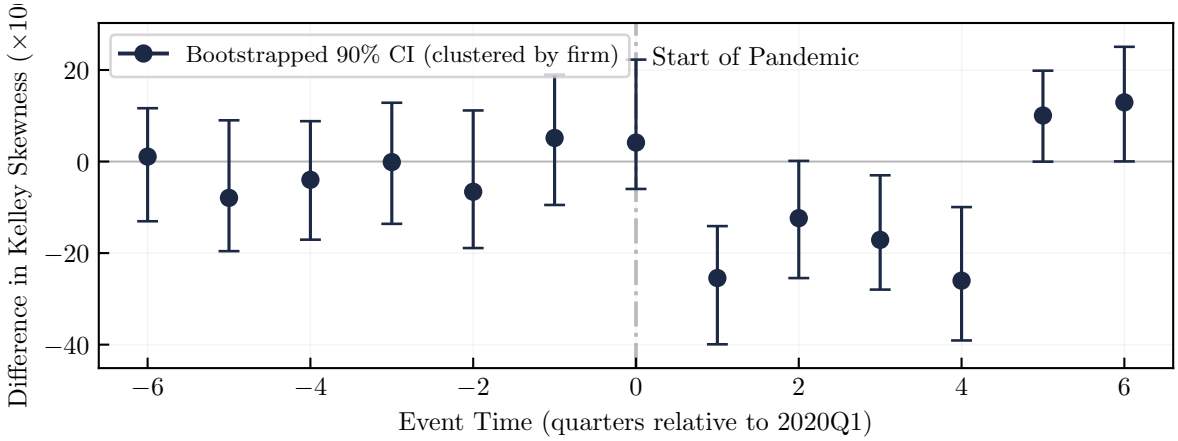


Note: The 90% confidence bands are based on Newey-West standard errors. Shock magnitudes are normalized to be one standard deviation. The signs of the sentiment and the TFP shock are reversed to be contractionary.

5.3 Cross-Sectional Exposure Amplifies Skewness Responses

Equation (19) implies that a firm's sensitivity to an aggregate shock $u_{l,t}$ is governed by its factor loading $\lambda_{i,l}$. Combined with Proposition 1, this yields another sharp cross-sectional prediction: firms with larger absolute exposures should exhibit greater responsiveness of cross-sectional skewness to aggregate shocks. In particular the cross-section of firms with $\lambda_{i,l}| > \phi$ for some $\phi > 0$ should exhibit a more pronounced decline in skewness on impact and a stronger rebound during the recovery. Intuitively, selecting firms with large absolute exposures widens the cross-sectional spread in how strongly firms react to the aggregate shock. In equation (8), this amounts to replacing the unconditional exposure moments, such as $\sigma_\lambda^2 = \mathbb{V}(\lambda)$ and $\mathbb{V}(\lambda^2)$, by their conditional counterparts within the selected sub-

Figure 7: Differential skewness response around the pandemic



Note: The figure plots event-time impulse responses of the difference in Kelley skewness of sales growth between high- and low-pandemic-exposure firms. Event-time dummies are relative to 2020Q1. Confidence bands are 90% bootstrap intervals based on resampling firms. Pandemic exposure measures come from Davis et al. (2025), to whom we are grateful for sharing their data.

sample, $\mathbb{V}(\lambda \mid |\lambda| > \phi)$ and $\mathbb{V}(\lambda^2 \mid |\lambda| > \phi)$. Because these firms have more extreme factor loadings, these conditional moments are larger than their unconditional counterparts, implying that cross-sectional skewness is more responsive to aggregate disturbances.

We test this prediction using heterogeneity in firms' exposure to pandemic risk. As an empirical proxy for $\lambda_{i,l}$ with respect to COVID-related disruptions, we use the pandemic exposure measure of Davis et al. (2025).[24] This measure combines information from firms' pre-2020 10-K risk disclosures with equity-market reactions on pandemic jump dates to identify firm-level exposure to pandemic shocks.

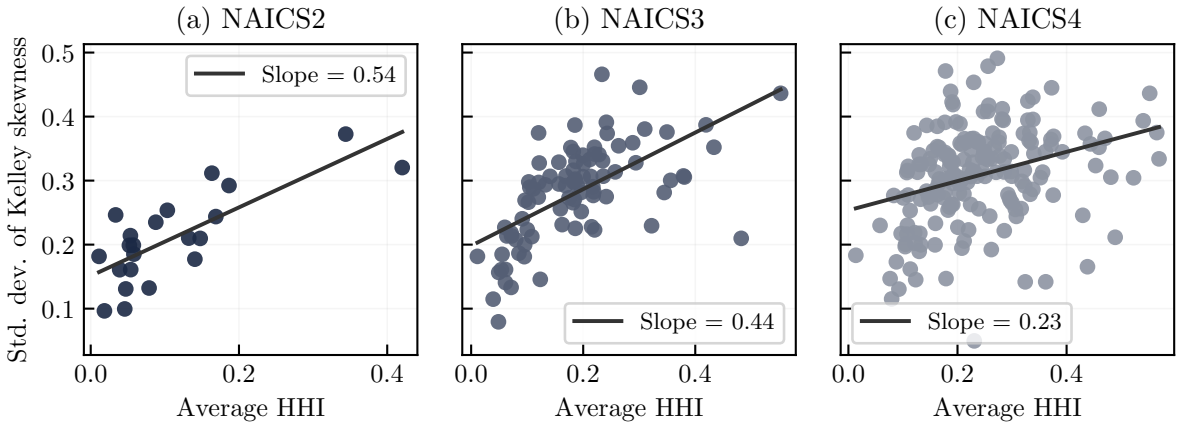
A key feature of this measure is that exposure is signed: some firms are adversely exposed, while others benefit from pandemic-related disruptions. To capture firms with the strongest sensitivity in either direction, we define the high-exposure group as firms in the bottom and top quartiles of the pandemic exposure distribution z_{pand}. The low-exposure group consists of firms in the middle 50 percent. This tail-versus-middle split maps directly into the theory, which predicts that cross-sectional skewness should react most for the subset of firms with large absolute exposures.

For each group, we compute quarterly Kelley skewness of sales growth and then take the difference between the high-exposure and low-exposure groups. We study the evolution of this difference in an event-time specification centered on 2020Q1, using event-time indicators to trace the dynamics around the onset of the pandemic. Inference is based on bootstrap confidence intervals obtained by resampling firms.

Figure 7 displays the results. Prior to the pandemic, skewness evolves similarly for high- and low-exposure firms. In the subsequent quarters, however, high-exposure firms experience a larger decline in skewness than low-exposure firms. This differential persists

[24] We are grateful to the authors for generously sharing these estimates with us.

Figure 8: Skewness Variation vs. Industry Concentration



Note: Figure shows the time-series variance of the skewness index, if the sample is split into NAICS-industries and sorted by industry concentration. We observe a positive relationship between the amplitude with which the skewness fluctuates (proxied by the variance) and the industry concentration, as predicted.

for roughly four quarters and then reverses, in line with the impact-and-recovery dynamics emphasized by the theory. Overall, both the timing and the magnitude of the response support the mechanism: firms with greater exposure to an aggregate shock exhibit larger movements in cross-sectional skewness.

5.4 Market Power Drives Skewness

So far, the empirical analysis has focused primarily on the size split, reflecting the close empirical relationship between firm size and market power Autor et al. (2020). We now consider additional proxies for market power based on industry concentration and profitability.

Concentration and Skewness Amplitude A higher HHI implies greater industry concentration and a larger role for firms with market power. In our framework, this should lead to stronger business-cycle fluctuations in the skewness index when the cross section is drawn from more concentrated industries. We capture these fluctuations by the time-series standard deviation of skewness. Figure 8 plots this measure against the HHI, with each observation corresponding to a NAICS industry, and shows a clear positive relationship. The relationship becomes weaker when industries are defined more narrowly, consistent with the idea that firms' exposures are more homogeneous within such groups.

Profitability and Procyclicality of Skewness We next use profitability as an additional proxy for market power. For each firm-quarter, we measure operating profitability

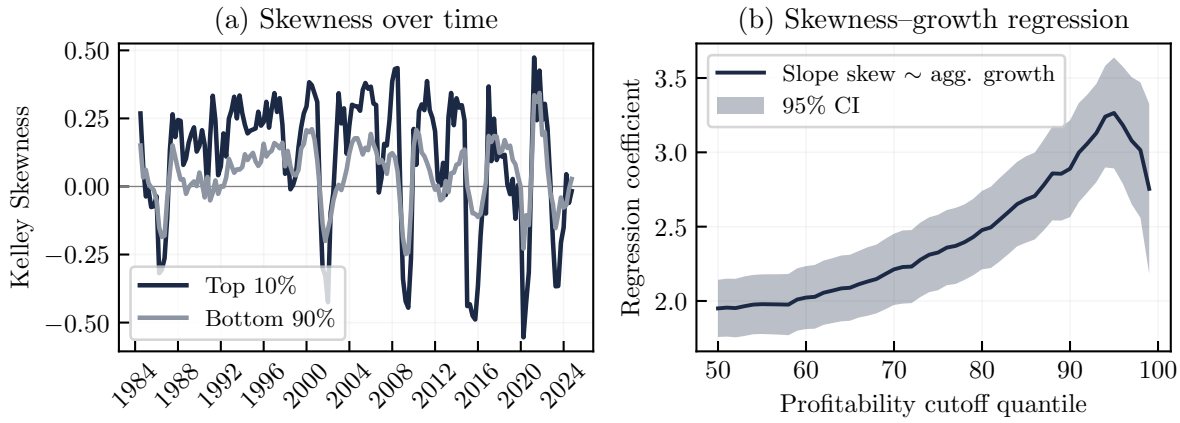


Figure 9: Profitability and skewness amplitude

as

$$\text{prof}_{i,t} \equiv \frac{\texttt{saleq}_{i,t} - \texttt{cogsq}_{i,t} - \texttt{xsgaq}_{i,t}}{\texttt{saleq}_{i,t}},$$

and assign firms to profitability groups based on the trailing three-year average of this measure. Similar to 3 we compute quarterly Kelley skewness of real sales growth and summarize the strength of skewness fluctuations over the business cycle by its relation to aggregate growth. Figure 9 shows a clear positive relationship between profitability and skewness amplitude: the most profitable firms exhibit substantially larger cyclical movements in skewness. However, as panel (b) reveals, the pattern is not perfectly monotone at the very top of the distribution as firms in the extreme upper tail display slightly lower amplitude than the next-highest group. Nevertheless, the coefficient is still much higher than, for example the top 30 percent.

Markups A natural alternative would be to sort firms directly by estimated markups. We do not pursue that approach here because, in our setting, markup measures are potentially contaminated by demand-side forces that are difficult to separate from the mechanism of interest. This concern applies to both main empirical approaches. Demand-side methods recover markups from an explicit demand system and detailed price-quantity data, often under functional forms such as CES or translog demand (Feenstra, 2003). Production-side methods, following Hall (1988) and De Loecker and Warzynski (2012) infer markups from cost-minimization conditions using output elasticities and input revenue shares. In Compustat-style data, however, these methods must be implemented with revenue rather than physical output, so under imperfect competition the resulting estimates need not isolate markup levels cleanly and may instead reflect both market power and demand-induced price variation (Bond et al., 2021). Indeed, Ridder et al. (2026) shows that revenue-based and quantity-based markup estimates are only modestly correlated in pooled firm-level data, although the correlation is higher within sectors. Since our objective is to isolate the link between market power and skewness on a broad level,

we therefore rely instead on more transparent proxies — industry concentration, firm size, and profitability.

6 Conclusion

This paper has documented new evidence on the shape of firm growth distributions over the business cycle. Using Compustat data, we confirmed that skewness is procyclical, becoming negative in recessions and positive in booms. We then showed that this pattern is strongly size dependent: large firms display much larger swings in skewness than smaller firms. Finally, we established that countercyclical variance amplifies these effects, with increases in volatility leading to disproportionately negative skewness among the largest firms. Taken together, these findings point to a systematic size gradient of skewness in the cross-section of firms.

Since this size gradient cannot be explained by theories in which firms are fundamentally homogeneous, we set out to develop a simple framework to rationalize all empirical observations jointly. When firms possess market power, Marshall's Second Law of Demand implies that symmetric shocks translate into concave output adjustments. This mechanism generates systematically skewed growth responses, with the effect increasing in the degree of market power and in the variance of shocks. The model rationalizes both the procyclicality of skewness and its dependence on firm size, and it matches the empirical impulse responses we estimate from the data. In this sense, skewness is not a primitive property of the shock distribution but an endogenous outcome of firm behavior under imperfect competition.

Appendix

Proofs

Sections 3.1 and 3.2

Proof of Proposition 1. Direct consequence of substituting into Lemma 1 below, with $\epsilon = 1$. ■

Lemma 3 (Small-scale third central moment and skewness)**.** *Let U_1, U_2, Z be independent real-valued random variables with continuous distributions such that:*

1. *$U_1 \overset{d}{=} U_2$, $\mathbb{E}[U_i] = 0$, and U_i is symmetric about 0;*
2. *Z is symmetric about 0 with $\mathbb{E}[Z] = 0$;*

3. $\mu_{4,U} := \mathbb{E}[U_1^4] < \infty$ *and* $\mu_{4,Z} := \mathbb{E}[Z^4] < \infty$.

Let $f \in C^2$ *in a neighborhood of* 0 *and denote* $f_1 := f'(0)$ *and* $f_2 := f''(0)$. *Fix* $a, b \in \mathbb{R}$, *and define for small* $\varepsilon > 0$

$$W_\varepsilon := f\big(\varepsilon(U_1 + Z)\big) - f\big(\varepsilon(bU_2 + aZ)\big).$$

Then,

$$\begin{aligned}
\mu_3(\varepsilon) := \mathbb{E}\big[(W_\varepsilon - \mathbb{E}W_\varepsilon)^3\big] &= \frac{3}{2} f_1^2 f_2 \, \varepsilon^4 \operatorname{Cov}(D^2, H) + o(\varepsilon^4), \\
D := (U_1 - bU_2) + (1-a)Z, \qquad & H := (U_1 + Z)^2 - (bU_2 + aZ)^2, \\
\operatorname{Cov}(D^2, H) &= (1 - b^4)\operatorname{Var}(U_1^2) - 4(a-1)(ab^2+1)\,\sigma_U^2\sigma_Z^2 - (a^2-1)(a-1)^2 \operatorname{Var}(Z^2).
\end{aligned}$$

Moreover,

$$\operatorname{Var}(W_\varepsilon) = f_1^2 \, \varepsilon^2 \Big((1+b^2)\sigma_U^2 + (a-1)^2\sigma_Z^2\Big) + o(\varepsilon^2),$$

and hence the skewness coefficient is

$$\gamma_1(\varepsilon) := \frac{\mu_3(\varepsilon)}{\operatorname{Var}(W_\varepsilon)^{3/2}} = \varepsilon \cdot \frac{f_2}{|f_1|} \cdot \frac{\frac{3}{2}\operatorname{Cov}(D^2, H)}{\Big(\big((1+b^2)\sigma_U^2 + (a-1)^2\sigma_Z^2\big)\Big)^{3/2}} + o(\varepsilon).$$

Proof. Set $X_1 := U_1 + Z$, $X_2 := bU_2 + aZ$, $D := X_1 - X_2$, and $H := X_1^2 - X_2^2$. Taylor-expand f at 0 to write $W_\varepsilon = f_1 \varepsilon D + \frac{f_2}{2}\varepsilon^2 H + o(\varepsilon^2)$. Since D is symmetric about 0, all odd moments of D vanish. Expanding the cube of the centered W_ε, the ε^3 term drops out by this symmetry and the leading contribution to μ_3 is $\frac{3}{2} f_1^2 f_2 \, \varepsilon^4 \operatorname{Cov}(D^2, H) + o(\varepsilon^4)$. Expanding $D^2 H$ and taking expectations term-by-term using independence, zero means, and $U_1 \overset{d}{=} U_2$ yields the stated formula for $\operatorname{Cov}(D^2, H)$. The variance follows from $\operatorname{Var}(W_\varepsilon) = f_1^2 \varepsilon^2 \operatorname{Var}(D) + o(\varepsilon^2)$. ■

The following lemma helps proving Proposition 2.

Lemma 4 (Skewness of Transformed RVs)**.** *Let* Z *be a random variable, continuously and symmetrically distributed about its mean* $\mathbb{E}[Z]$. *Let* $X = \sigma_X Z$ *with* $0 < \sigma_X < \infty$. *Suppose* g *is concave and increasing over the support of* X. *Then:*

1. *It holds that* $\operatorname{Sk}_r[g(X)] \le 0$, *and strictly if* g *is strictly concave.*
2. *If* h *is concave relative to* g, *i.e.* $h \prec g$, *then* $\operatorname{Sk}_r[h(X)] < \operatorname{Sk}_r[g(X)]$.
3. *Skewness locally decreases for larger* σ_X: $\frac{\partial}{\partial \sigma_X}\operatorname{Sk}_r[g(X)] < 0$, *which also holds strictly if* g *is strictly concave around* 0.

Proof of Lemma 4. (1) and (2) are simple corollaries of the notion of 'c-comparability' of Groeneveld and Meeden (1984). (3) follows from (2), locally around 0, since $\rho[f(\sigma_X \cdot x)](0) < \rho[f](0) \iff \sigma_X > 1$. ■

Proof of Proposition 2. Let $f = Q^\star$ be strictly concave and decreasing, λ symmetric, $\kappa_t = 0$.

Impact ($t = 1$). Since $Q^\star$ is decreasing, the quantiles of $Q^\star(\lambda)$ reverse those of λ. By symmetry of λ and strict concavity, $Q^\star(\lambda_r) + Q^\star(-\lambda_r) < 2Q^\star(0)$ (Jensen), so the Sk_r numerator is negative while the denominator is positive.
Rebound ($t > 1$). Write $Z = \beta^{t-1}\lambda$, $a = 1/\beta$, so $g_t = f(Z) - f(aZ)$, which is increasing in Z. By symmetry, $\mathrm{Sk}_r[g_t] = (1-F)/(1+F)$ with $F[f] \equiv [f(aZ_r) - f(Z_r)]/[f(-Z_r) - f(-aZ_r)]$. Concavity of f gives $0 < F < 1$, so $\mathrm{Sk}_r[g_t] > 0$. As $t \to \infty$, $Z \to 0$ and the ratio $F \to f'(0)(a-1)/[f'(0)(a-1)] = 1$, so $\mathrm{Sk}_r[g_t] \to 0$.
Amplitude. For $t = 1$, this follows from Lemma 4, part 3. For $t > 1$, $\mathrm{Sk}_r[g_t]$ is decreasing in F, so it suffices to show $F[\varphi \circ f] < F[f]$ for any increasing, concave φ. Bounding $\varphi \circ f$ by its tangent at $f(x)$ in the numerator and at $f(-x)$ in the denominator yields $F[\varphi \circ f] < \frac{\varphi'(f(x))}{\varphi'(f(-x))} F[f]$. Since $x < 0$ implies $f(x) > f(-x)$, concavity of φ gives $\varphi'(f(x)) < \varphi'(f(-x))$, so the prefactor is less than one. ■

Section 4

Proof of Lemma 1. See Parenti et al. (2017), note that log-concavity implies quasi-concave profits and decreasing marginal revenue. ■

For reference in further proofs, we restate a longer version of Lemma 2 and prove it.

Lemma 2 (extended) (Concavity of Output). *Consider eq. (12) and let $H(\widehat{q}) := (\ln \circ \mathfrak{mr})(e^{\widehat{q}}) + [\eta - (\eta - 1)\widehat{q}]$ denote the left hand side of the FOC. Then, $Q^\star = H^{-1}$. Then:*

$$Q^\star \text{ is concave } \Leftrightarrow (\ln \circ \mathfrak{mr})(e^{\widehat{q}}) \text{is concave in } \widehat{q}$$
$$\Leftrightarrow \mathcal{E}\mathfrak{mr}(q) = \mathcal{E}p + \frac{\mathcal{E}p}{\mathcal{E}p + 1}\mathcal{E}^2 p \text{ is decreasing in } q \;\Leftrightarrow MSLD'$$

Proof of Lemma 2. Since the marginal cost part of H is linear in $\widehat{q}$, H is concave iff $\ln \circ \mathfrak{mr}(e^{\widehat{q}})$ is concave in $\widehat{q}$. Concavity of H implies concavity of $Q^\star = H^{-1}$ because the inverse of a concave, strictly decreasing function is concave (Boyd and Vandenberghe, 2004, Ex. 3.3). For the remaining equivalences, define $\widetilde{p}(\widehat{q}) := p(e^{\widehat{q}})$ and use $\mathcal{E}p = \widetilde{p}'/\widetilde{p}$ to rewrite the derivative of $\ln \circ \mathfrak{mr}(e^{\widehat{q}})$ as $(\widetilde{p}' + \widetilde{p}'')/(\widetilde{p} + \widetilde{p}')$. Expressing this in elasticities yields $\mathcal{E}\mathfrak{mr}(q) = \mathcal{E}p + \frac{\mathcal{E}p}{\mathcal{E}p+1}\mathcal{E}^2 p$, which is decreasing in q iff MSLD' holds. ■

Proof of Proposition 4. Note first that any twice continuously differentiable, function $f : \mathbb{R} \supset D \to \mathbb{R}$ is either locally s-convex or s-concave in some point x_0, unless it is affine. Therefore, any inverse demand function which guarantees globally unskewed growth rates must be such that $\ln \circ \mathfrak{mr} \circ e^x$ is linear in x, which implies that $\mathfrak{mr}(q) = e^{A + B \ln q} = \bar{A} q^B$. Since marginal revenue is $p(q) + p'(q)q$, we are given the linear, first-order ODE: $p(q) + qp'(q) = Aq^B$. Noting that $p(q) + qp'(q) = \frac{\partial}{\partial q} qp(q)$, integrate both sides to obtain $p(q) = \frac{\bar{A}}{B+1} q^B + \frac{C}{q}$ for $B \neq -1$. Suppose the integrating constant, C, is zero. Then we must have $B \in (-1, 0)$ because $B = \mathcal{E}p$. This is indeed the only possible case, since otherwise one can show that p is not log-concave around 0. ■

Before we prove *monotone concavity* under *ISID* and Proposition 3, we prove monotone concavity under a weaker property: *Monotone Markup-to-Price Elasticities* (MMPE_α). It is implied by *ISID* and equivalent to eq. (15) for a given α. MMPE_α states that given some degree of α, markups change more strongly than the price elasticity, if quantities are perturbed.

Property 6 (Monotone Markup-to-Price Elasticities (MMPE_α))**.** *The ratio of markup elasticity to price elasticity is strictly increasing:*

$$\frac{\partial}{\partial q}\frac{\mathcal{E}\mu_\alpha(q)}{\mathcal{E}p^*(q)} > 0.$$

Proposition 5. *Consider two firms which have the same long-run output, $\widehat{q}_1 = \widehat{q}_0 = \widehat{q}$, but different degrees of market power, $1 \geq \alpha_1 > \alpha_0 \geq 0$, and otherwise face the same demand curves, p^*, up to differences in $\bar{p}$*[25]*. If $MMPE_{\alpha_0}$ or $MMPE_{\alpha_1}$ holds, then $Q^\star_{\alpha_1} \prec Q^\star_{\alpha_0}$, hence $\text{Sk}_r(\widehat{q}_1) < \text{Sk}_r(\widehat{q}_0) \leq 0$.*

We need a series of Lemmas to prove Proposition 5.

Lemma 5 (Relative Concavity of Twice Differentiable Functions)**.** *If $f, g \in \mathcal{C}^2$ then the following are equivalent: $g \prec f \iff \frac{g''}{|g'|} < \frac{f''}{|f'|}$.*

Proof. See Palmer (2003). ■

We now relate MMPE_α to this notion of relative concavity:

Lemma 6 (MMPE_α implies relative concavity, $\phi_\alpha \prec \ln p^*$)**.** *As before, denote marginal profits in terms of log sales, $\widehat{q}$, by H, but add a subscript for market power. Write*

$$\begin{aligned} H_\alpha(\widehat{q}) &= \overbrace{\alpha \ln(p^*(e^{\widehat{q}}))}^{\textit{log inverse demand}} + \overbrace{\ln\left(1/\mu(e^{\widehat{q}})\right)}^{\textit{log inverse markup, } \equiv \phi_\alpha(\widehat{q})} + \overbrace{\ln\eta - (\eta-1)\widehat{q}}^{\textit{log marginal cost, } c'(e^{\widehat{q}})} \\ &= \alpha \ln(\widetilde{p}^*(\widehat{q})) + \phi_\alpha(\widehat{q}) + c'(e^{\widehat{q}}). \end{aligned}$$

Suppose, ϕ_α and $\ln p^*$ are both concave and decreasing. Then:

$$\phi_\alpha \prec \ln p^*(\exp(\cdot)) \iff MMPE_\alpha.$$

Proof of Lemma 6. Suppress α subscripts and the $*$ superscript; the proof holds at any fixed α for which MMPE_α is satisfied. Since $f \prec g$ iff $f \circ \exp \prec g \circ \exp$, the relation $\phi_\alpha \prec \ln p(\exp(\cdot))$ is equivalent to $\ln(1+\mathcal{E}p(q)) \prec \ln p(q)$ in levels. By the characterization of relative concavity for decreasing, concave functions, this becomes $\rho[\ln(1/\mu)] < \rho[\ln p]$, i.e. the curvature ratio f''/f' is larger for $\ln\mu$ than for $\ln p$. Using $f''/f' = \frac{\partial}{\partial q}\ln f'$ for $f = \ln\mu$ and $f = \ln p$, adding and subtracting $\ln q$ inside the derivatives converts the inequality into $\frac{\partial}{\partial q}\ln(\mathcal{E}\mu/\mathcal{E}p) > 0$, which is precisely MMPE_α. ■

[25] Note that $\bar{p}$ has to be different for both firms in order to equalize their output and facilitate the local comparison.

The next Lemma on relative concavity is new and applies directly to our setting.

Lemma 7. *Let $f, g, h \in \mathcal{C}^2$, $X \subseteq \mathbb{R}$, $f : X \to \mathbb{R}$, $g : X \to D \subset \mathbb{R}$ and $h : D \to \mathbb{R}$. Let $a > 0$, $g \prec f \prec \mathrm{Id}$, and $f' < 0, g' < 0$. Then*

1. *multiply-and-transform induces concavity:*

$$(h \prec \mathrm{Id} \wedge h' > a) \implies af + h \circ g \prec f + g.$$

2. *multiply-and-transform induces convexity:*

$$(h \succ \mathrm{Id} \wedge h' < a) \implies af + h \circ g \succ f + g.$$

Proof. We prove only the first claim; the second follows with reversed inequalities. By Lemma 5, we must show that $\rho[af + h \circ g] < \rho[f + g]$. Since h is concave, the $h''g'^2$ term in the numerator of $\rho[af + h \circ g]$ is negative and can be dropped, leaving a stronger inequality involving only $af'' + h'g''$ in the numerator. Cross-multiplying this against $\rho[f + g]$ and cancelling symmetric terms reduces the condition to $h'(g''f' - f''g') > a(g''f' - f''g')$. Because $g \prec f$, we have $g''f' - f''g' > 0$, so the inequality holds iff $h' > a$, which is assumed. ■

Lemma 8. *Let $f, g, h \in \mathcal{C}^2$ and $f \prec g \prec \mathrm{Id}$ with $f'' < g''$ as well as $f' < 0$, $g' < 0$. Let $v(x) := bx + c$ with $b < 0$, $c \in \mathbb{R}$. Then*

$$f + v \prec g + v$$

Proof. By the differentiability criterion, $f + v \prec g + v$ iff $f''/(f' + b) > g''/(g' + b)$. Cross-multiplying and rearranging isolates $f''g' - g''f' > (g'' - f'')b$. The left side is positive by $f \prec g$, the factor $(g'' - f'')$ is positive by assumption, and $b < 0$, so the inequality holds. ■

We are now ready to prove skewness monotonicity in α.

Proof of Proposition 5. Write the LHS of the FOC as $H_\alpha(\widehat{q}) = \alpha \ln \widetilde{p}^*(\widehat{q}) + \phi_\alpha(\widehat{q}) + \ln \eta - (\eta - 1)\widehat{q}$, where $\phi_\alpha = \ln(1/\mu_\alpha)$ is the log-inverse markup term. Consider $\alpha_1 > \alpha_0$. The key observation is that $\phi_1 = h \circ \phi_0$ where $h(y) := \ln(1 + \frac{\alpha_1}{\alpha_0} e^y)$, which is strictly concave and satisfies $h' > \alpha_1/\alpha_0$. Under MMPE_{α_0}, Lemma 6 gives $\phi_0 \prec \ln \widetilde{p}^*$. Lemma 7 (part 1) then yields $\alpha_1 \ln \widetilde{p}^* + \phi_1 \prec \alpha_0 \ln \widetilde{p}^* + \phi_0$. (Under MMPE_{α_1}, part 2 gives the same conclusion.) Since the linear marginal-cost terms cancel, Lemma 8 extends this to $H_1 \prec H_0$.

It remains to pass from the FOC to its inverse. Since $H_1 \prec H_0$, we have $H_0 = \phi \circ H_1$ for a strictly increasing, convex ϕ. Using the inverse-function differentiation formulas $(f^{-1})' = 1/f'$ and $(f^{-1})'' = -f''/f'^3$, a direct calculation shows $\rho[H_0^{-1}] = \phi'' + \rho[H_1^{-1}]$. Since $\phi'' > 0$, we obtain $H_1^{-1} \prec H_0^{-1}$, i.e. the high-market-power firm has a more concave policy. The skewness ordering then follows from Lemma 4 (2). ■

We now work towards proving Proposition 3. We begin with two lemmas which *(i)* show that MMPE_α is implied by a technical condition, and *(ii)* that the technical condition is equivalent to an increasing superelasticity (*ISID*). This will help us prove the implications in eq. (17).

Lemma 9. *For any $\alpha \in (0,1]$: $\mathcal{E}p^*(\cdot) \prec \ln(p^*(\cdot)) \implies \ln(1+\alpha\mathcal{E}p^*(\cdot)) \prec \ln(p^*(\cdot))$ (that is, MMPE_α).*

Proof of Lemma 9. By Lemma 6, MMPE_α is equivalent to $\frac{1}{\alpha}\ln(1+\alpha\,\mathcal{E}p^*) \prec \ln p^*$, and the concavity of the LHS is increasing in α. It therefore suffices that the relation holds at the limit $\alpha \to 0$, which is exactly the hypothesis $\mathcal{E}p^* \prec \ln p^*$. ■

Lemma 10. *$\mathcal{E}p^*(\cdot) \prec \ln(p^*(\cdot)) \iff \mathcal{E}^2p^*$ is strictly increasing.*

Proof. As in the proof of Lemma 6, the relative-concavity condition $\mathcal{E}p^* \prec \ln p^*$ is equivalent to $\frac{\partial}{\partial q}\ln\left(\frac{(\mathcal{E}p^*)'}{(\ln p^*)'}\right) > 0$. Writing $(\ln p^*)' = \mathcal{E}p^*/q$ and simplifying, this becomes $\frac{(\mathcal{E}p^*)'}{\mathcal{E}p^*}\cdot q$ increasing, which is precisely $\mathcal{E}^2p^*$ increasing. ■

We follow the approach of Mrázová and Neary (2017), who express local properties of (inverse) demand functions in terms of three unit-free statistics:

$$\varepsilon(x) \equiv \mathcal{E}p^*(x) = \frac{xp^{*\prime}(x)}{p(x)}, \quad \zeta(x) \equiv \frac{xp^{*\prime\prime}(x)}{p^{*\prime}(x)}, \quad \chi(x) \equiv \frac{xp^{*\prime\prime\prime}(x)}{p^{*\prime\prime}(x)}.$$

Coefficient ζ measures the curvature of demand; $\zeta > -2$ by log-concavity of p^*. Coefficient χ is known as the *coefficient of relative temperance* of Eeckhoudt et al. (1995).

Proof of Proposition 3 Eq. (17). Eq. (17) is a direct consequence of Lemma 11. ■

Lemma 11. *(1) If MSLD' holds strictly for all $\alpha \in (0,1]$, then*

$$(\zeta\chi - \zeta^2 + \zeta\varepsilon) \geq 0.$$

(2) An increasing superelasticity, $\mathcal{E}^2p^*$, implies that MMPE_α holds for all $\alpha \in (0,1]$. The former is equivalent to

$$(\zeta\chi - \zeta^2 + \zeta\varepsilon) + (\varepsilon^2 - \varepsilon - 2) > 0.$$

(3) An increasing superelasticity, $\mathcal{E}^2p^*$, implies MSLD' for any $\alpha \in (0,1]$.

Proof. Part 1: Marginal revenue under the parametrized demand is $\mathfrak{mr}(q) = p^*(q)^\alpha(1+\alpha\mathcal{E}p^*(q))$. Computing $\mathcal{E}\mathfrak{mr}$ in terms of $\varepsilon = \frac{p^{*\prime}}{p^*}q$, $\zeta = \frac{p^{*\prime\prime}}{p^{*\prime}}q$, $\chi = \frac{p^{*\prime\prime\prime}}{p^{*\prime\prime}}q$, and differentiating with respect to q using $\varepsilon' = \frac{1}{q}(\zeta\varepsilon + \varepsilon - \varepsilon^2)$ and $\zeta' = \frac{1}{q}(\zeta\chi + \zeta - \zeta^2)$, the MSLD' condition $|(\mathcal{E}\mathfrak{mr})'| > 0$ can be rearranged to

$$(\zeta\varepsilon + \varepsilon - \varepsilon^2) + \frac{\Big[-(\zeta\varepsilon + \varepsilon - \varepsilon^2) + (\zeta\chi + \zeta - \zeta^2)\Big]}{(1+\alpha\varepsilon)} + \frac{-\alpha\varepsilon(1+\zeta-\varepsilon)^2}{(1+\alpha\varepsilon)^2} > 0. \tag{21}$$

Taking $\alpha \to 0$ yields $(\zeta\chi + \zeta - \zeta^2) \geq 0$.

Part 2: By Lemmas 9 and 10, it suffices to show that eq. (16) is equivalent to the stated inequality. The relative-concavity condition $\mathcal{E}p^* \prec \ln p^*$ translates via the differentiability criterion into $\frac{(\ln p^*)''}{(\ln p^*)'} < \frac{(\mathcal{E}p^*)''}{(\mathcal{E}p^*)'}$. Using $\frac{(\ln p^*)''}{(\ln p^*)'} = \frac{1}{x}(\zeta - \varepsilon)$ and computing $(\mathcal{E}p^*)' = \frac{\varepsilon}{x}(\zeta + 1 - \varepsilon)$, $(\mathcal{E}p^*)'' = \frac{\varepsilon}{x^2}\big[\zeta(\chi + \varepsilon + 2) - 2\varepsilon(1 + \varepsilon)\big]$, the curvature-ratio comparison reduces (after cross-multiplication and collecting terms; note $p^{*\prime} < 0$ flips the inequality) to

$$\zeta\chi - \zeta^2 + \zeta\varepsilon + \varepsilon^2 - \varepsilon - 2 > 0.$$

Part 3: The last term in (21) is negative (since $\varepsilon < 0$), so dropping it strengthens the inequality. The remaining two terms are positive whenever $(\zeta\chi + \zeta - \zeta^2) - (\zeta\varepsilon + \varepsilon - \varepsilon^2) > 0$, i.e. $\zeta > 2/\varepsilon$. Since $\varepsilon \in (-1, 0)$ by regularity, $2/\varepsilon < -2 < \zeta$, so this holds. Therefore ISID implies MMPE_α for all α. Comparing the ISID bound from Part 2 with the Part 1 bound, the difference is $(\varepsilon^2 - \varepsilon - 2) < 0$, so ISID also implies MSLD' for all α. ■

Proof of Proposition 3, Eq. (18). Suppress the superscript $*$ on p^*. Define the (negative) elasticity $\varepsilon(q) \equiv \frac{q\,p'(q)}{p(q)} < 0$, curvature $\zeta(q) \equiv \frac{q\,p''(q)}{p'(q)}$, and superelasticity $\psi(q) \equiv \frac{q}{\varepsilon(q)}\,\varepsilon'(q)$. Direct differentiation gives the identity

$$\psi(q) = 1 - \varepsilon(q) + \zeta(q). \tag{22}$$

We show that MSLD ($\varepsilon' < 0$, hence $\psi \geq 0$) and IPT jointly imply $\psi'(q) > 0$.

Step 1 (Reduced-form pass-through). Let $q^\star(\bar{c})$ solve $\max_q p(q)q - \bar{c}\,q^\eta$. The FOC is $p(q) + qp'(q) = \eta\bar{c}\,q^{\eta-1}$. By the implicit function theorem, $\tau(\bar{c}) = p'(q)\frac{dq^\star}{d\bar{c}}$. Eliminating $\bar{c}$ via the FOC and substituting $qp' = \varepsilon p$, $qp'' = \zeta p'$, then using (22) to eliminate ζ, yields

$$\tau(q) = \frac{\eta\,\varepsilon(q)\,q^{\eta-1}}{\varepsilon(q)\big(\varepsilon(q) + \psi(q) + 2 - \eta\big) - (\eta - 1)}. \tag{23}$$

Step 2 (Sign of $\tau'(q)$). Since $q^\star$ is decreasing in $\bar{c}$, IPT ($\frac{d\tau}{d\bar{c}} > 0$) implies $\frac{d\tau}{dq} < 0$ by the chain rule.

Step 3 (Extract ψ'). Write $\tau = N/D$ with $N = \eta\varepsilon q^{\eta-1}$ and D the denominator in (23). Differentiating using $\varepsilon' = \psi\varepsilon/q$, the condition $\tau' < 0$ becomes (after factoring out $\eta q^{\eta-2}\varepsilon/D^2 < 0$):

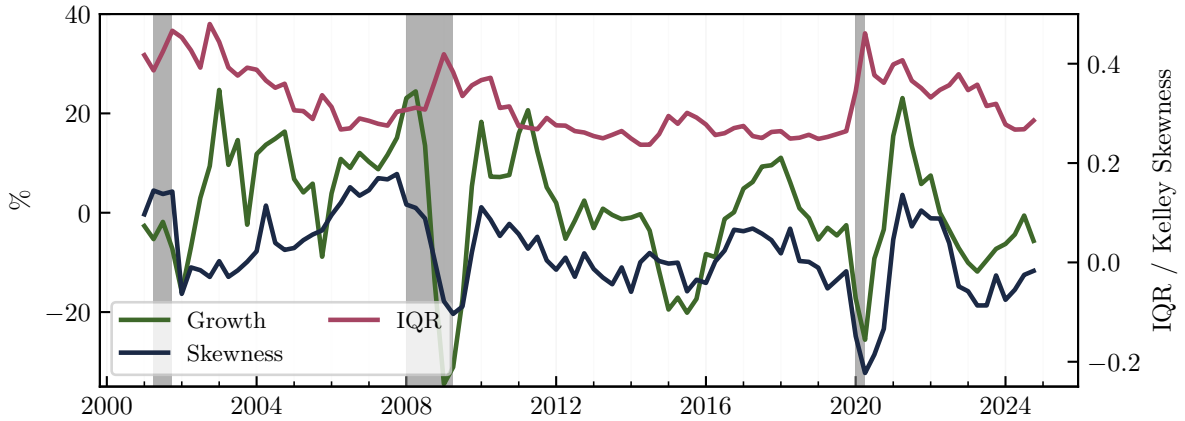
$$(\psi + \eta - 1)D - \varepsilon\big[\psi(2\varepsilon + \psi + 2 - \eta) + q\psi'\big] > 0. \tag{24}$$

Substituting for D and collecting terms isolates $q\psi'(q)$ on the left. The resulting lower bound on $q\psi'$ decomposes as

$$q\,\psi'(q) > \underbrace{\tfrac{(\eta-1)\psi}{-\varepsilon}}_{\geq 0} + \underbrace{(\eta - 1)^2(\tfrac{1}{-\varepsilon} - 1)}_{\geq 0} + \underbrace{(\eta - 1)(1 - \varepsilon\psi)}_{\geq 0} + \underbrace{\psi(-\varepsilon)}_{>0}, \tag{25}$$

where every term is nonnegative because $\eta \geq 1$, $\varepsilon < 0$, and $\psi \geq 0$ (by MSLD). The last

Figure 10: Skewness and mean of sales growth in Compustat Global



Note: YoY sales growth is computed as $\log(s_{i,t}) - \log(s_{i,t})$ where $s_{i,t}$ denotes Compustat item `saleq`, first converted into US dollar using the average monthly exchange rate of the reporting date. Then we deflate the sales series by the US GDP deflator. Skewness is measured by Kelley Skewness and NBER recessions are shaded in gray.

term is strictly positive, so $q\psi' > 0$, hence $\psi'(q) > 0$. ■

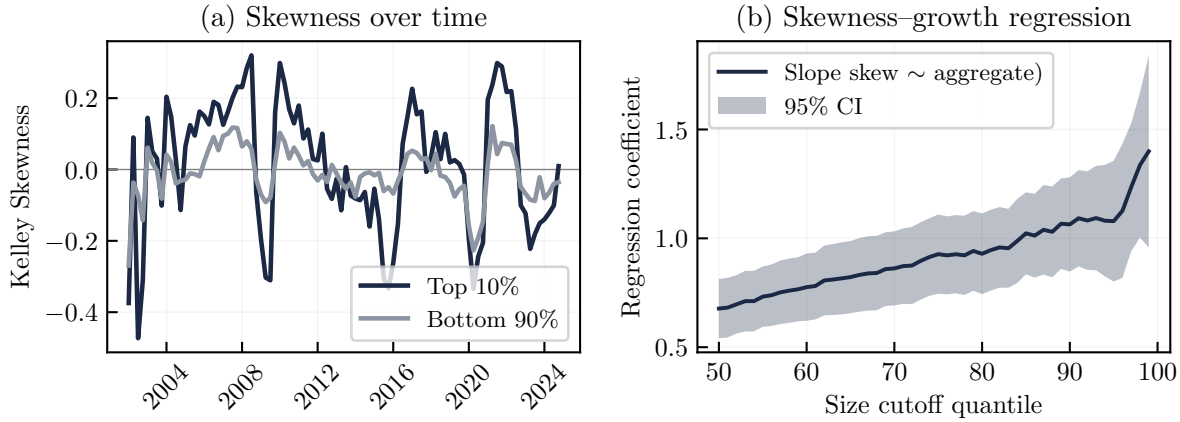
Additional Empirical Results

We present some complementary analyses to establish robustness of our results. For questions regarding sample construction or shock sequences please consult the supplementary online appendix. The online appendix also includes a bottom-up estimation of impulse-response functions, a growth decomposition documenting skewed aggregate (as opposed to idiosyncratic) components and simulation evidence.

In order to ensure that our main stylized facts are not only present in the US, but more broadly, we present the stylized facts for the Compustat Global Sample (which *excludes US companies*), between 2000Q1 and 2024Q4. We conduct a similar cleaning procedure as described in Appendix C in order to stay consistent between the two datasets.

First, Figure 10 plots the Kelley skewness of year-over-year sales growth against aggregate real sales growth in the Compustat Global Sample and confirms our first main stylized fact; Skewness is procyclical. Second, similar to the main text, panel (a) of Figure 11 plots the evolution of Kelley skewness over time for large and small firms in the Compustat Global dataset. The definition is the same as the main text. The figure confirms that skewness for large firms is more procyclical in amplitude than for smaller firms, also in the Global Sample. Panel (b) shows the estimated sensitivity of Kelley skewness with respect to aggregate growth for increasing size thresholds for large firms. Again, the coefficient is increasing, similar to the main text.

Figure 11: Size-dependent skewness



Note: Size groups are defined based on average real sales over previous three years. The standard deviation of Kelley skewness for large firms is about 0.23 — more than twice the corresponding value of 0.11 for small firms.

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Supplemental Online Appendix

Monopolistically Skewed Business Cycles

A Supply Side Skewness

Up until now, we have focused on (negative) skew driven by the shape of the demand curve, neglecting contributions of the cost function to skewness. We discuss two types of cost functions which can contribute to skewed growth rates. Either approach rests on the assumption that adjustment of some status quo (or the 'current state') is costly in one direction and cheap in the other. This leads to a cost function with a kink located at previous-period output, which immediately implies local log-convexity of the cost function and locally left-skewed values of $\widehat{q}$.

We start from a price taking firm's static profit maximization problem (suppressing price taker subscripts),

$$\max_q q\,[\bar{p} - c(q)], \qquad c(q) = e^{\epsilon}\Psi(q).$$

Here, $\bar{p}$ is the given output price, $\Psi(q)$ is the baseline cost function, and the exponential term e^{ϵ} is a multiplicative cost shifter. The first-order condition equates marginal revenue and marginal cost. Differentiating, we obtain

$$0 = \bar{p} - e^{\epsilon}\big(\Psi(q) + q\Psi'(q)\big) \equiv \bar{p} - e^{\epsilon}\mathfrak{mc}(q),$$

where $\mathfrak{mc}(q)$ is the slope of total cost. Taking logs yields $\ln \mathfrak{mc}(q) = \log \bar{p} - \epsilon$. This condition allows us to study how equilibrium output responds to the cost shifter. Differentiating the FOC, one can show that

$$\frac{d\widehat{q}}{d\epsilon} = -\frac{1}{\mathcal{E}\mathfrak{mc}(q)}, \qquad \frac{d^2\widehat{q}}{d\epsilon^2} = \frac{1}{(\mathcal{E}\mathfrak{mc}(q))^2}\,(\mathcal{E}\mathfrak{mc})'(q)\,\frac{dq}{d\epsilon}.$$

Since $dq/d\epsilon < 0$ (higher costs reduce output), concavity of $\widehat{q}$ in ϵ requires that $(\mathcal{E}\mathfrak{mc})'(q) \geq 0$, i.e. that the elasticity of marginal cost is increasing in quantity. Put differently: if $\mathfrak{mc}(q)$ is log-convex, then $\widehat{q}$ is locally concave in the cost shifter ϵ. Now consider how different frictions shape $\Psi(q)$, and thereby $\mathfrak{mc}(q)$.

Capacity adjustments. If producing beyond installed capacity $\bar{q}$ incurs a convex penalty on top of baseline costs, marginal cost jumps at $\bar{q}$ and rises steeply beyond it, making $\mathcal{E}\mathfrak{mc}(q)$ increasing and thus $\widehat{q}$ locally concave in ϵ. Expansions beyond capacity are disproportionately costly, so output falls quickly in response to adverse cost shocks but rises only sluggishly. Empirical support for such asymmetric adjustment costs comes from the investment literature (Cooper and Haltiwanger, 2006), and their inclusion in quantitative macro models is standard since Christiano et al. (2005).

Customer acquisition. If sales depend on a pre-existing customer base b plus new customers acquired at convex cost, then expanding output beyond b requires increasingly expensive acquisition effort, while contracting merely lets the base erode. This again makes $\mathcal{E}\mathfrak{mc}(q)$ increasing beyond b, ensuring local concavity of $\widehat{q}$. Models embedding customer bases into the firm problem include Gourio and Rudanko (2014), Roldan-Blanco and Gilbukh (2021), and Ignaszak and Sedlácek (2025).

In both cases, the critical feature is that the marginal cost function becomes increasingly steep as output rises. This makes the mapping from cost shocks ϵ to equilibrium quantities concave in logs, and thereby generates the negative skewness in firm-level growth rates. Existence of log-convexities in firms' cost functions can explain some pattern in Figure 3 (b). For a small size cutoff quantile, the plotted regression coefficients taper off and become constant at about 2.2. This suggests that growth rate skewness is still procyclical, even if market power, α, becomes small. Part of the reason may be that shocks are inherently skewed, but a more structural explanation is existence of log-convex cost functions even for price takers.

Yet, cost functions are unable to generate size dependence of skewness. To this observation, a demand-side oriented explanation caters nicely. Second, the approaches discussed above—capacity adjustments and customer acquisition cost—each only guarantee local skewness: growth rates relative to the steady state are skewed locally around long run output, q_0. That means, if the shock is not mean zero but, say, has a positive support such that *every* firm i in the cross-section adjusts to some $q_i < q_0$, then the negative skewness in growth rates is no longer guaranteed. In contrast, invoking ISID as a demand curve property immediately guarantees global left-skewness of growth rates, $\ln q/q_0$.

B Homothetic Demand Industry Equilibrium

B.1 A Separation Result

Consider a representative consumer with fixed nominal expenditure $E > 0$ and homothetic preferences $U(\mathbf{q}) = F\big(h(\mathbf{q})\big)$, where F is strictly increasing and h is homogeneous of degree one, over a continuum of differentiated varieties supplied under monopolistic competition. The expenditure function factorizes as $e(\mathbf{p}, \bar{h}) = P(\mathbf{p})\,\bar{h}$, where $P(\mathbf{p}) \equiv e(\mathbf{p}, 1)$ is the unit-cost function (homogeneous of degree one in $\mathbf{p}$). By Shephard's lemma the Hicksian demand is $h_i(\mathbf{p}, \bar{h}) = \bar{h}\,\partial P/\partial p_i$; substituting $\bar{h} = E/P$ gives Marshallian demand

$$q_i(\mathbf{p}, E) = \frac{E}{P(\mathbf{p})}\,\frac{\partial P(\mathbf{p})}{\partial p_i} = \frac{E}{P(\mathbf{p})}\,D_i\!\left(\frac{\mathbf{p}}{P(\mathbf{p})}\right),$$

where $D_i(\mathbf{r}) \equiv \partial P(\mathbf{r})/\partial r_i$ depends on the full normalized price vector $\mathbf{r} = \mathbf{p}/P(\mathbf{p})$ (using homogeneity of degree zero of $\partial P/\partial p_i$).

Firms have isoelastic variable costs $C_i(q) = e^{u(\xi+\lambda_i)}q^{\eta}$ with $\eta > 1$, where u is an aggregate shock, ξ is common exposure, and λ_i is idiosyncratic exposure. There is a measure-zero set of "heterogeneous firms" (type A, $\lambda_i \neq 0$) and a unit mass of "fringe" firms (type B, $\lambda_i = 0$). Since type-A firms are measure zero they do not affect the price index or any aggregate object pinned down by the fringe.

Proposition A.6 (Common-scaling separability under homotheticity)**.** *Suppose (i) preferences are homothetic with Marshallian demands*

$$q_i(\mathbf{p}, E) = \frac{E}{P(\mathbf{p})}\,D_i\!\left(\frac{\mathbf{p}}{P(\mathbf{p})}\right),$$

(ii) heterogeneous firms (type A) have measure zero, while the fringe (type B) has unit mass and $\lambda = 0$, and (iii) for each u there exists a symmetric interior equilibrium for fringe firms with well-defined perceived elasticity at the symmetric point. Then log output of any heterogeneous firm i admits the decomposition

$$\log q_i(u, \lambda_i) = -\frac{\xi}{\eta}\,u + Q^{\star}(\lambda_i u),$$

where $Q^{\star}(\cdot)$ is independent of the common exposure ξ.

Proof. In the symmetric fringe equilibrium all fringe prices coincide at $p_B(u)$; since type-A firms have measure zero, $P(u) = \bar{P}\,p_B(u)$ for a constant $\bar{P} > 0$ (by HD1 of P). The normalized fringe price $r_B = p_B/P = 1/\bar{P}$ is independent of u. A fringe firm's profit at relative price r is $\pi_B = E\,r\,D(r) - e^{u\xi}(E/P(u))^{\eta}D(r)^{\eta}$. Revenue $E\,r\,D(r)$ is u-free, so for the FOC at r_B to hold for all u the cost prefactor $e^{u\xi}P(u)^{-\eta}$ must be constant, giving

$$P(u) = P(0)\,e^{\xi u/\eta}, \qquad \frac{E}{P(u)} = \frac{E}{P(0)}\,e^{-\xi u/\eta}.$$

Firm i takes $P(u)$ and all other prices as given. Writing $r_i = p_i/P(u)$ and $\mathbf{r}_{-i} =$

$\mathbf{p}_{-i}/P(u)$, revenue is $E\, r_i\, D_i(r_i; \mathbf{r}_{-i})$ (u-free) and cost is

$$e^{u(\xi+\lambda_i)}\left(\tfrac{E}{P(0)}\, e^{-\xi u/\eta}\, D_i\right)^{\eta} = e^{u(\xi+\lambda_i)} \cdot e^{-\xi u} \cdot \left(\tfrac{E}{P(0)}\right)^{\eta} D_i^{\eta} = e^{u\lambda_i}\left(\tfrac{E}{P(0)}\right)^{\eta} D_i^{\eta}.$$

All ξ-dependence cancels; the FOC w.r.t. r_i depends on u only through $e^{u\lambda_i}$. Every other type-A firm faces the same structure, so $\mathbf{r}^{\star}_{-i}$ depends on the respective $\lambda_j u$, not on ξu. Hence $\log q_i = -(\xi/\eta)\, u + Q^{\star}(\lambda_i u)$, where $Q^{\star}$ absorbs the equilibrium dependence on $r_i^{\star}$, $\mathbf{r}^{\star}_{-i}$, and D_i. ■

The key cancellation is $e^{u(\xi+\lambda_i)} \cdot (E/P(u))^{\eta} = e^{u\lambda_i}(E/P(0))^{\eta}$: the common cost factor $e^{u\xi}$ is absorbed exactly by $P(u)^{\eta} \propto e^{\xi u}$. The residual common scaling $e^{-\xi u/\eta}$ in output reflects the erosion of real expenditure as the price level rises with the common cost shock.

B.2 HSA demand and monopolistic competition

We illustrate the previous proposition within a symmetric homothetic single-aggregator (HSA) demand system by Matsuyama and Ushchev (2020). Such is characterized by a *market share* function $s : \mathbb{R}_+ \to \mathbb{R}_+$, allowing to write each variety ω's revenue share as $s_\omega = s\left(\frac{p_\omega}{A(\mathbf{p})}\right)$, where the *aggregate price* $A(\mathbf{p})$ is defined implicitly by the *adding-up constraint* $\int_\Omega s\left(\frac{p_\omega}{A(\mathbf{p})}\right) d\omega = 1$. Thus, under HSA, all cross-variety interactions are summarized by the scalar $A(\mathbf{p})$, and firm-level demand depends on own price only through the normalized price $z_\omega \equiv p_\omega/A(\mathbf{p})$. With sectoral expenditure normalized to $E = 1$, a firm charging price p sells $x(p) = \frac{s(z)}{p} = \frac{s(z)}{Az}$.

Retaining the type-A/B structure and cost shocks from Section B, marginal cost is $c_i = \bar{c}\, e^{e_i}$ with $e_B = u\xi$ (type B) and $e_A = u(\xi + \lambda_i)$ (type A). We additionally impose symmetric distribution of the demeaned exposures, i.e. $\lambda_i \sim \Lambda$, $\Lambda(\lambda) = \Lambda(-\lambda)$. For nice analytical expressions, we consider linear costs, seting $\eta = 1$. Like above, since type-A firms have measure zero, $A(u)$ is determined by the symmetric type-B equilibrium alone. We now consider the CoPaTh sub-family of HSA to analyze firm policies.

CoPaTh specialization. Under the CoPaTh family (Matsuyama and Ushchev, 2020), $s(z) = \gamma[\sigma - (\sigma-1)(z/\beta)^k]^{1/k}$ with $k \equiv (1-\rho)/\rho$, $\rho \in (0,1)$, $\sigma > 1$, $\beta, \gamma > 0$. CoPaTh implies constant pass-through in normalized prices: $z_i = z_0\, e^{\rho e_i}$, where $z_0 = \bar{z}^{\,1-\rho} A^{-\rho}$ for a constant $\bar{z} > 0$.

Under CoPaTh, type-A firms respond to aggregate shocks at first and second order in the following fashion:

Proposition A.7 (Type-A log output under CoPaTh–HSA to second order)**.** *In the environment above with $E = 1$ and no fixed costs, type-A firm i's log output admits the decomposition of Proposition 1,*

$$\log x_i(u, \lambda_i) \;=\; -\xi\, u \;+\; Q^{\star}(\lambda_i u),$$

where $Q^\star(0) = -\log A^\star - \log z^\star$ (with $A^\star \equiv A(0)$ and $s(z^\star) = 1$), and the derivatives of $Q^\star$ at the origin are

$$Q_1 \equiv Q^{\star\prime}(0) = -\rho\,\sigma\,\gamma^k, \qquad Q_2 \equiv Q^{\star\prime\prime}(0) = -\rho(1-\rho)\,\big(\sigma\gamma^k - 1\big)\big(\sigma\gamma^k\big).$$

Proof. Under $E = 1$, $\log x_i = -\log A(u) - \log z_i(u) + \log s(z_i(u))$. The adding-up constraint with measure-zero type-A firms gives $s(z_B(u)) = 1$, so $z_B(u) = z^\star$ for all u. CoPaTh pricing then yields $z_0(u) = z^\star e^{-\rho u\xi}$, and using $z_0 = \bar{z}^{1-\rho}A^{-\rho}$ gives $A(u) = A^\star e^{u\xi}$. For type-A firm i, $z_i(u) = z_0(u)e^{\rho u(\xi+\lambda_i)} = z^\star e^{\rho u\lambda_i}$ (common exposure ξ cancels). Hence $\log x_i = -\log A^\star - u\xi - \log z^\star - \rho u\lambda_i + \log s(z^\star e^{\rho u\lambda_i})$, confirming the $-\xi u + Q^\star(\lambda_i u)$ decomposition.

Define $B(z) \equiv (\sigma-1)(z/\beta)^k$, $g(z) \equiv \sigma - B(z)$, $b(z) \equiv B(z)/g(z)$. Then $d\log s/d\log z = -b(z)$ and $d^2\log s/d(\log z)^2 = -kb(1+b)$. At $z^\star$: $g(z^\star) = \gamma^{-k}$, so $b^\star = \sigma\gamma^k - 1$ and $1+b^\star = \sigma\gamma^k$. Since $(\log z_i)'(0) = \rho\lambda_i$ and $(\log z_i)''(0) = 0$:

$$Q^{\star\prime}(0) = -\rho\lambda_i - b^\star\rho\lambda_i/\lambda_i = -\rho(1+b^\star) = -\rho\sigma\gamma^k,$$

$$Q^{\star\prime\prime}(0) = -k\,b^\star(1+b^\star)\rho^2 = -\rho(1-\rho)(\sigma\gamma^k - 1)(\sigma\gamma^k). \qquad \blacksquare$$

Proposition A.7 states the local derivatives $Q_1^\star$ and $Q_2^\star$ of Proposition 1 for the CoPaTh demand system. We see that CoPaTh satisfies local concavity, hence it qualitatively agrees with the stylized facts up to the size gradient. We show next, by specializing the example further, that taking CoPaTh inverse demand as a primitive for our market power parametrization does not satisfy the sufficient conditions for monotone concavity.

An α-Parametrization under CoPaTh Inverting the HSA demand $x = s(z)/(Az)$ with $z \equiv p/A$ and $g(z) \equiv \sigma - (\sigma-1)(z/\beta)^k$ yields the CoPaTh inverse demand[26]

$$p^*(x;A) = A\beta\left(\frac{\sigma}{\sigma - 1 + (\beta Ax/\gamma)^{\frac{1-\rho}{\rho}}}\right)^{\frac{\rho}{1-\rho}}.$$

By construction, the CoPaTh family yields *constant* log pass-through: $d\log p^*/d\log c = \rho$ for all cost levels. Since IPT (Property 5) requires pass-through to be *strictly increasing*, CoPaTh demand does not satisfy IPT.

Next, set $\sigma = 2$, $\rho = \frac{1}{2}$ (so $k = 1$), $\beta = \gamma = 1$, and normalize $A = 1$, i.e. move to a partial equilibrium setting. The CoPaTh inverse demand becomes $p^*(x) = 2/(1+x)$, with $z = p^*$ since $A = 1$. For $\alpha \in (0,1]$ define the *α-power inverse demand*

$$p(x) \;\equiv\; p^*(x)^\alpha \;=\; \left(\frac{2}{1+x}\right)^\alpha.$$

[26] *Derivation:* Write $Ax = \gamma\,g(z)^{1/k}/z$, raise to power k, and set $y \equiv (z/\beta)^k$:

$$\left(\frac{Ax}{\gamma}\right)^k = \frac{\sigma - (\sigma-1)y}{\beta^k y} \quad\Longrightarrow\quad y = \frac{\sigma}{\sigma - 1 + (\beta Ax/\gamma)^k},$$

so $z(x;A) = \beta\,y^{1/k}$ and $p^* = Az$.

The case $\alpha = 1$ recovers CoPaTh; as $\alpha \to 0$ the inverse demand $p(x) \to 1$ becomes constant.

Since p is a power of p^*, the elasticity of inverse demand scales by α while the superelasticity is invariant:

$$\mathcal{E}p(x) = \alpha\,\mathcal{E}p^*(x) = -\frac{\alpha x}{1+x}, \qquad \mathcal{E}^2 p(x) = \mathcal{E}^2 p^*(x) = \frac{1}{1+x}.$$

We immediately see that p^* and p both satisfy MSLD. Because $\mathcal{E}^2 p > 0$ everywhere, the elasticity of inverse demand is increasing in output (in absolute value) for every α. However, $\mathcal{E}^2 p = 1/(1+x)$ is *decreasing* in x, so ISID fails. Since ISID and IPT both fail, neither of the sufficient conditions for monotone concavity are satisfied. Yet, if one were to plot $Q_\alpha^{\star\prime\prime}(0)/|Q_\alpha^{\star\prime}(0)|$, the resulting plot would still be downward sloping, with (local) monotone concavity *de facto* holding true.

For completeness, we also examine MSLD'. Marginal revenue under p is $\mathfrak{mr}_\alpha(x) = p(x)[1+\mathcal{E}p(x)] = p(x)\cdot(1+(1-\alpha)x)/(1+x)$, and its elasticity evaluates to

$$\mathcal{E}\mathfrak{mr}_\alpha(x) = -\frac{(1+\alpha)x}{1+x} + \frac{(1-\alpha)x}{1+(1-\alpha)x}.$$

At $\alpha = 1$ this simplifies to $|\mathcal{E}\mathfrak{mr}_\alpha| = |2x/(1+x)|$ and MSLD' holds for p^*, which is increasing. Hence, log-output is globally concave under this CoPaTh demand.

For general α, *MSLD'* requires $\partial_x|\mathcal{E}\mathfrak{mr}_\alpha(x)| > 0$. Differentiating yields

$$\frac{d}{dx}\mathcal{E}\mathfrak{mr}_\alpha(x) = -\frac{1+\alpha}{(1+x)^2} + \frac{1-\alpha}{(1+(1-\alpha)x)^2}.$$

which is negative only if x is small enough, i.e., if the marginal cost is large enough. Hence, the MSLD' only holds locally for general α. Thus, policies are log-concave precisely if the equilibrium value of output is small enough.

This example shows that IPT $\wedge$ MSLD $\Rightarrow$ ISID (Proposition 3) relies on *strict* IPT. In the knife-edge case of constant pass-through rates, inverse demand satisfies weak IPT and MSLD, yet ISID fails. It also illustrates that the sufficient conditions constructed from ISID and IPT are not necessary, so MSLD' (and hence concavity of $Q^\star$) can hold even when both IPT and ISID fail.

C Data Preparation

This appendix describes the construction of the firm-level panel used throughout the paper. The backbone of the analysis is quarterly data file from Compustat. In some exercises, we merge this sample with additional sources (CRSP, Worldscope Fundamentals, and I/B/E/S); these merges are described in separate subsections. Unless stated otherwise, the unit of observation is a firm-quarter, firms are identified by `gvkey`, and time is indexed by the calendar quarter reported in Compustat (`datacqtr`).

C.1 Compustat Quarterly Sample

Sample restrictions. We start from the full Compustat North America database at quarterly frequency. We restrict the sample to U.S. firms by requiring that the firm is both incorporated in the United States (`fic`=`USA`) and headquartered in the United States (`loc`=`USA`). We exclude non-operating and public establishments by dropping observations with SIC codes in $[9000, 9999]$. We drop firm-quarters with non-positive nominal sales and remove duplicate firm-quarter observations.

Real sales and growth rates. Nominal quarterly sales are deflated using the GDP price deflator (FRED series `GDPDEF`) to obtain real sales,

$$s_{i,t} \equiv \frac{\texttt{saleq}_{i,t}}{\texttt{GDPDEF}_t}.$$

and compute the year-on-year growth rate of real sales as

$$g_{i,t} \equiv \log s_{i,t} - \log s_{i,t-4}.$$

When real sales are missing in a quarter that is surrounded by non-missing observations for the same firm (i.e., observed at both $t-1$ and $t+1$), we fill the missing value by linear interpolation. If missing observations occur in adjacent quarters, we do not impute.

Aggregate and industry growth. We construct an aggregate real sales growth series by value-weighting firm-level growth rates using lagged real sales:

$$g_t \equiv \frac{\sum_i s_{i,t-4}\, g_{i,t}}{\sum_i s_{i,t-4}}. \tag{A.1}$$

This construction ensures that the aggregate series is based only on firms with observed sales in both t and $t-4$, and is therefore not mechanically affected by entry or exit. We analogously compute value-weighted sales growth at different levels of NAICS aggregation.

Firm characteristics. We construct several firm-level characteristics commonly used in the literature. Total debt is the sum of short-term and long-term debt (`dlcq`+`dlttq`), and total assets are `atq`. Leverage is defined as total debt divided by total assets. Net leverage subtracts net current assets from total debt before scaling by assets, where net current assets are current assets minus current liabilities (`actq`-`lctq`). Liquidity is measured as cash and short-term investments (`cheq`) divided by total assets. We additionally construct standard controls such as Tobin's Q, book-to-market, profitability proxies (EBITDA and cash flow), and sales-to-assets.

Cleaning and outlier treatment. We define the *full sample* as the set of firm-quarters that satisfy the geographic and industry restrictions above and for which year-on-year real sales growth can be computed. We then apply the following additional filters to obtain the *cleaned sample* on which most of our analysis is based, unless otherwise stated.

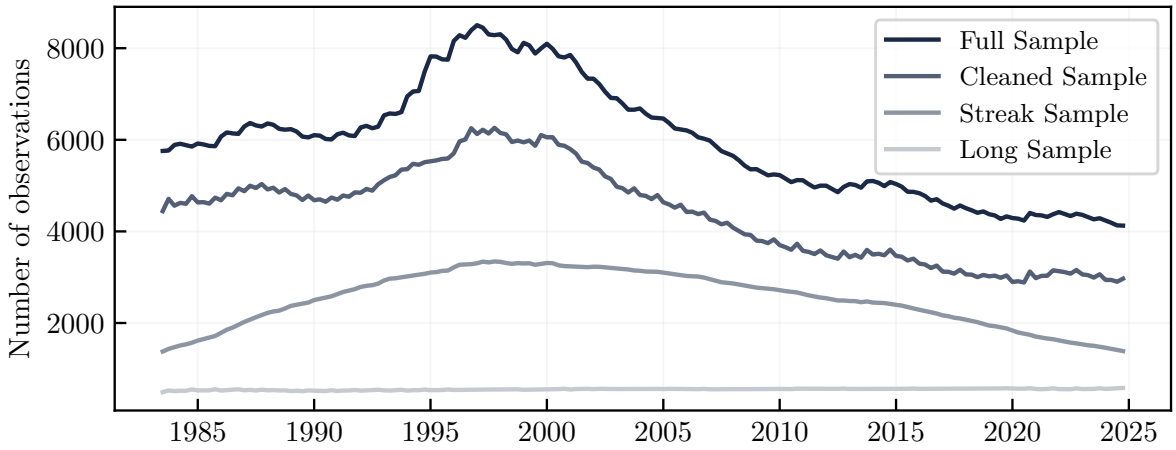
1. We drop firm-quarters with non-positive total assets or current assets. We also drop firm-quarters with non-positive cash and short-term investments (so that the liquidity ratio is well-defined).
2. We drop observations with extreme net current assets relative to total assets. Specifically, we keep only firm-quarters for which net current assets (current assets minus current liabilities) lie between -10 and $+10$ times total assets.
3. We drop observations with leverage below zero or above 10, where leverage is defined as total debt (short-term plus long-term) divided by total assets.
4. To remove extreme outliers in accounting levels, we drop firm-quarters in the top 0.1% of the sales-to-assets distribution.
5. We trim the top/bottom 1% of the cross-sectional distribution of year-on-year percent real sales growth within each quarter, and we drop observations with percent growth below -100%. We do not trim log growth

Sample period. Our baseline analysis focuses on the post-1983 period. We restrict the sample to 1983Q3–2024Q4.

Streak and long samples. Some parts of the analysis require substantial within-firm time-series variation. Following Ottonello and Winberry (2020), we construct a *streak sample* by retaining only sequences of at least 40 consecutive quarters for which year-on-year sales growth is observed for a given firm. In addition, to approximate a balanced panel, we define a *long sample* consisting of firms in the cleaned sample that are observed both near the beginning and near the end of the sample window.

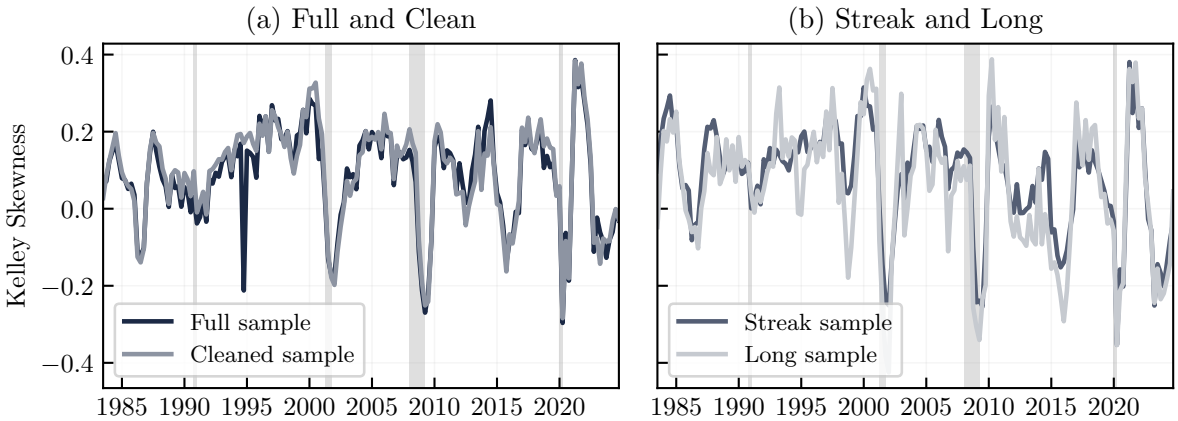
Figure A.1 shows the number of firm-level observations per quarter for the different samples. Despite differences in the number of observations, the skewness pattern across samples is very similar, see Figure A.2.

Figure A.1: Number of sales growth observations per quarter



Note: The full sample of growth rates covers 1962Q1-2022Q3. The other samples cover 1983Q3-2021Q4.

Figure A.2: Kelley skewness for different samples



Note: Skewness is computed using 90% Kelley skewness. The sample period is 1983Q3-2021Q4.

C.2 Compustat Global Quarterly Sample

Universe and currency conversion. We complement the North America sample with quarterly accounting data from Compustat Global. We drop non-operating and public establishments by excluding SIC codes in $[9000, 9999]$, remove duplicate firm-quarter observations, and drop firm-quarters with non-positive nominal sales. Because Compustat Global reports accounting items in local currencies, we first convert all key nominal balance-sheet and income-statement items to U.S. dollars using monthly exchange rates. Specifically, we match each firm-quarter to the average exchange rate over the calendar quarter (constructed from monthly averages by currency) and multiply the relevant flow and stock variables (e.g., sales, assets, debt, cash, costs) by this exchange rate. For observations already reported in U.S. dollars, the exchange rate is set to one.

Deflation, real sales, and growth rates. After converting nominal values into U.S. dollars, we deflate nominal sales using the GDP price deflator (FRED series `GDPDEF`) to obtain real sales. We compute year-on-year real sales growth in both logs and percent changes, and we treat isolated missing values in real sales by linear interpolation when a missing quarter is surrounded by non-missing observations for the same firm. As in the North America sample, we expand the panel to make missing firm-quarters explicit for the purposes of interpolation and streak construction.

Firm characteristics and cleaning. We construct the same set of firm characteristics as in the North America sample, including leverage, net current assets, liquidity, and standard profitability measures. We then apply the same outlier and plausibility filters: we drop firm-quarters with non-positive assets, current assets, or liquidity; drop observations with extreme net current assets relative to assets; drop observations with leverage outside a wide admissible range; trim the upper tail of sales-to-assets; and trim the top and bottom 1% of year-on-year percent real sales growth within each quarter while dropping percent growth below -100%.

Sample period and subsamples. The Global sample is restricted to the post-2000 period. We construct (i) a cleaned sample, (ii) a streak sample that retains firm-level sequences of at least 40 consecutive quarters with observed year-on-year percent sales growth, and (iii) a long sample intended to capture firms with observations both near the beginning and near the end of the sample window.

D Supplementary Empirical Results

D.1 Aggregate Factors Explain Skewness Fluctuations

Although we argue that market power is a key driver of procyclical skewness, skewed idiosyncratic shocks may also contribute. To assess their relative importance, consider the case where market power is absent ($\alpha = 0$). The concave policy $Q^{\star}_{\alpha_i}$ in eq. (19) then becomes linear, and growth rates reduce to a factor model (up to an affine transformation):

$$g_{i,t} = a + \widetilde{\boldsymbol{\lambda}}_i^T \boldsymbol{u}_t + w_{i,t}. \tag{A.2}$$

Under the maintained assumption of symmetrically distributed $\boldsymbol{\lambda}_i$, skewness of growth rates at a given time t would reside exclusively in $w_{i,t}$. Reintroducing market power would channel skewness into the aggregate component by skewing the distribution of exposures. A central question is therefore whether skewness originates primarily in the aggregate or the idiosyncratic component. Confirming that the aggregate component drives skewness would provide direct evidence for our propagation mechanism, complementing a growing literature on heterogeneous firm responses to aggregate shocks (Ottonello and Winberry, 2020; Cloyne et al., 2023).

We use eq. (A.2) to decompose sales growth rates into an aggregate and an idiosyncratic component via principal component analysis, following Herskovic et al. (2016). The aggregate component captures the contribution of common business cycle factors to individual growth rates; the idiosyncratic component captures all residual variation.

The existing evidence on the relative importance of these components for cross-sectional skewness is mixed: Ilut et al. (2018) find no significant skewness in establishment-level TFP shocks, while Salgado et al. (2025) find strong procyclical skewness in TFP shocks. Neither approach is conclusive about the role of idiosyncratic shocks: even if TFP shocks are not skewed (Ilut et al., 2018), other idiosyncratic shocks with skewed distributions could drive skewness in sales growth; even if TFP shocks are skewed (Salgado et al., 2025), their contribution to sales growth may be negligible because the shocks are small.[27] Focusing on a particular idiosyncratic shock therefore cannot be conclusive unless the shock is both skewed and explains a significant share of variation in sales growth.

Our decomposition is conservative in the sense that the idiosyncratic component may still contain aggregate fluctuations to which firms respond nonlinearly. However, it is certain to capture all firm-specific sources of variation.[28] If skewness in idiosyncratic

[27] Panel regressions in Salgado et al. (2025) confirm this intuition. The skewness in TFP shocks explains virtually none of the variation in firm-level sales, employment, or investment growth, as evident from the R^2 values of zero reported in Table 2 of their paper.

[28] This is true except under a network perspective in which idiosyncratic shocks cause comovement across firms that is perceived as aggregate fluctuations by PCA. See Foerster et al. (2011) for a discussion.

shocks matters for cross-sectional skewness, it must show up in this component. The estimates therefore provide an upper bound for the importance of idiosyncratic skewness.

Table A.1: Common vs idiosyncratic drivers of skewness

No. Factors:	1	4	8
Correlations with skewness:			
$\rho\left(skew_u, skew_g\right)$	0.89	0.76	0.72
$\rho\left(skew_a, skew_g\right)$	0.68	0.72	0.81
Decomposition of variation in skewness:			
R_u^2	0.21	0.26	0.32
R_a^2	0.79	0.74	0.68
Fit of aggregate component:			
R_i^2 q(0.25)	0.01	0.09	0.16
R_i^2 q(0.5)	0.06	0.20	0.30
R_i^2 q(0.75)	0.18	0.36	0.47
Observations	398,316		

Note: Each column refers to a decomposition using a different number of principal components. The decomposition uses the weighted PCA algorithm of Delchambre (2015) with zero weights for missing values and unit weights for all other observations. The first two rows measure the correlation of 90% Kelley skewness in sales growth rates with the skewness in the idiosyncratic components ($skew_u$) or the aggregate components ($skew_a$). The following two rows decompose the variation in Kelley skewness into the contributions by skewness in the idiosyncratic part and skewness in the aggregate part. The last three rows show the 25, 50, and 75% quantile of the distribution across R^2 from firm-level time series regressions of the sales growth rate onto the aggregate component. The number of observations refers to the actual firm-quarter observations.

Factor Decomposition Results

Table A.1 shows the results. Skewness across aggregate components $a_{i,t}$ correlates closely with skewness in growth rates even with only one factor (row 1). The comovement between idiosyncratic skewness and growth rate skewness decreases with the number of factors, though it remains sizeable even with eight factors (row 2).

Correlations can be misleading because comovement may be strong even when magnitudes of variation differ. To assess which component explains most of the variation in Kelley skewness, we decompose the numerator of the skewness measure—the component capturing distributional asymmetry—while the denominator serves solely as a scaling factor ensuring Kelley skewness lies in $(-1, 1)$.

Let the numerator of the Kelley skewness expression be $\eta\left(X\right) \equiv [X]_{0.9} - 2[X]_{0.5} + [X]_{0.1}$, where $[X]_r$ denotes the r-quantile of $X := \{x_i\}_{i=1,\ldots,N}$. The decomposition of demeaned growth rates $\bar{g}_{i,t} := g_{i,t} - \gamma_i$ is

$$\frac{\eta\left(\bar{g}_t\right)}{[\bar{g}_t]_{0.9} - [\bar{g}_t]_{0.1}} = \frac{\eta\left(a_t\right)}{[\bar{g}_t]_{0.9} - [\bar{g}_t]_{0.1}} + \frac{\eta\left(w_t\right)}{[\bar{g}_t]_{0.9} - [\bar{g}_t]_{0.1}} + \Delta_{a_t} + \Delta_{w_t}, \tag{A.3}$$

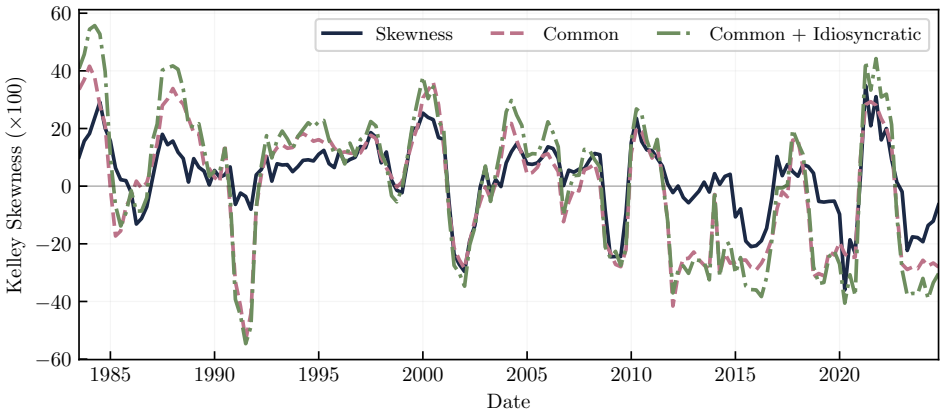
where a_t and w_t refer to the distributions of the aggregate and idiosyncratic components,

respectively. Because the ordering of firms within these distributions may differ from the ordering of sales growth rates, the decomposition is not exact. The difference is captured by approximation errors

$$\Delta_a = \left([\widetilde{a}_t]_{0.9} - [a_t]_{0.9} + 2\left([\widetilde{a}_t]_{0.5} - [a_t]_{0.5}\right) + [\widetilde{a}_t]_{0.1} - [a_t]_{0.1}\right) / \left([\bar{g}_t]_{0.9} - [\bar{g}_t]_{0.1}\right) \tag{A.4}$$

with $[\widetilde{a}_t]_r$ denoting the aggregate component of the r-quantile of the growth rate distribution g_t, and by Δ_w, defined analogously. Given these objects, we compute partial contributions to the explained variance of growth rate skewness. Of the skewness unexplained by the approximation error, the idiosyncratic component explains only 25% (R^2_w). The remaining 75% is attributed to skewness in the common factors (R^2_a). This decomposition is broadly stable across the number of aggregate factors. Because the skewness of the components is not orthogonal, the explained variance attributed to each depends on the ordering of variables. The results presented here order the idiosyncratic component first, yielding conservative estimates for the aggregate component. Reversing the ordering attributes between 92% and 96% to the aggregate component (not shown). Figure A.3 underscores this finding: skewness in the idiosyncratic component adds little information beyond the procyclical pattern present in the common component. The figure also demonstrates that the approximation error of eq. (A.3) is small, as the sum of common and idiosyncratic contributions (orange line) closely tracks the skewness measure (blue line).

Figure A.3: Skewness in common vs idiosyncratic component



Note: The blue line is the skewness in demeaned growth rates. The red line shows the contribution of skewness in the aggregate component ($\eta(a_t)$ in equation (A.3)) to skewness in demeaned sales growth rates. The green line adds the contribution of skewness in the idiosyncratic component ($\eta(w_t)$ in equation (A.3)).

The weak contribution of the idiosyncratic component is not due to its small size. For most firms, the idiosyncratic component remains large after removing aggregate factors. The last three rows of Table A.1 report the 25%, 50%, and 75% quantiles of the distribution of R^2 values from firm-level time series regressions of demeaned sales growth onto the

aggregate factors. Even with eight factors, the aggregate component explains no more than 30% of time series variation for half the firms (R_i^2 q(0.5)), and exceeds 47% for only 25% of firms (R_i^2 q(0.75)). To emphasize: a single aggregate factor explains 79% of the variation in skewness (R_a^2) while explaining only 6% of firm-level sales growth variation on average.

D.2 Robustness: Bottom-up impulse responses

In the main text we have used measures of aggregate growth and cross-sectional skewness as inputs to the local projections to study their impulse responses. Instead, this section estimates impulse responses of firm-level sales growth rates to aggregate shocks and then constructs the response of cross-sectional skewness and aggregate sales growth bottom-up from the distribution of firm-level IRFs. The local projection specification at the firm level is

$$g_{i,t+h} = \alpha_{i,h} + \beta_{i,h}\text{shock}_t + \sum_{\ell=1}^{L} \gamma'_{i,\ell,t}\text{controls}_{i,t-\ell} + e_{i,t+h}, \tag{A.5}$$

where $\beta_{i,h}$ is the response of firm i's year-on-year sales growth rate at horizon h to a shock at horizon 0. All firm-level regressions control for lagged values of the shock and lagged GDP, as well as sales growth at the firm and the 2-digit NAICS level. In addition, we include shock-specific controls: shadow rate and leverage (monetary shock), GDP deflator (oil supply), excess bond premium (credit shock), Jurado et al. (2015) financial uncertainty (uncertainty), ICE consumer sentiment, macroeconomic uncertainty, and S&P500 stock prices (sentiment), and GDP per capita, labor productivity, and S&P500 stock prices per capita (TFP news). All controls are included with two lags. The only exception is a contemporaneous control for GDP growth in the credit shock regression, mirroring the specification in Gilchrist and Zakrajšek (2012).[29]

The summary statistics for the distribution of firm-level impulse response estimates are reported in Table A.2. Varying sample periods across shocks and missing values for firm-specific controls (in particular leverage for the monetary shock) imply differences in sample sizes. The number of unique streaks is above 4,000 for all shocks except credit. The sample for the credit shock is smaller since we only consider streaks covering the Great Financial Crisis, which turns out to be crucial to identify the effects of credit shocks using the Gilchrist and Zakrajšek (2012) specification. The number of streaks can be larger than the number of unique firms in the sample since some firms can have multiple streaks in the data, although this does not happen frequently. The average time series is roughly 70 quarters long. The firm-level regressions have average R^2 values of at least 25% and over 77% of impulse response estimates have statistically significant

[29]See Table A.3 for variable definitions.

Table A.2: Firm-level local projections – Summary statistics

	Monetary	Oil	Credit	Uncertainty	Sentiment	TFP News
# Streaks	4,120	5,332	2,813	5,115	5,125	4,829
# Firms	4,017	5,061	2,813	4,893	4,902	4,651
Avg. # Obs.	64	70	84	66	66	63
Avg. R^2	0.33	0.29	0.25	0.26	0.32	0.31
Sign. IRFs (%)	77	80	83	78	82	80
$Q_{0.1}^{\text{IRF}}$ (%)	-5.4	-14.3	-2.1	-5.0	-4.5	-5.4
$Q_{0.5}^{\text{IRF}}$ (%)	-0.16	-0.14	-0.33	-0.30	-0.33	-0.39
$Q_{0.9}^{\text{IRF}}$ (%)	5.0	13.5	1.2	4.1	3.8	4.7

Note: The number of streaks can be larger than the number of unique firms. The average number of time series observations is measured for impact effect regressions and rounded to the nearest integer. The adjusted R^2 values are averaged across horizons and firms. The share of significant IRFs is the relative frequency of statistically significant IRFs for the peak of the impulse response estimates, measured using 90% confidence intervals based on Newey–West standard errors. Quantiles across firm-level IRFs are averaged across horizons. IRFs for the credit shock are only estimated for firms existing during the Great Financial Crisis.

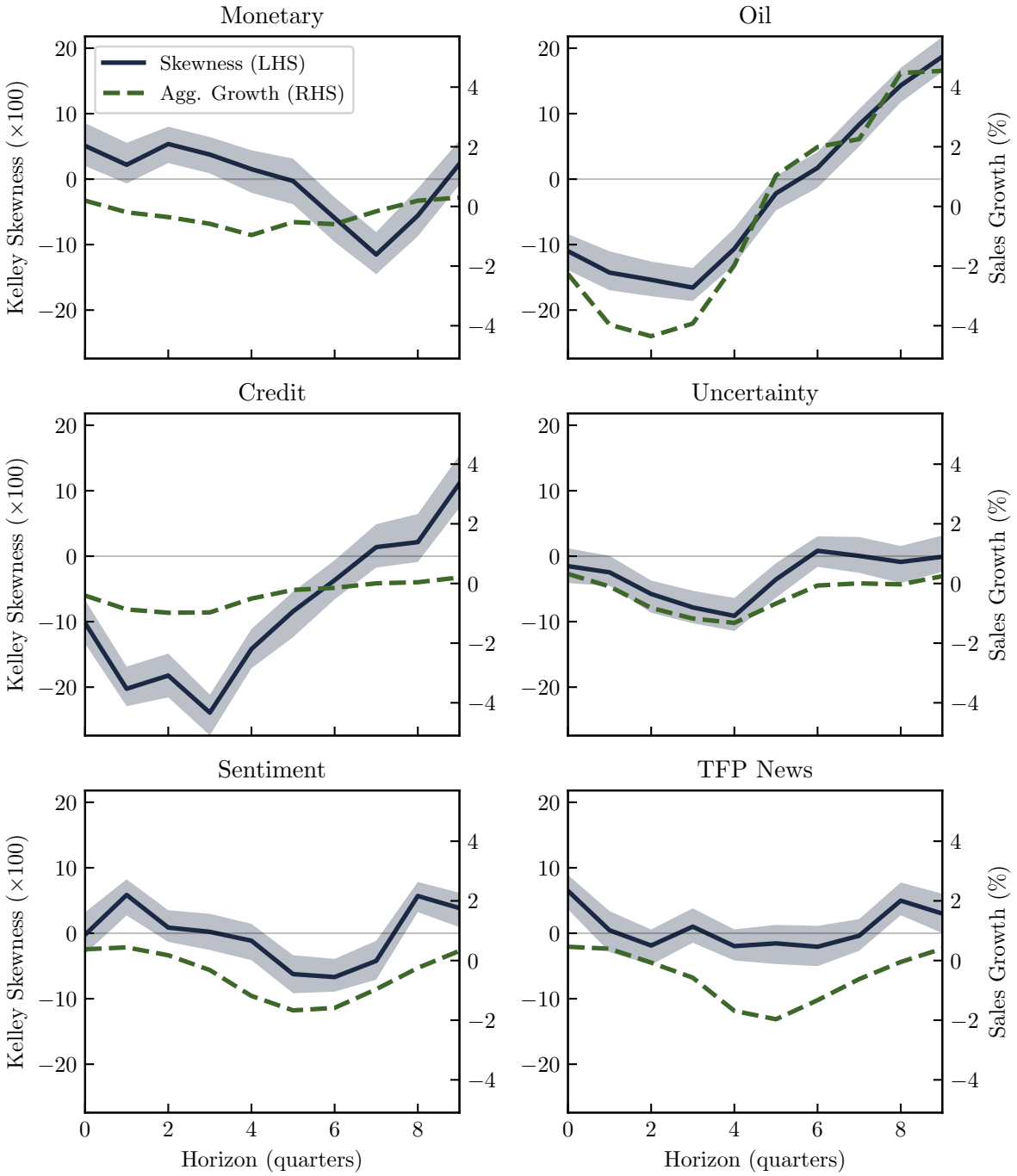
peak effects for each shock. The distribution of IRF estimates is widely dispersed with negative (unweighted) mean estimates, reflecting the contractionary nature of the shocks, and large heterogeneity in terms of firm responses. These distributions look very similar when only considering IRFs with significant peak effects (results not shown).

Based on the firm-level IRF estimates $\widehat{\beta}_{i,h}$ from equation (A.5), we construct the response of skewness and aggregate sales growth from the bottom up. The aggregate sales growth IRF is the size-weighted average of the firm IRFs: $\widehat{\beta}_h^{agg} = \sum_i \omega_{\beta,i}\widehat{\beta}_{i,h}$, where $\omega_{\beta,i}$ is the average real sales of firm i divided by the sum of average real sales across firms. The response of cross-sectional skewness is estimated from the cross section of firm IRFs: $\widehat{\beta}_h^{ksk} = ksk\left(\widehat{\mathcal{B}}_h\right)$, where $\widehat{\mathcal{B}}_h = \{\widehat{\beta}_{i,h}\}_{i=1,\ldots,N}$ is the set of firm IRF estimates at horizon h. Testing for procyclical skewness in this exercise is significantly harder than when using a top-down skewness index since individual firm IRFs are much more volatile than aggregate sales and the only source of procyclical skewness in response to a properly identified aggregate shock are heterogeneous responses across firms.

The results are shown in Figure A.4, where shaded areas are 90% confidence intervals based on a simple bootstrap with 2000 replications.[30] Following a contractionary ag-

[30]The bootstrap procedure resamples from the set of impulse responses with replacement. This takes the IRF estimates as given and does not consider the sampling uncertainty surrounding these estimates, thus understating the width of the confidence intervals (Pagan, 1984). Using a parametric bootstrap to estimate bottom-up statistics based on a distribution of firm-level IRFs across firms and bootstrap samples would correct the confidence intervals but is infeasible because of the computational burden involved. A proper estimation requires to run 1000 bootstrap replications for 5000 firms for 6 shocks, resulting in 30 million regressions. In addition, the parametric bootstrap would reduce the effective sample size even further due to the lag structure of the data-generating process, which is undesirable given the already short time series samples for some firms.

Figure A.4: Comovement of cross-sectional growth and skew after aggregate shocks



Note: The figure shows the response of aggregate sales growth ($\widehat{\beta}_h^{agg}$; green dashed line) and cross-sectional skewness ($\widehat{\beta}_h^{ksk}$; blue solid line) to different aggregate shocks. The blue shaded areas are 90% confidence bands for the skewness response, based on a bootstrap with 2000 replications. Shock magnitudes are normalized to be one standard deviation. The signs of the sentiment and the TFP shock are reversed to be contractionary.

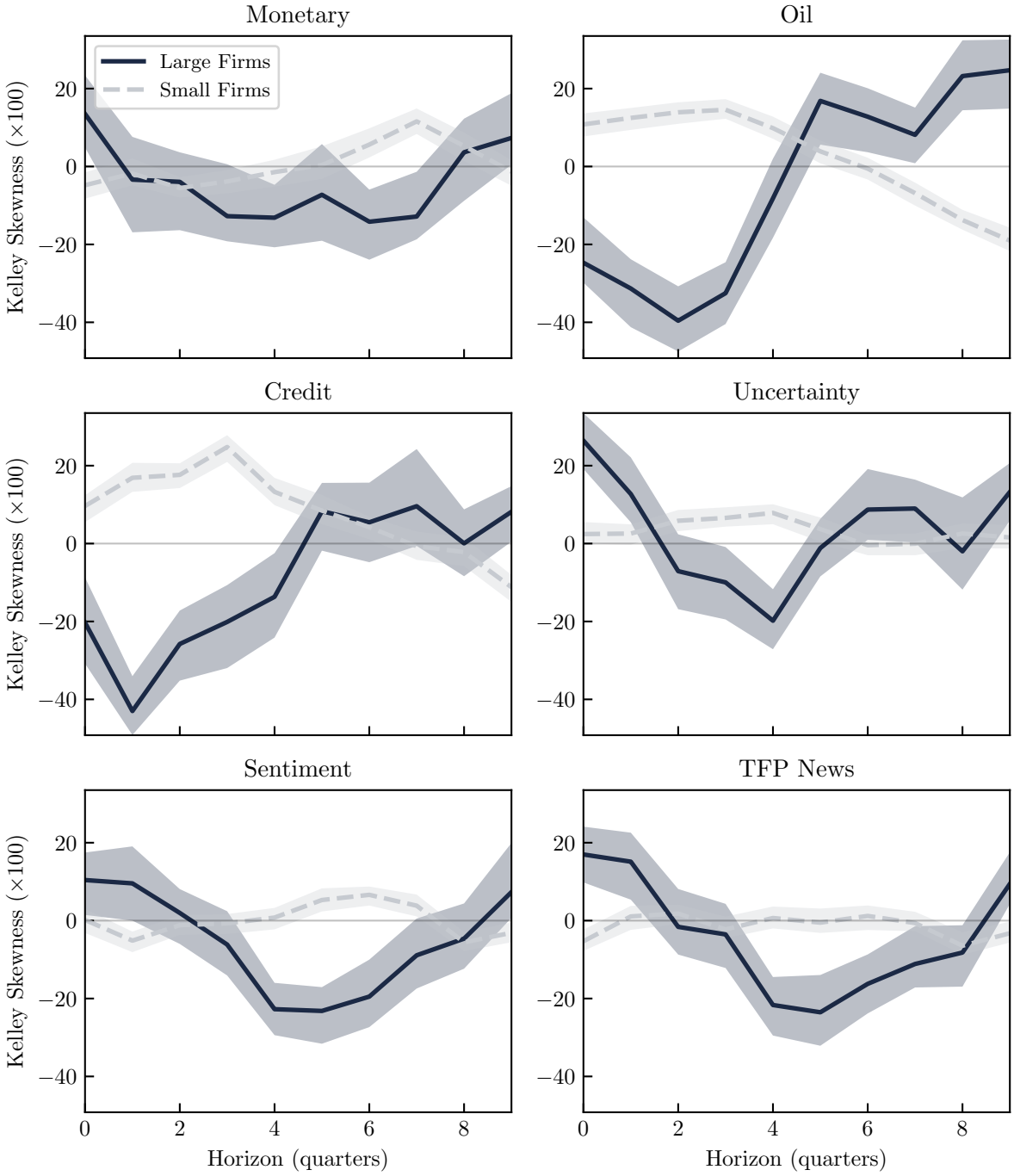
gregate shock, cross-sectional skewness (solid blue) and aggregate sales growth (dashed black) show a closely correlated decline. This is especially true for the oil, credit and uncertainty shocks. The correlations of skewness and growth following a sentiment or TFP shock are also close but the evidence for a negative skewness response is less clear. The monetary shock leads to a severe contraction in skewness but only after eight quarters, with a positive resonse on impact. Sales growth declines earlier and is recovering while skewness bottoms out. Without putting too much emphasis on any individual impulse response estimate, the findings across the six different shocks confirm that 1) cross-sectional skewness declines following contractionary aggregate shocks and 2) aggregate sales growth and cross-sectional skewness are strongly correlated following aggregate shocks.

How does the response of large firms differ from the response of small firms? We split the sample into two size groups (largest firms versus the rest) to study the impulse response of skewness across large vs small firms and compute their contribution to aggregate sales growth. Figure A.5 shows the IRFs for the largest 10% of firms (defined by average real sales) and the IRFs for the bottom 90% of firms. The black dotted line shows the contribution of large firms to the impulse response of aggregate sales growth, and the black dashed line shows the contribution of small firms. By construction, the sum of the two lines equals the impulse response of aggregate sales growth shown in Figure A.4. The red (blue) line shows the impulse response of skewness across large (small) firms. The shaded areas are 90% confidence intervals.

The bottom-up skewness response of large firms is significantly negative across shocks and in line with the impulse responses for the skewness index (Figure 5). The response of the largest firms is more skewed than the response of the rest of the firms. The differences in skewness can be large. For example, the minimum of the skewness IRF in response to a one standard deviation sentiment shock is around -0.2 for the largest firms but only -0.04 for the smaller firms. In response to an oil shock, large firms' skewness declines by over 0.3 points, while smaller firms' skewness falls by 0.1 points at most. The differences are also large for the monetary and the TFP shock and less pronounced for the credit and the uncertainty shock. In any case, the response across large firms is *not less* skewed than the response of small firms.

Summarizing the results of this section so far, aggregate shocks induce procyclical skewness through heterogeneous responses across firms. Aggregate shocks explain most of the business cycle variation of cross-sectional skewness. The set of the largest firms in the US economy also shows skewed responses to aggregate shocks, suggesting that some large firms respond strongly to those shocks.

Figure A.5: Large vs small firms: Cross-sectional skewness responses



Note: The figure shows the responses of cross-sectional skewness ($\widehat{\beta}_h^{ksk}$; blue solid line for large and light blue dashed for small) to different aggregate shocks, split by large and small firms. Large firms are the top 10% of the sales distribution, which averages real sales over time for each firm. Small firms are all other firms. The shaded areas are 90% confidence bands for the skewness responses, based on a bootstrap with 2000 replications. Shock magnitudes are normalized to be one standard deviation. The signs of the sentiment and the TFP shock are reversed to be contractionary

D.3 Cross-Sectional Skewness v. Aggregate Shocks

D.3.1 Shock Series for IRFs

Monetary shock We use the Bu et al. (2021) shock series, which are constructed to bridge periods of conventional and unconventional monetary policy. This is useful because the skewness series only starts in the mid-1980s while unconventional monetary policy became an important policy tool from 2008 onwards. Being restricted to a 1985-2008 sample period would make identification difficult, especially with quarterly data. The shock is estimated with Fama-MacBeth regressions using changes in interest rates at different maturities around FOMC announcements such that the second-stage coefficient estimates are the monetary shock series. In our local projection specification, we include lags of real GDP and the GDP deflator (both as detrended log levels) as well as the Wu and Xia (2016) shadow rate and the excess bond premium as controls. The excess bond premium captures financial conditions and is a useful control for the predictable component of the business cycle. We also include lags of the dependent variable and the shock as controls.

Oil supply shock The oil supply shocks are identified following Baumeister and Hamilton (2019), who use carefully selected priors for demand and supply elasticities in the oil market (among priors for other coefficients) in a Bayesian VAR. Their identification scheme allows them to relax some identifying assumptions previously imposed in the literature, for example that oil supply does not respond on impact to shocks to the oil price. Under the new identification strategy, the authors find oil supply shocks to be a more important determinant of historical oil price movements than found in the previous literature. The shock series we use is the median of the posterior distribution. We add lags of the shock, GDP, the GDP deflator, the crude petroleum producer price index, and the dependent variable as controls.

Credit shock. The credit shock uses innovations in the excess bond premium (EBP) following Gilchrist and Zakrajšek (2012). The EBP is constructed from corporate bond spreads to proxy investor risk appetite and is orthogonal to the risk of corporate default. Gilchrist and Zakrajšek (2012) use a recursive identification strategy in a VAR to study the effect of EBP innovations on macroeconomic variables. They assume that indicators of economic activity do not respond to EBP shocks within the same quarter while financial variables can respond immediately. We replicate their VAR, extract the shock, and use it in a local projection controlling for lags of the shock and the dependent variable, GDP, the GDP deflator, and the EBP.

Uncertainty shock The identification of the uncertainty shock follows Ludvigson et al. (2021), who use restrictions on the time series of the structural shocks to jointly identify financial uncertainty, macroeconomic uncertainty, and output shocks. Given the VAR residuals, the authors randomly draw many candidates for the time series of the structural shocks and only retain those that satisfy restrictions motivated from economic theory and narratives of historical events. For example, financial uncertainty should be high in October 1987 ('Black Monday') and September 2008 (Lehman collapse).[31] The remaining shocks series can be used for set identification of the impulse responses. The authors find that financial uncertainty shocks are a source of business cycle fluctuations, while macroeconomic uncertainty is more likely to be an endogenous response to output shocks. To obtain a single shock series for the financial uncertainty shock, we use the 'maxG' solution, which jointly maximizes the inequalities associated with a subset of the constraints. The controls are lags of the shock, GDP, the GDP deflator, VXO, and the dependent variable.

Sentiment shock While the previous shocks are related to economic fundamentals or financial conditions, business cycles may also be affected by fluctuations in consumer sentiment that are unrelated to economic conditions. Lagerborg et al. (2023) show that exogenous changes in consumer confidence can be recessionary. Their identification strategy relies on mass shootings in the United States, which are widely reported in the media and are shown to be predictors of downturns in sentiment. The authors show that the number of fatalities in mass shooting events can be viewed as exogenous to the state of the economy and used as a valid instrument to identify the effect of consumer confidence shocks on the business cycle. The authors estimate impulse responses in a proxy SVAR, and we extract the shock series from this system using the authors' replication codes. Similar to Lagerborg et al. (2023), we include lags of the shock, the University of Michigan Index of Consumer Expectations, real GDP, the Jurado et al. (2015) 12-month macroeconomic uncertainty index, real stock prices, and the dependent variable in the local projections.

TFP news shock News about future productivity can explain a significant share of business cycle variation, as shown in Beaudry and Portier (2006). We use shocks following the identification strategy of Ben Zeev and Khan (2015), who impose medium-run restrictions to identify news about investment-specific technology. Their shock is chosen to maximize the explained variance in (the inverse of) the relative price of investment in the medium term, while being orthogonal to both current TFP and the current relative price of investment. The authors find TFP news to account for a significant share

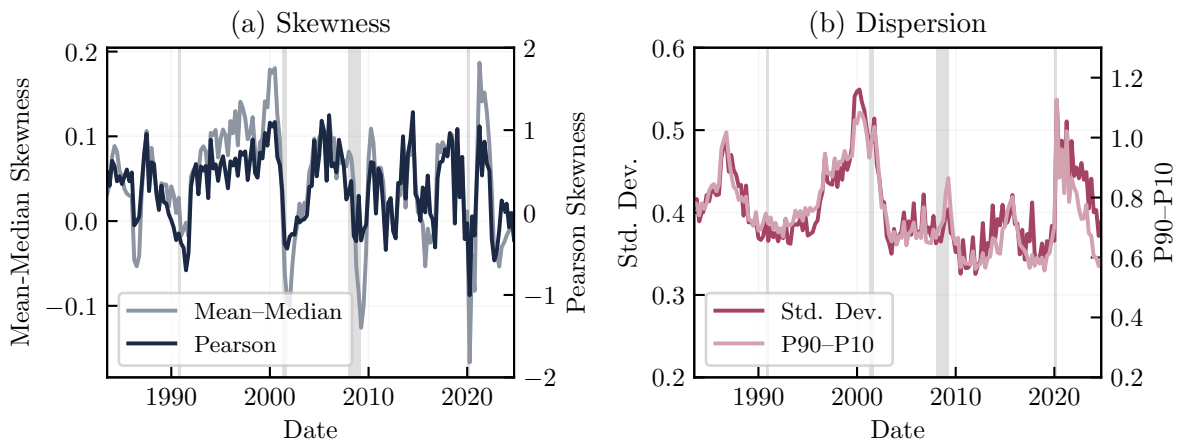
[31] The idea behind the identification scheme is similar to the classic sign restrictions, except that the restrictions are directly imposed on the time series of the structural shocks as opposed to the shape or magnitude of the impulse response estimates.

of business cycle fluctuations. The impulse responses are estimated similar to the local projections of Ramey (2016), controlling for lags of the shock, real GDP per capita, real stock prices per capita, labor productivity, and the dependent variable.

D.3.2 Local projection specifications and robustness.

Table A.3 describes the construction of all data entering the local projections, including the shocks. We use existing data or the authors' replication codes for all shock series. For the baseline specifications, cross-sectional skewness and aggregate sales growth are computed using the streak sample as described in Appendix C. We confirm that the results for the impulse responses of cross-sectional skewness to aggregate shocks (Figure 5) are robust to several robustness checks: Using four lags instead of two, including lagged values of aggregate sales growth as controls, or using the cleaned sample instead of the streak sample to compute the skewness index. Results are not shown here to conserve space, but all alternative specifications yield very similar results, sometimes so close that the different impulse responses are indistinguishable from each other because they agree up to the third decimal.

Figure A.6: Comparison of different skewness measures.



Note. The first three measures are plotted on the LHS axis. The Pearson Skewness measure is plotted on the RHS axis.

Table A.3: Data for local projections

Variable	Transformation from raw data	Data source
Real GDP	Log level	FRED (GDPC1)
GDP Deflator	Log level	FRED (GDPDEF)
Real oil price	Quarterly average of monthly data, deflated	FRED (WPU0561, GDPDEF)
GDP per capita	Real GDP ('rgdp') per population ('civpop')	Ramey (2016) TFP data
Labor productivity	Real GDP ('rgdp') per hours worked ('tothours')	Ramey (2016) TFP data
Shadow rate	Quarterly average of monthly data	Atlanta Fed*
Stock prices	Shiller stock prices divided by GDP deflator	Ramey (2016), FRED
Stock prices per capita	Stock prices per population ('civpop')	Ramey (2016)
VXO	Quarterly average of daily data	FRED (VXOCLS)
Uncertainty Index	Log level of Jurado et al. (2015) index	Lagerborg et al. (2023)
Consumer Expectations	Log level	Lagerborg et al. (2023)
Monetary shock	Quarterly sum of monthly data	Fed Board**
Oil shock	Quarterly average of monthly data	Baumeister***
Credit shock	Quarterly average of monthly data	Favara et al. (2016)†
Uncertainty shock	Quarterly average of monthly 'maxG' shock	Ludvigson et al. (2021)
Sentiment shock	Quarterly average of monthly data	Lagerborg et al. (2023)‡
TFP News Shock	Level of Ben Zeev and Khan (2015) shock	Ramey (2016) TFP data
Cross-sectional skewness	Own construction based on streak sample	–
Sales growth	Own construction based on streak sample	–

Note:
(*) Shadow rate: https://www.atlantafed.org/cqer/research/wu-xia-shadow-federal-funds-rate.
(**) Bu et al. (2021) monetary shocks: https://www.federalreserve.gov/econres/feds/a-unified-measure-of-fed-monetary-policy-shocks.htm.
(***) Baumeister and Hamilton (2019) shocks: https://sites.google.com/site/cjsbaumeister/datasets?authuser=0.
(†) Credit shock from eight-variable VAR of Gilchrist and Zakrajšek (2012), 1973Q1–2019Q4.
(‡) Sentiment shock from Lagerborg et al. (2023) proxy SVAR, 1965:1–2018:11. Instrument is number of fatalities ($\leq$7), excluding 2017 Las Vegas shooting.

E Simulation

We show that the stylized environment of our formal results carries over into a 'messier' environment. Applying the implied recipe of our theorems, we generate time-series of (1) countercyclical variance and (2) procyclical skewness from a single series of aggregate impulses. The goal is to generate aggregate series that qualitatively look like Figures 2 and 3(a). We use Algorithm 1 which follows the logic of Section 3. The simulation results are shown in Figure 7. They qualitatively agree with our stylized facts. Parameter choices for the simulation algorithm are given by Table A.4.

Table A.4: Simulation Parameters and Functional Definitions

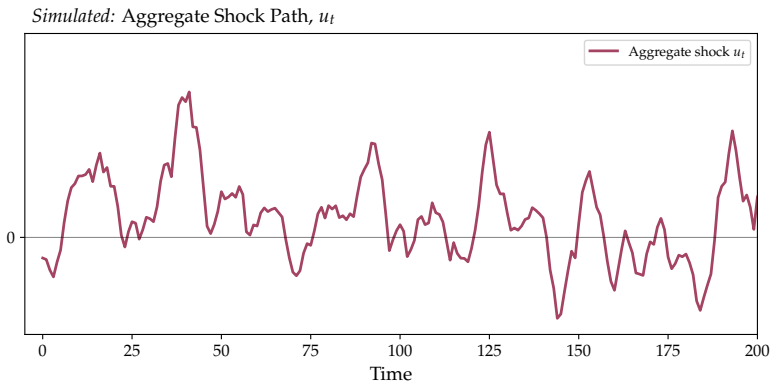
Symbol / Function	**Description**	**Value / Definition**
n	Time-series sample size	500
burn	Burn-in observations	5000
I	Cross-sectional sample size per t	50,000
$\phi_1,\ \phi_2$	AR(2) coefficients	$1.4,\ -0.5$
p	Probability of Pareto innovation	0.04
b	Pareto shape parameter	1.4
s_P	Pareto scale parameter	0.6
$\mu_N,\ \sigma_N$	Mean, std. dev. of normal innovation	$0.1,\ 0.8$
amp	Amplitude of concave map	5
$z_1,\ z_2$	Curvature parameters for $Q_1^\star,\ Q_2^\star$	$6,\ 12$
a	Skew-normal shape parameter	10
loc	Skew-normal location parameter	-0.04
s_S	Skew-normal scale parameter	0.1
$Q_h^\star(x; m, M)$	Concave decreasing map ($h = 1, 2$); m, M are min/max of sample	$\frac{\text{amp}}{z_h}\left[1 - \left(\frac{x-m}{M-m} + 0.5\right)^{z_h}\right]$

Note: $Q_h^\star$ acts elementwise on vectors $\boldsymbol{x}_t$ within the algorithm (vectorized over i).

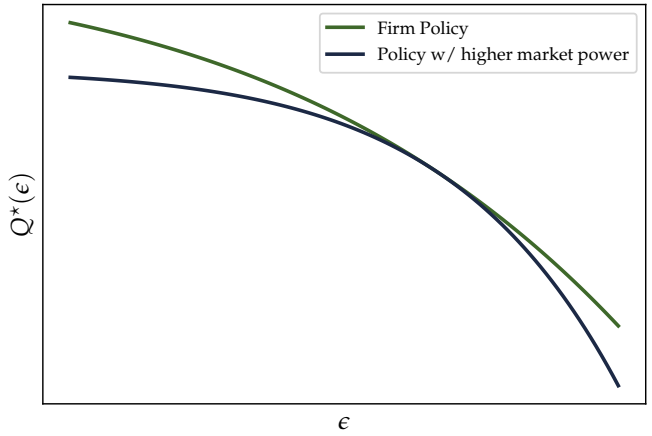
Algorithm 1: DGP for y_t and Cross-Sectional Statistics (vectorized over i)

Aggregate shocks, y_t: AR(2) with normal-Pareto mixture innovations
for $t \leftarrow 1$ **to** n+burn **do**
  Draw $u_t \sim \text{Unif}(0,1)$
  if $u_t < p$ **then** $\varepsilon_t \sim \text{Pareto}(b, \text{scale}{=}s_P)$
  else $\varepsilon_t \sim \mathcal{N}(\mu_N, \sigma_N^2)$
  $y_t \leftarrow \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$
Discard first burn and relabel t=1:n
Firm-level exposures (vectorized)
for $t \leftarrow 1$ **to** n **do**
  Draw $\boldsymbol{x}_t \in \mathbb{R}^I \sim \mathcal{N}\big((y_t/8)\mathbf{1},\ |y_t|^2\mathbf{I}\big)$
$m \leftarrow \min_{t,i} x_{t,i}, \quad M \leftarrow \max_{t,i} x_{t,i}$
Log-output through concave transforms $Q_1^\star, Q_2^\star$
for $t \leftarrow 1$ **to** n **do**
  $\boldsymbol{\ell}_t \leftarrow Q_1^\star(\boldsymbol{x}_t; m, M), \quad \boldsymbol{\ell}'_t \leftarrow Q_2^\star(\boldsymbol{x}_t; m, M)$
Growth rates through quarterly differencing
for $t \leftarrow 5$ **to** n **do**
  $\boldsymbol{g}_t \leftarrow \Delta_4 \boldsymbol{\ell}_t = \boldsymbol{\ell}_t - \boldsymbol{\ell}_{t-4}, \quad \boldsymbol{g}'_t \leftarrow \Delta_4 \boldsymbol{\ell}'_t = \boldsymbol{\ell}'_t - \boldsymbol{\ell}'_{t-4}$
Idiosyncratically skewed trend noise (constant across t per unit)
Draw $\boldsymbol{\eta} \in \mathbb{R}^I \sim \text{SkewNormal}(a, \text{loc}, s_S)$
for $t \leftarrow 5$ **to** n **do**
  $\boldsymbol{g}_t \leftarrow \boldsymbol{g}_t + \boldsymbol{\eta}, \quad \boldsymbol{g}'_t \leftarrow \boldsymbol{g}'_t + \boldsymbol{\eta}$
Cross-sectional stats
for $t \leftarrow 5$ **to** n **do**
  $\bar{g}_t \leftarrow mean(\boldsymbol{g}_t), \quad \bar{g}'_t \leftarrow mean(\boldsymbol{g}'_t)$
  $\text{KellySkew}_t \leftarrow skew(\boldsymbol{g}_t), \quad \text{KellySkew}'_t \leftarrow skew(\boldsymbol{g}'_t)$
Volatility and volatility growth
for $t \leftarrow 1$ **to** n **do**
  $v_t \leftarrow y_t^2 / \max_{s \le n} y_s^2$
for $t \leftarrow 5$ **to** n **do**
  $\Delta_4 v_t \leftarrow v_t - v_{t-4}$

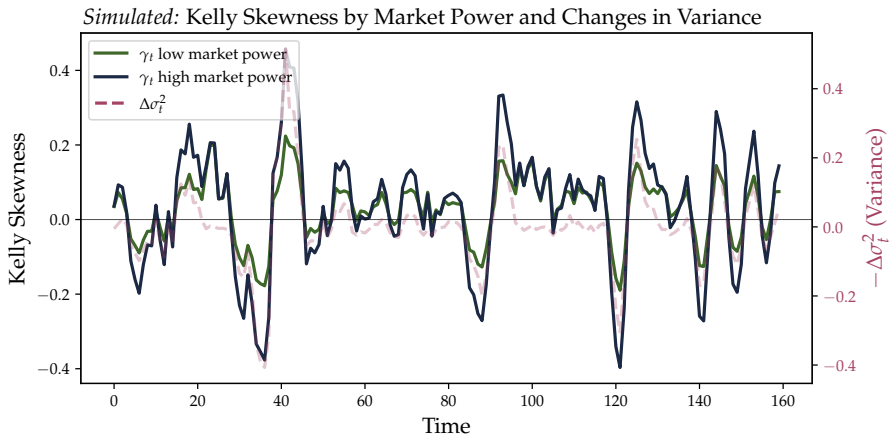
Figure 7: Simulation



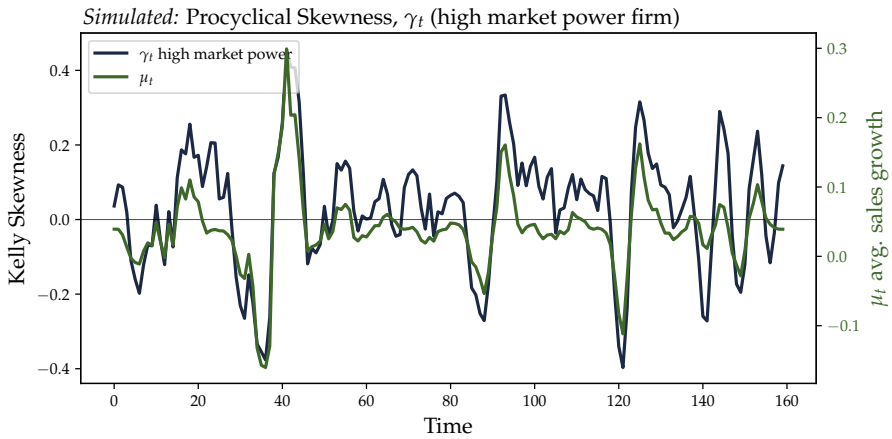
(a) Simulated time-series of a single aggregate factor.



(b) Two concave firm policies.



(c) Co-movement of cross-sectional skewness and changes in variance.



(a) Comovement of growth and skewness.

Note: Simulation procedure of procyclical skewness in pictures. Panel (a) shows an AR(2) process with jumps and slowly decaying IRF as aggregate shock. The aggregate shock scales the mean and the standard deviation of the cross sectional idiosyncratic input shocks, ϵ_i, t (not shown). Panel (b) displays the policy functions for log-output of two firms with different market power. Policy functions map shocks $\epsilon_{i,t}$ to $\widehat{q}$. Growth rates are subject to an additive idiosyncratic growth trend component (secular trend not shown, see algorithm). Panel (c) shows the simulated growth skewness across high- versus low-market power firms, and correlates those with changes in cross-sectional variance of input shocks. Panel (d) shows co-movement of mean and skewness.